

Sampling-law separation and finite nonlinear corrections in additive coagulation–fragmentation

Abstract

Additive coagulation with constant per-particle fragmentation can conserve particle count while the number and mass distributions move toward opposite size boundaries. We prove fractional-moment inequalities that quantify this separation, with coefficients that are sharp as uniform instantaneous constants, for measurable time controls and parent-dependent expected daughter measures. These coefficients are Lyapunov bounds, not asymptotic rates: the explicit Golovin–Borel solution decays several times faster. An auxiliary process with the normalized number marginals accumulates only a finite upward log-size correction and undergoes finitely many coagulations almost surely, even when its expected total number of coagulations is infinite. For parent-independent daughter fractions, this gives an exact finite transport cost to a fragmentation reference and transfers compound-Poisson scaling limits. We bound the last auxiliary coagulation time, prove daughter-law envelopes that are extremal over all mean-conserving daughter laws, and obtain exact representations of critical last-event observables in terms of the unknown law. The open critical exponent reduces to the decay rate of the deterministic number–mass overlap, and archived simulations suggest it is near half the coagulation rate. At critical balance $\sigma = \lambda m$, a physical finite population’s normalized count remains accurate on sublinear time horizons although mass-distribution error approaches its maximum by logarithmic times from identical initial data; this is a corollary of the moment bound and the material ceiling of a finite vessel. Finally, a Fourier factorization with an exponentially small remainder in time yields structural fragmentation identification from ideal late-time distributions on a low-frequency window, at a certified rate whose asymptotic optimality is not asserted. A separate independent sampling model gives a pointwise fixed-frequency ratio bound and a sufficient observation-time balance. These results distinguish bulk observability, auxiliary path asymptotics, and finite-population validity.

1 Introduction

Count, total mass, and mean size do not determine the evolution of a coagulating and fragmenting population. For the additive kernel $K(x, y) = \lambda(x + y)$ and constant per-particle fragmentation rate σ , mass conservation gives

$$N'(t) = (\sigma - \lambda m)N(t).$$

At the critical balance $\sigma = \lambda m > 0$, all three bulk observables remain fixed. Nevertheless, almost all particles by number become small, while almost all material is carried by increasingly large particles. The distinction between selecting a particle and selecting a unit of material is therefore central to both asymptotic analysis and observation. Throughout, “constant fragmentation rate” means constant per-particle selection rate; a constant binary fragmentation density can instead have a size-dependent integrated selection rate.

Population-balance modeling and the recovery of aggregation and breakage rates from dynamic size distributions have a long history; see Ramkrishna [33, Chapters 2, 6, and 7] for the modeling,

inverse, and stochastic viewpoints, and Aldous [2], Wattis [37] for mean-field surveys. Pure additive coagulation is explicitly solvable [23], has a probabilistic interpretation through branching structures and the additive coalescent [1, 3, 10], and a complete scaling theory [29]. Fractional-moment estimates are standard tools in the analysis of Smoluchowski equations [16, 17, 19, 20]. Constant count in an evolving population with aggregation and fragmentation is also established. Patil and Andrews [32] and Lage [27] provide constant-count examples, as explicitly discussed in the introduction of McCoy and Madras [28], whose solution allows count to vary. The results here concern quantitative separation of sampling laws, its pathwise consequences for the additive model, and the implications for finite populations and inverse observations.

1.1 Results and their relation to established ingredients

The main estimate is a decay inequality for every normalized fractional moment $\mathcal{A}_p = M_p/(N^{1-p}m^p)$, $0 < p < 1$. These quantities compare the number law $\eta_t = n_t/N(t)$ with the mass law $\pi_t = xn_t/m$; at $p = 1/2$ they give their Hellinger affinity. [Theorem 3.3](#) proves

$$\mathcal{A}_p(t) \leq \mathcal{A}_p(0) \exp \left[-a_p \int_0^t \lambda(s) m \, ds - a_{1-p} \int_0^t \sigma(s) \, ds \right], \quad a_p = 1 + p - 2^p.$$

The coefficients are sharp as uniform instantaneous coefficients over the class of solutions and daughter laws, in the explicit sense stated in the theorem. They are not asymptotic rates: for the Golovin–Borel solution the exact exponent of $\mathcal{A}_{1/2}$ is about six times $a_{1/2}$, and the critical simulations of [section 7](#) show a similar gap. The pair inequality behind the estimate is a weighted variant of the subadditivity-defect estimates used in the works cited above; the contribution is the explicit constant with its equality case and its uniform use for parent-dependent daughters. At criticality the estimate proves number concentration at zero, mass escape to infinity, and nonstationarity without a logarithmic moment assumption. Mass remains conserved at each finite time.

The same estimate controls the nonlinear contribution to log size. Probabilistic representations of coagulation equations and changes between sampling weights are established [10, 11]. We construct directly an auxiliary number process with marginals η_t , including measurable parent-dependent daughters. Its accumulated upward log increment is finite in mean and almost surely. A daughter-product martingale then proves that its total number of coagulation jumps is finite almost surely, although the mean can be infinite. With selfsimilar daughters, coupling to the independent fragmentation reference gives an exact extended Wasserstein cost. Classical compound-Poisson limit theory transfers to the nonlinear number law, with the daughter log moments needed for each limit stated separately. Pure-fragmentation logarithmic asymptotics already appear in Bertoin [4], Doumic and Escobedo [13]; the finite coagulation correction supplies the transfer here.

The last auxiliary coagulation gives a stronger path-law comparison. [Section 6](#) bounds its tail under arbitrary controls, proves all-constant-rate daughter-law envelopes by a convexity argument, and compares them with the number–mass overlap. At criticality the future hazard is an exponential functional of a compound-Poisson subordinator [6], and the equal-split formula is the Poissonian exponential functional of Bertoin et al. [7]. We derive exact representations of last-event observables and a density in terms of the unknown law, together with upper and lower calendar-time tail bounds. The exponent between these bounds is not determined; the envelopes reduce it to the decay rate of the deterministic overlap, and the archived simulations suggest the value $b/2$ of the pure-coagulation benchmark. The Golovin–Borel solution [23], recorded by Bertoin [5], provides that benchmark. These established formulas are used with explicit attribution.

A finite physical population must obey a total-material ceiling that the continuum equation does not share. Differences between finite stochastic coagulation–fragmentation systems and mean-field

predictions are already documented, for example by D’Orsogna et al. [12] in a finite-cluster assembly model. For the present critical additive model, [section 7](#) records an explicit distinction in growing time: starting from exactly the same empirical initial measure, mass cumulative distributions have nearly maximal discrepancy at least by times proportional to the logarithm of population scale, while normalized count is uniformly accurate on every sublinear horizon. The discrepancy statement is a corollary of the moment bound and the material ceiling of a finite vessel; only the count statement uses the particle dynamics. Nine reproducible particle runs illustrate separation diagnostics and the gap between certified and observed decay rates. They do not compute a continuum comparison.

Finally, the integrable nonlinear log correction gives a Fourier factorization whose remainder is exponentially small in time, at the certified rate $\kappa(b + \sigma)$. Local late-time quotients on a low-frequency window identify the fragmentation exponent and, by one-sided transform uniqueness, the expected daughter fraction measure. The second step is analytic continuation from a short interval and is severely ill-posed. This sits within established inverse population-balance work: Doumic et al. [14] uses asymptotic fragmentation profiles, Doumic et al. [15] uses short-time distributions, and Mirzaev et al. [30] estimates daughter distributions in an aggregation–fragmentation model. Two-time quotient cancellation and distinguished logarithms have a direct precedent in Garnier [21, Section 2.2]. The contribution of [section 9](#) is the proved nonlinear remainder and its identification consequence for additive coagulation. The statistical application is a pointwise bound under independent sampling, with no claim of stable recovery of a whole daughter measure or of statistical optimality.

1.2 Scope, assumptions, and organization

Three different probabilistic objects occur. The auxiliary process represents one-time continuum number laws; its jumps are not a physical material genealogy. The finite particle system describes an actual mass-conserving vessel with interacting particles. The empirical Fourier sampling bound assumes independent draws from deterministic continuum marginals. No independence or approximation between these objects is implicit in the results.

[Table 1](#) records the main changes of assumptions. All deterministic estimates are conditional on the mass-conserving weak-solution framework in [section 2](#). [Theorem 2.3](#) and [appendix A](#) prove existence and uniqueness with finite initial second moment, while most asymptotic estimates use only finite number and mass once a solution is supplied. A finite second size moment does not imply an integrable negative log size. No logarithmic moment is silently inherited from that existence theorem.

[Section 8](#) adds the complete fixed-mass count-neutral classification, rigidity from one further moment, and the failure of fractional-moment monotonicity beyond the additive kernel. These results clarify which conclusions depend on the additive structure. Elementary finite-grid preparation and initial-rate algebra, including dilution and source ambiguities, are collected in a supplementary note distributed with the numerical supplement. The appendices contain forward well-posedness, product error certification, finite-count formulas, and unbounded-test justifications. [Section 10](#) separates the proved conclusions from the remaining asymptotic and statistical questions.

2 Model, solution framework, and sampling laws

The size variable $x \in E := (0, \infty)$ is proportional to particle mass. We fix its reference unit throughout, so that logarithms of size and sums such as $1 + x$ below are dimensionless. The finite

Results	Additional scope or assumptions
Sampling separation and finite auxiliary correction	Locally integrable time controls; measurable parent-dependent expected daughters with count two and conserved mean mass.
Transport and logarithmic limits	Parent-independent fractions for the independent reference; fixed rates and fraction law for stationary increments; daughter log moments as stated.
Daughter envelopes and critical representations	Constant rates; critical formulas require $\sigma = \lambda m$; equal splitting only for the formulas labeled as such.
Finite-population comparison	Critical rates and eventwise complementary positive binary splits; continuum and particles start from the identical empirical measure.
Fourier identification and sampling	Constant $\lambda > 0$, $\sigma \geq 0$, and fixed parent-independent B ; exact distributions for identification, independent samples for the sampling bound.

Table 1: Assumption roadmap. Individual statements specify all moment, observation, and solution hypotheses.

nonnegative measure n_t describes particle number per unit volume. For $p \geq 0$, write

$$M_p(t) = \int_E x^p n_t(dx), \quad N(t) = M_0(t), \quad m = M_1(0). \quad (2.1)$$

We assume $0 < N_0 := N(0) < \infty$ and $0 < m < \infty$.

Assumption 2.1 (Rates and daughters). The functions $\lambda, \sigma : [0, \infty) \rightarrow [0, \infty)$ are measurable and locally integrable. Coagulation has the additive kernel $K_t(x, y) = \lambda(t)(x + y)$. Fragmentation has per-particle selection rate $\sigma(t)$. Its expected daughter measure is a measurable kernel $(t, x) \mapsto b_{t,x}$ on E , satisfying

$$b_{t,x}(E \setminus (0, x)) = 0, \quad \int_E b_{t,x}(dz) = 2, \quad \int_E z b_{t,x}(dz) = x. \quad (2.2)$$

These conditions hold for every (t, x) ; the values of the model at a common null set of times may be chosen arbitrarily subject to these conditions.

Only the expected daughter count and mass enter the deterministic equation. An actual binary split has expected measure $\mathbb{E}[\delta_{Vx} + \delta_{(1-V)x}]$ for a random $V \in (0, 1)$. Conditions (2.2) allow a larger class of expected measures: they do not require symmetry about $x/2$, which an eventwise complementary binary split would impose. Results about a finite system of particles will explicitly require eventwise binary mass conservation. Unless stated otherwise, the deterministic results use only (2.2).

Write $\langle f, \mu \rangle = \int_E f d\mu$ whenever this integral is defined. For a Borel test f , define

$$\Delta f(x, y) = f(x + y) - f(x) - f(y), \quad \mathcal{F}_t f(x) = \int_E f(z) b_{t,x}(dz) - f(x). \quad (2.3)$$

The weak population balance is

$$\begin{aligned} \langle f, n_t \rangle - \langle f, n_0 \rangle &= \frac{1}{2} \int_0^t \lambda(s) \iint_{E^2} (x + y) \Delta f(x, y) n_s(dx) n_s(dy) ds \\ &\quad + \int_0^t \sigma(s) \int_E \mathcal{F}_s f(x) n_s(dx) ds. \end{aligned} \quad (2.4)$$

The factor $1/2$ counts each unordered coagulation pair once.

Definition 2.2 (Mass-conserving solution). A solution is a curve of nonnegative finite measures n_t on E such that $t \mapsto n_t(A)$ is measurable for each Borel set $A \subset E$, with $\sup_{s \leq T} N(s) < \infty$ for every $T < \infty$, constant first moment $M_1(t) = m$, and (2.4) for every bounded Borel f and every $t \geq 0$.

All terms in this definition are absolutely integrable on bounded time intervals: their absolute values are bounded using $\iint (x+y)n_s(dx)n_s(dy) = 2mN(s)$ and $|\mathcal{F}_s f| \leq 3\|f\|_\infty$. In particular the solution is continuous in total variation. For a signed measure μ , write

$$\|\mu\|_w = \int_E (1+x)|\mu|(dx), \quad \|\mu\|_{\text{var}} = |\mu|(E). \quad (2.5)$$

For probability measures the distance $\|\mu - \nu\|_{\text{TV}} := \sup_A |\mu(A) - \nu(A)|$ equals $\frac{1}{2} \|\mu - \nu\|_{\text{var}}$.

Theorem 2.3 (A well-posed solution class). *Under assumption 2.1, suppose additionally that $M_2(0) < \infty$. There is a unique mass-conserving solution with locally bounded second moment. It is continuous in the weighted-variation norm $\|\cdot\|_w$ and satisfies*

$$M_2(t) \leq M_2(0) \exp\left(2m \int_0^t \lambda(s) ds\right). \quad (2.6)$$

No continuity of the daughter kernel in the parent variable is required.

A self-contained proof appears in [appendix A](#). It uses a weighted-variation stability estimate and approximations that converge in that norm. This avoids passing a merely measurable parent-dependent kernel through a narrow limit. The proof also avoids treating weak integrals of moving atomic measures as Bochner integrals. We use the second moment as a sufficient condition for this construction, not as a necessary condition for the population balance or the results below. In the time-independent selfsimilar setting, weaker initial-moment assumptions are available for fixed relative-fragment laws satisfying the dislocation hypotheses of Cepeda [9]. That work treats homogeneous-like coagulation and multiple fragmentation using a Wasserstein-type distance. The finite first moment is the relevant homogeneity moment for the additive kernel. Neither that result nor the proof here is a claim of novelty for existence methods.

The estimates that follow require only [definition 2.2](#), unless a theorem explicitly adds moment assumptions. Thus they also apply to any separately constructed finite-count, finite-mass solution outside [theorem 2.3](#). This distinction is useful for logarithmic observables: neither a finite second size moment nor finite count and mass implies a finite negative logarithmic moment.

Set

$$b(t) = m\lambda(t), \quad B_c(t) = \int_0^t b(s) ds, \quad F(t) = \int_0^t \sigma(s) ds, \quad C(t) = B_c(t) + F(t). \quad (2.7)$$

Here $b(t)$ is a scalar rate; the two-index symbol $b_{t,x}$ always denotes the daughter measure. Testing (2.4) with $f = 1$ shows that N is locally absolutely continuous, with

$$N'(t) = (\sigma(t) - b(t))N(t) \quad \text{for almost every } t, \quad N(t) = N_0 e^{F(t) - B_c(t)} \quad (t \geq 0). \quad (2.8)$$

In particular $N(t) > 0$ at each finite time. The number and mass sampling laws are respectively

$$\eta_t(dx) = \frac{n_t(dx)}{N(t)}, \quad \pi_t(dx) = \frac{x n_t(dx)}{m}, \quad \frac{d\pi_t}{d\eta_t}(x) = \frac{N(t)x}{m}. \quad (2.9)$$

The first law samples particles uniformly by number; the second samples a uniform unit of material and records the size of its containing particle. Their distinction is central even when their underlying

population is the same. Probabilistic representations of mass sampling in pure coagulation are established; see, for example, Deaconu et al. [11].

For $0 < p < 1$, define the normalized fractional moment

$$\mathcal{A}_p(t) = \frac{M_p(t)}{N(t)^{1-p} m^p} = \int_E \left(\frac{d\pi_t}{d\eta_t} \right)^p d\eta_t. \quad (2.10)$$

Hölder's inequality gives $0 < \mathcal{A}_p \leq 1$. At $p = 1/2$ it is the Hellinger affinity of the two sampling laws. No density of n_t with respect to Lebesgue measure is assumed anywhere in these definitions.

3 Fractional-moment decay and sampling-law separation

3.1 The pair inequality and admissible moment tests

Lemma 3.1 (Sharp fractional pair inequality). *For $x, y > 0$ and $0 < p < 1$,*

$$(x + y)[x^p + y^p - (x + y)^p] \geq (2 - 2^p)(xy^p + yx^p). \quad (3.1)$$

Equality holds exactly when $x = y$.

Proof. Put $u = x/(x + y)$, $v = 1 - u$, and $A = 2^p - 1 \in (0, 1)$. After division by $(x + y)^{p+1}$, (3.1) is equivalent to $g(u) \geq 1$, where

$$g(u) = A(u^p + v^p) + (1 - A)(u^{p+1} + v^{p+1}).$$

By symmetry it suffices to consider $0 < u < 1/2$. The continuous endpoint values are $g(0) = g(1/2) = 1$, whereas $g'(0+) = +\infty$ and $g'(1/2) = 0$. The sign of $g''(u)$ is the sign of

$$(1 - A)(p + 1) - A(1 - p)R(u), \quad R(u) = \frac{u^{p-2} + v^{p-2}}{u^{p-1} + v^{p-1}} = \frac{u^{2-p} + v^{2-p}}{uv(u^{1-p} + v^{1-p})}.$$

On $(0, 1/2)$ the numerator in the final expression strictly decreases, and each positive factor in its denominator strictly increases. Hence R strictly decreases and g'' changes sign at most once, from negative to positive. This change must occur. Since $R(u) \rightarrow +\infty$ as $u \downarrow 0$, $g'' < 0$ near 0; equivalently, $g'(0+) = +\infty$ is incompatible with g' increasing on $(0, 1/2)$ to the value $g'(1/2) = 0$. Nor can g'' remain negative: then g' would decrease to zero while remaining positive, contradicting the equal endpoint values of g . On the final interval where $g'' > 0$, the derivative approaches zero from below. On the initial interval it therefore crosses zero exactly once. Thus g first increases and then decreases to its endpoint value 1, and $g(u) > 1$ in $(0, 1/2)$. This proves the inequality and its equality case. $\square$

Estimates of the subadditivity defect $x^p + y^p - (x + y)^p$ are standard tools in the fractional-moment analysis of Smoluchowski equations; see Escobedo et al. [16], Fournier and Laurençot [19, 20], and Escobedo et al. [17]. Inequality (3.1) is a weighted variant of these estimates, with the additive weight $x + y$ built in. The contribution here is the explicit constant $2 - 2^p$ with its equality case, and its use below as a uniform coefficient for parent-dependent daughters.

Lemma 3.2 (Fractional moment balance). *For every solution in definition 2.2 and every $0 < p < 1$, M_p is locally absolutely continuous. Its weak balance is (2.4) with $f(x) = x^p$, with absolutely integrable right-hand side on finite intervals.*

Proof. Hölder's inequality gives $M_p(t) \leq N(t)^{1-p} m^p$. Use the bounded Borel tests $f_R(x) = \min(x^p, R)$. They are increasing, concave, and subadditive, with $f_R(0) = 0$. Consequently

$$0 \leq f_R(x) + f_R(y) - f_R(x+y) \leq \min(x^p, y^p).$$

After multiplication by $x+y$ this is bounded by $xy^p + yx^p$, whose integral against $n_t \otimes n_t$ is $2mM_p(t)$. For daughters, $f_R(z)/z$ is nonincreasing in z ; since $z < x$ and daughter mass is conserved,

$$0 \leq \int_E f_R(z) b_{t,x}(dz) - f_R(x) \leq \int_E z^p b_{t,x}(dz) \leq 2^{1-p} x^p.$$

The last inequality is Jensen's inequality for the probability $b_{t,x}/2$, whose mean is $x/2$. These bounds are uniform in R and integrable in time after multiplication by the locally integrable rates. Monotone convergence on the left and dominated convergence on the right of (2.4) yield the asserted identity. Its integrable right-hand side also proves local absolute continuity. $\square$

3.2 A uniform estimate for controlled rates

Write

$$a_p = 1 + p - 2^p > 0 \quad (0 < p < 1), \quad \kappa = 3 - 2\sqrt{2} = 2a_{1/2} > 0. \quad (3.2)$$

Positivity of a_p follows from strict convexity of 2^p and its chord between $p = 0$ and $p = 1$.

Theorem 3.3 (Separation of the sampling laws). *Under [assumption 2.1](#) and [definition 2.2](#), for every $0 < p < 1$,*

$$M'_p(t) \leq [-b(t)(2 - 2^p) + \sigma(t)(2^{1-p} - 1)]M_p(t) \quad \text{for almost every } t, \quad (3.3)$$

$$\mathcal{A}_p(t) \leq \mathcal{A}_p(0)e^{-a_p B_c(t) - a_{1-p} F(t)}. \quad (3.4)$$

In particular, writing $H(t) = M_{1/2}(t)$,

$$\frac{H(t)^2}{N(t)} \leq \frac{H(0)^2}{N_0} e^{-\kappa C(t)}. \quad (3.5)$$

The coefficients are instantaneously sharp in the following sense: if constants c_1, c_2 satisfy $\mathcal{A}_p(t) \leq \mathcal{A}_p(0)e^{-c_1 B_c(t) - c_2 F(t)}$ for all $t \geq 0$, all solutions of [theorem 2.3](#), and all daughter kernels satisfying (2.2), then $c_1 \leq a_p$ and $c_2 \leq a_{1-p}$.

Proof. By [lemma 3.1](#), the coagulation term in the fractional moment balance is at most $-\lambda(t)(2 - 2^p)mM_p(t)$. Jensen's inequality in the proof of [lemma 3.2](#) bounds the fragmentation contribution by $\sigma(t)(2^{1-p} - 1)M_p(t)$. This proves (3.3). Differentiate the number normalization using (2.8). Its contribution is $-(1-p)(\sigma - b)$, and the resulting coefficients are $-a_p b - a_{1-p} \sigma$. An integrating factor proves (3.4) without dividing by M_p . Taking $p = 1/2$ and squaring gives (3.5).

For the sharpness statement take constant rates, initial data $n_0 = N_0 \delta_{x_0}$, and equal daughters $b_x = 2\delta_{x/2}$, so that $\mathcal{A}_p(0) = 1$. For $f(x) = x^p$, weighted-variation continuity from [theorem 2.3](#), the bound $(x+y)|\Delta f(x, y)| \leq xy^p + yx^p \leq 2(1+x)(1+y)$, and the equal-split identity $\mathcal{F}f(x) = (2^{1-p} - 1)x^p$ make the moment-balance integrand continuous at zero. Thus M_p has a right derivative at zero, equal to the value of the vector field at n_0 . There both the pair inequality and Jensen's inequality are equalities, so $M'_p(0+) = [-b(2 - 2^p) + \sigma(2^{1-p} - 1)]M_p(0)$, and by (2.8) $\mathcal{A}'_p(0+) = -a_p b - a_{1-p} \sigma$. Now let c_1, c_2 satisfy the stated bound. With $\sigma = 0$ and $\lambda > 0$ the bound gives $\mathcal{A}_p(t) \leq 1 - c_1 b t + o(t)$ as $t \downarrow 0$, while $\mathcal{A}_p(t) = 1 - a_p b t + o(t)$; hence $c_1 \leq a_p$. With $\lambda = 0$ and $\sigma > 0$ the same comparison gives $c_2 \leq a_{1-p}$.

For equal splitting the identity $\int z^p b_x(dz) = 2^{1-p} x^p$ holds for every parent size x , not only at a monodisperse state. Hence with $\lambda = 0$, [lemma 3.2](#) gives $M_p(t) = M_p(0)e^{(2^{1-p}-1)F(t)}$ and, by (2.8), $\mathcal{A}_p(t) = \mathcal{A}_p(0)e^{-a_{1-p}F(t)}$ for every initial measure and every $t \geq 0$. The fragmentation coefficient is then exact for all times, not only instantaneously. $\square$

Remark 3.4 (What the estimate compares). If $C(t) \rightarrow \infty$, the sampling laws separate in total variation, because

$$1 - \|\eta_t - \pi_t\|_{\text{TV}} = \int_E \min\left(1, \frac{N(t)x}{m}\right) \eta_t(dx) \leq \mathcal{A}_p(t) \rightarrow 0. \quad (3.6)$$

This concerns relative number and mass sampling. Outside the critical case below it does not imply that raw mass moves to arbitrarily large sizes.

Instantaneous sharpness does not identify the long-time exponent for a prescribed daughter law or initial distribution, and the gap can be large. For pure additive coagulation ($\sigma = 0$, constant λ , unit monodisperse initial data with $N_0 = m = 1$, so that $b = \lambda$) the explicit solution of Golovin [23], recorded in Bertoin [5], has $N(t) = e^{-bt}$ and number law η_t equal to the Borel law with parameter $\mu_t = 1 - e^{-bt}$, that is $\eta_t(\{k\}) = e^{-\mu_t k} (\mu_t k)^{k-1}/k!$; for its probabilistic interpretation see Bertoin [3], Deaconu and Tanré [10], and for the scaling theory of this kernel see Menon and Pego [29]. As $t \rightarrow \infty$ the number law converges to the Borel law with parameter one, whose tail is of order $k^{-3/2}$. Consequently M_p/N stays bounded for $p < 1/2$ and grows like $e^{(2p-1)bt}$ for $p > 1/2$, and $\mathcal{A}_p = (M_p/N)N^p$ decays at the exact exponential rate $\min(p, 1-p)b$, in the sense that $-t^{-1} \log \mathcal{A}_p(t) \rightarrow \min(p, 1-p)b$. For $p = 1/2$ this is about six times the coefficient $a_{1/2} \approx 0.086$ in (3.4). The critical equal-split particle simulations of [section 7](#) show a gap of the same kind: for the largest population the fitted late-time decay rate of $\mathcal{A}_{1/2}$ is about $0.45b$, against the guaranteed $\kappa b \approx 0.17b$. The estimate is therefore a uniform Lyapunov bound, valid for every admissible daughter law and initial measure, not an asymptotic rate.

3.3 Criticality: constant count and motion toward both boundaries

Suppose now that $\lambda > 0$ is constant and $\sigma = b$, so that both rates are constant. The daughter measure may still depend measurably on time and parent size. Count remains exactly N_0 . Set

$$\kappa_p = 3 - 2^p - 2^{1-p} = a_p + a_{1-p} > 0. \quad (3.7)$$

Corollary 3.5 (Two-boundary separation). *At criticality, for every $0 < p < 1$,*

$$M_p(t) \leq M_p(0)e^{-b\kappa_p t}. \quad (3.8)$$

For each fixed p the coefficient is instantaneously sharp in the sense of [theorem 3.3](#): if $M_p(t) \leq M_p(0)e^{-c b t}$ for all $t \geq 0$, all critical solutions of [theorem 2.3](#), and all daughter kernels satisfying (2.2), then $c \leq \kappa_p$. Among these coefficients the unique maximum is $\kappa_{1/2} = \kappa$. For every $\varepsilon, R > 0$,

$$n_t([\varepsilon, \infty)) \leq \varepsilon^{-p} M_p(0) e^{-b\kappa_p t}, \quad (3.9)$$

$$\int_{(0, R]} x n_t(dx) \leq R^{1-p} M_p(0) e^{-b\kappa_p t}. \quad (3.10)$$

Viewed as measures on $[0, \infty)$, n_t converges weakly to $N_0 \delta_0$. On the compactified space $[0, \infty]$, π_t converges weakly to δ_∞ . No nonzero stationary solution with finite count and positive finite mass exists.

Proof. Equation (3.8) follows from (3.3); its sharpness follows from the monodisperse equal-split example in the proof of theorem 3.3, where $M_p'(0+) = -b\kappa_p M_p(0)$, so $M_p(t) = M_p(0)(1 - b\kappa_p t + o(t))$ and $c \leq \kappa_p$. The strict convexity and symmetry of $2^p + 2^{1-p}$ give its unique minimum $2\sqrt{2}$ at $p = 1/2$, and its value is strictly below the common endpoint value 3 in $(0, 1)$. On $[\varepsilon, \infty)$, $x^p \geq \varepsilon^p$; on $(0, R]$, $x \leq R^{1-p} x^p$. These prove the window bounds. For a bounded continuous function on $[0, \infty)$, its oscillation near zero is small and the number outside that neighborhood tends to zero. This proves the first weak limit. A continuous function on $[0, \infty]$ is uniformly close to its value at infinity outside a sufficiently large bounded interval; (3.10) proves the second limit. A stationary measure would have $M_p > 0$ constant in time, contradicting (3.8). $\square$

Both limits are long-time statements. All material mass is conserved at every finite time. The particles that become rare in number carry most of the material. There is no assertion of finite-time shattering or gelation. On a fixed bounded size interval with an upper cutoff R , positive mass m forces $M_p \geq m/R^{1-p}$; hence a discretization that confines all mass there cannot satisfy (3.8) indefinitely.

Corollary 3.6 (Exponentially moving observation windows). *For $\varepsilon_0, R_0 > 0$ and $0 < v < b \log 2$, both*

$$n_t([\varepsilon_0 e^{-vt}, \infty)) \longrightarrow 0, \quad \int_{(0, R_0 e^{vt}]} x n_t(dx) \longrightarrow 0 \quad (3.11)$$

at exponential rates.

Proof. The two respective bounds, for arbitrary $p \in (0, 1)$, are

$$\varepsilon_0^{-p} M_p(0) e^{-[b\kappa_p - pv]t}, \quad R_0^{1-p} M_p(0) e^{-[b\kappa_p - (1-p)v]t}.$$

Since $\kappa_p/p \rightarrow \log 2$ as $p \downarrow 0$ and $\kappa_p/(1-p) \rightarrow \log 2$ as $p \uparrow 1$, a suitable exponent exists for each bound. They generally use different exponents p . Since κ_p is concave with $\kappa_0 = 0$, the quotient κ_p/p decreases in p , so $\sup_{0 < p < 1} \kappa_p/p = \log 2$; by the symmetry $\kappa_p = \kappa_{1-p}$ the same holds for $\kappa_p/(1-p)$. Thus $v < b \log 2$ is exactly the range this argument reaches. $\square$

These guaranteed velocities do not give a matching asymptotic front speed. The endpoint $v = b \log 2$ is not included.

3.4 A nonstationarity criterion under rate perturbations

Proposition 3.7 (A sufficient open rate condition). *Consider a jointly measurable symmetric kernel $(t, x, y) \mapsto K_t(x, y)$ and jointly measurable selection rate $(t, x) \mapsto S_t(x)$ satisfying*

$$a(x + y) \leq K_t(x, y) \leq A(x + y), \quad 0 \leq S_t(x) \leq s_*, \quad 0 < a \leq A < \infty, \quad (3.12)$$

with daughters as in assumption 2.1. Use definition 2.2 with $\lambda(s)(x + y)$ replaced by $K_s(x, y)$ in the coagulation term of (2.4), and $\sigma(s)\mathcal{F}_s f(x)$ replaced by $S_s(x)\mathcal{F}_s f(x)$ in the fragmentation term. In particular, the balance holds for every bounded Borel test, count is locally bounded, and mass is conserved. For every such solution of mass m and every $0 < p < 1$,

$$M_p(t) \leq M_p(0) e^{-\gamma_p t}, \quad \gamma_p = (2^{1-p} - 1)(am 2^p - s_*). \quad (3.13)$$

If $s_ < 2am$, no stationary measure of mass m has finite positive particle count.*

Proof. The count balance gives $N(t) \leq N_0 e^{s_* t}$. The upper bounds in (3.12) and the proof of lemma 3.2 justify the fractional test. Its coagulation contribution is at most $-am(2 - 2^p)M_p$ by the lower kernel bound, and fragmentation contributes at most $s_*(2^{1-p} - 1)M_p$. Their difference equals $-\gamma_p M_p$ because $2 - 2^p = 2^p(2^{1-p} - 1)$. If $s_* < 2am$, choose $p < 1$ close enough to one that $am2^p > s_*$. A stationary measure would then have a strictly positive moment that decays exponentially, a contradiction. $\square$

The threshold here is sufficient. We make no claim of its optimality or of existence of stationary solutions above it. Count need not be constant under (3.12).

4 An auxiliary number process and its finite coagulation correction

Fix a solution in definition 2.2. In this section the rates and daughter kernel retain the full generality of assumption 2.1. The construction uses the deterministic curve n_t as its environment. Its one-time laws are the number sampling laws η_t , but its paths are not physical mass tags or chronological particle lineages.

4.1 Construction and identification of the marginals

Theorem 4.1 (Number-weighted jump representation). *There is a nonexplosive time-inhomogeneous Markov jump process X_t on E , with initial law η_0 , whose law at every t is η_t . Its generator, meaning its jump kernel acting on bounded Borel functions f , is*

$$\begin{aligned} \mathcal{L}_t f(x) = & \sigma(t) \int_E [f(z) - f(x)] b_{t,x}(dz) \\ & + \lambda(t) N(t) x \int_E [f(x+y) - f(x)] \eta_t(dy). \end{aligned} \quad (4.1)$$

Thus fragmentation occurs at rate $2\sigma(t)$ and selects a daughter with law $b_{t,x}/2$. Coagulation occurs at rate $\lambda(t)N(t)x$ and adds a fresh sample from the prescribed law η_t . The fragmentation term has a direct interpretation: a fragmentation event replaces the tagged particle by one of its two daughters chosen uniformly, which has law $b_{t,x}/2$; number weighting counts both daughters, so the number-weighted process sees this daughter law at twice the selection rate $\sigma(t)$.

Proof. Symmetry rewrites the coagulation term of (2.4) as

$$\lambda(t) \iint x [f(x+y) - f(x) - f(y)] n_t(dx) n_t(dy).$$

For each bounded Borel f , divide its scalar, absolutely continuous balance by $N(t)$ and use (2.8). The result is

$$\langle f, \eta_t \rangle - \langle f, \eta_0 \rangle = \int_0^t \langle \mathcal{L}_s f, \eta_s \rangle ds. \quad (4.2)$$

All these integrals are finite on bounded time intervals.

For the construction use two independent Poisson random measures, independent also of X_0 . The fragmentation driver has intensity $2\sigma(t) dt du$, $0 < u < 1$; the quantile map $(t, x, u) \mapsto \inf\{z : b_{t,x}((0, z])/2 \geq u\}$ chooses its daughter using u . This map is jointly measurable: the set where it is at most z equals $\{(t, x, u) : b_{t,x}((0, z])/2 \geq u\}$, by right-continuity in z of the distribution function, and $(t, x, z) \mapsto b_{t,x}((0, z])$ is measurable because b is a measurable kernel. The coagulation driver has intensity $dt dv du$, $v > 0$, $0 < u < 1$; retain a point when $v \leq \lambda(t)N(t)X_{t-}$ and use the quantile

map of η_t , jointly measurable in (t, u) for the same reason, to choose its added size. Let $(\mathfrak{F}_t)_{t \geq 0}$ be the filtration generated by X_0 and the two driving Poisson random measures, augmented with the $\mathbb{P}$ -null sets; it satisfies the usual conditions. Because the two drivers are independent of each other and of X_0 , each driver keeps its deterministic compensator in this joint filtration. All stopping times, compensators, and local martingales in this section refer to $(\mathfrak{F}_t)$. Recursive construction up to successive jumps gives the minimal process, initially defined up to a possible explosion time. Its jump kernel $Q_t(x, dz)$ is the sum of the two positive kernels in (4.1), and its total rate is

$$q_t(x) = 2\sigma(t) + \lambda(t)N(t)x. \quad (4.3)$$

These rates are bounded by an integrable time function on every bounded size set and finite time interval.

Here is a first-moment nonexplosion argument. Daughter mass conservation and $\langle x, \eta_t \rangle = m/N(t)$ give, for $V(x) = x$,

$$\mathcal{L}_t V(x) = (b(t) - \sigma(t))x \leq b(t)x. \quad (4.4)$$

The generator was defined on bounded functions; its application to the unbounded V is justified by the stopping that follows, which keeps every term of the Dynkin formula finite. Stop at the k th jump τ_k and at τ_R , the first time the state exceeds R . Before these stops rates are integrably bounded. An upward jump can overshoot R , but its mark has finite conditional mean $m/N(t)$; the expected positive and negative increments in the stopped Dynkin formula are therefore finite. Gronwall's inequality gives $\mathbb{E}X_{t \wedge \tau_k \wedge \tau_R} \leq \mathbb{E}X_0 e^{B_c(t)}$. For fixed k , only finitely many finite states are visited, so the size stopping can be removed as $R \rightarrow \infty$. Fatou's lemma yields

$$\mathbb{E}X_{t \wedge \tau_k} \leq \frac{m}{N_0} e^{B_c(t)}. \quad (4.5)$$

The jump-count compensator is the time integral of the conditional jump rate; after integrable stopping, its expectation equals the expected jump count. The compensator of the stopped total jump count has expectation at most

$$L_T := 2F(T) + \frac{m}{N_0} \int_0^T \lambda(s)N(s)e^{B_c(s)} ds < \infty. \quad (4.6)$$

With the size stop in place all rates are integrably bounded, so the compensator identity is exact; as $R \rightarrow \infty$ the stopped counts and their compensators increase to those stopped at $\tau_k \wedge T$ alone, and monotone convergence in R justifies the compensator calculation without the size stop. Since $k\mathbb{P}(\tau_k \leq T) \leq L_T$, there is no finite-time explosion. It also proves nonexplosion from each deterministic finite starting state. Recursive sampling with a prescribed environment gives the Markov property.

It remains to identify the marginals; a formal forward equation alone would not suffice. The prescribed probability curve η_t has integrable jump flux, since

$$\int_E q_t(x)\eta_t(dx) = 2\sigma(t) + b(t). \quad (4.7)$$

Put $G_s(dz) = \int_E Q_s(x, dz)\eta_s(dx)$ and $S_{s,t}(z) = \exp(-\int_s^t q_u(z) du)$. The weak loss identity (A.3) of [appendix A](#), used with weight one, turns (4.2) into

$$\eta_t(dz) = S_{0,t}(z)\eta_0(dz) + \int_0^t S_{s,t}(z)G_s(dz) ds. \quad (4.8)$$

What is used is the following. Restrict the output variable to $z \in (0, R]$; there the loss multiplier $q_s(z)$ is bounded by the integrable function $2\sigma(s) + \lambda(s)N(s)R$, and the incoming gain is the restriction of the *full* G_s , including arrivals from parents larger than R . The restricted loss equation has a unique solution with bounded total variation on bounded time intervals, given by the exponential formula: existence is the convergent loss-operator series, and uniqueness is Gronwall's inequality for the variation of the difference of two solutions. This proves the formula on the output set $(0, R]$, and positivity allows $R \rightarrow \infty$. Both gain and loss have finite integrated total mass by (4.7). This argument uses weak kernel integrals, not a Bochner derivative.

The right-hand side of (4.8) is positive. Its successive substitutions show that η_t dominates the sum of the zero-jump, one-jump, and all subsequent finite-jump contributions of the minimal construction. More explicitly, start with $\nu_t^{(0)} = S_{0,t}\eta_0$ and define

$$\nu_t^{(j+1)}(dz) = \int_0^t S_{s,t}(z)(Q_s^* \nu_s^{(j)})(dz) ds.$$

Then $\eta_t \geq \sum_{j=0}^k \nu_t^{(j)}$ for every k . Nonexplosion makes $\sum_{j \geq 0} \nu_t^{(j)}$ a probability measure, the law of X_t . Domination between probability measures forces equality. No additional uniqueness theorem for an unbounded forward operator, or second size moment, is needed. $\square$

Remark 4.2 (Relation to mass tagging and to classical constructions). None of the ingredients of [theorem 4.1](#) is new, and we state plainly what is classical. The fragmentation part, rate 2σ with daughter law $b_{t,x}/2$, is the classical many-to-one or intensity-measure computation for finite-activity conservative binary fragmentation: the expected number of particles in a Borel set evolves linearly, and its normalization by the total count is the law of a tag that follows a uniformly chosen daughter. The coagulation part is the number-weighted counterpart of the mass tagging of Deaconu et al. [11]: the number-weighted process is the $h = 1/x$ Doob transform of that mass-weighted process, as the identity below shows. For pure coagulation the auxiliary path is the cluster growth chain associated with the additive coalescent of Aldous and Pitman [1], seen through the total progeny of Poisson Galton–Watson trees (cf. Bertoin [3] and [corollary 5.8](#)). The new ingredient of this section is only the finite correction bound of [theorem 4.3](#).

Jump representations of mass sampling in coagulation are established; see Deaconu et al. [11]. The corresponding mass-tag generator, with the present fragmentation added, is

$$\mathcal{L}_t^{\text{mass}} f(x) = \lambda(t) \int_E (x+y)[f(x+y) - f(x)] n_t(dy) + \frac{\sigma(t)}{x} \int_E z[f(z) - f(x)] b_{t,x}(dz).$$

For $h(x) = 1/x$, direct integration gives $\mathcal{L}_t^{\text{mass}} h = (\sigma(t) - b(t))h$ and the algebraic inverse-size Doob-transform identity

$$h^{-1} \mathcal{L}_t^{\text{mass}}(hf) - (\sigma(t) - b(t))f = \mathcal{L}_t f.$$

This calculation explains the change of sampling weight and the fragmentation rate 2σ . It is an identity of generators; the preceding direct construction supplies the required path law. In particular the mass-tag coagulation rate is $\lambda(t)N(t)x + b(t)$, whereas the number process has rate $\lambda(t)N(t)x$. Conclusions about the number process below are specific to this representation.

4.2 A finite logarithmic correction under arbitrary controls

Let s_j be the fragmentation times and $\Theta_j = X_{s_j}/X_{s_j-} \in (0, 1)$ their retained fractions. For a coagulation time s , write $U_s = X_s - X_{s-} > 0$ and set

$$A_t = \sum_{\substack{s \leq t \\ s \text{ coagulation}}} \log(1 + U_s/X_{s-}), \quad a(t) = \frac{\lambda(t)}{N(t)} \iint_{E^2} x \log(1 + y/x) n_t(dx) n_t(dy). \quad (4.9)$$

Theorem 4.3 (Finite correction and finite total coagulations). *For the process in [theorem 4.1](#),*

$$\log X_t = \log X_0 + \sum_{s_j \leq t} \log \Theta_j + A_t. \quad (4.10)$$

The nonnegative process A_t increases to a finite A_∞ almost surely and in L^1 , with $\mathbb{E}A_t = \int_0^t a(s) ds$ and

$$\begin{aligned} \mathbb{E}(A_\infty - A_t) &\leq \frac{H(0)^2}{mN_0} \int_t^\infty b(s)e^{-\kappa C(s)} ds \\ &\leq \frac{H(0)^2}{\kappa mN_0} e^{-\kappa C(t)}. \end{aligned} \quad (4.11)$$

In particular $\mathbb{E}A_\infty \leq 1/\kappa$. The total number K_∞ of auxiliary coagulation jumps is finite almost surely, although its mean is exactly

$$\mathbb{E}K_t = B_c(t), \quad \mathbb{E}K_\infty = B_c(\infty), \quad (4.12)$$

which is infinite whenever $B_c(\infty) = \infty$, in particular for every constant $\lambda > 0$. No initial or daughter logarithmic moment is assumed.

Proof. Nonexplosion makes [\(4.10\)](#) an identity of finite sums at each finite time. The elementary inequality $\log(1+r) \leq \sqrt{r}$ for $r > 0$ follows by differentiating their difference: the derivative is $-(\sqrt{r}-1)^2/[2\sqrt{r}(1+r)] \leq 0$, and the difference tends to zero at zero. Thus [theorem 3.3](#) gives

$$0 \leq a(t) \leq \frac{\lambda(t)H(t)^2}{N(t)} \leq \frac{H(0)^2}{mN_0} b(t)e^{-\kappa C(t)}. \quad (4.13)$$

The nonnegative jump compensator and marginal identification give $\mathbb{E}A_t = \int_0^t a(s) ds$. Since $C' = b + \sigma \geq b$,

$$\int_t^\infty b(s)e^{-\kappa C(s)} ds \leq \frac{e^{-\kappa C(t)} - e^{-\kappa C(\infty)}}{\kappa} \leq \frac{1}{\kappa} e^{-\kappa C(t)},$$

where $e^{-\infty} = 0$. Monotone convergence proves finiteness of $\mathbb{E}A_\infty$ and [\(4.11\)](#); it also gives $\mathbb{E}|A_\infty - A_t| \rightarrow 0$. When $C(\infty) < \infty$, the final exponential bound need not tend to zero, but the preceding integral bound does. Finally $H(0)^2 \leq mN_0$.

To prove finite total coagulation, define

$$\mathcal{M}_t = e^{F(t)} \prod_{s_j \leq t} \Theta_j. \quad (4.14)$$

Conditional on the full past and on a fragmentation at (t, x) , the multiplier has mean

$$\frac{1}{2x} \int_E z b_{t,x}(dz) = \frac{1}{2}.$$

The predictable drift of [\(4.14\)](#) is therefore $\sigma(t)\mathcal{M}_{t-} + 2\sigma(t)\mathcal{M}_{t-}(1/2 - 1) = 0$. It is a local martingale in the construction filtration $(\mathfrak{F}_t)$, which includes the coagulation randomness. Since $0 < \mathcal{M}_t \leq e^{F(T)}$ on $[0, T]$, it is a true mean-one martingale on every finite horizon. Doob's maximal inequality gives $\mathbb{P}(\sup_{t \leq T} \mathcal{M}_t > L) \leq 1/L$; first let $T \rightarrow \infty$ and then $L \rightarrow \infty$ to obtain $\sup_{t \geq 0} \mathcal{M}_t < \infty$ almost surely.

Combining [equations \(2.8\), \(4.10\) and \(4.14\)](#),

$$N(t)X_t = N_0X_0 e^{-B_c(t)} \mathcal{M}_t e^{A_t}.$$

Hence the cumulative predictable coagulation intensity $\Lambda_t := \int_0^t \lambda(s)N(s)X_{s-} ds$ satisfies

$$\begin{aligned}\Lambda_\infty &:= \lim_{t \rightarrow \infty} \Lambda_t \\ &\leq \frac{N_0 X_0}{m} e^{A_\infty} \sup_{t \geq 0} \mathcal{M}_t \int_0^\infty b(t) e^{-B_c(t)} dt < \infty \quad \text{almost surely,}\end{aligned}\tag{4.15}$$

because the last integral is $1 - e^{-B_c(\infty)} \leq 1$. All other factors are finite pathwise; their expectations are not needed. For completeness, stop the coagulation count at τ_L , the first time its continuous compensator Λ_t reaches a positive level L ; this is an $(\mathfrak{F}_t)$ -stopping time. The expected stopped count over all time is at most L , by compensation and monotone convergence, so it is finite almost surely. On $\{\Lambda_\infty < L\}$ this stop changes no coagulation jump. The union over integer L proves $K_\infty < \infty$ almost surely. On the other hand, marginal identification and Tonelli give

$$\mathbb{E}K_t = \int_0^t \lambda(s)N(s)\mathbb{E}X_{s-} ds = \int_0^t b(s) ds = B_c(t).$$

There are no fixed-time jumps under the absolutely continuous driving intensities, so the left limits do not change this integral. Monotone convergence as $t \rightarrow \infty$ gives $\mathbb{E}K_\infty = B_c(\infty)$, proving (4.12); for constant $\lambda > 0$ this is $\mathbb{E}K_\infty = \infty$. $\square$

After its finite last coagulation, A_t is exactly constant. That last time generally is not a stopping time, so this assertion does not make the subsequent fragmentation independent of it. Nor does finite K_∞ imply that coagulation ceases in the population. Since $\mathbb{E}K_\infty = B_c(\infty)$, the mean total count is infinite whenever $B_c(\infty) = \infty$, in particular for every constant $\lambda > 0$: rare auxiliary paths account for an infinite mean even though each path has only finitely many coagulation events.

Corollary 4.4 (A pathwise dichotomy for the constructed auxiliary process). *For the auxiliary process constructed in [theorem 4.1](#): if $F(t) \rightarrow \infty$, then $X_t \rightarrow 0$ almost surely and $\eta_t \Rightarrow \delta_0$ on $[0, \infty)$; if $F(\infty) < \infty$, then X_t is eventually constant at a positive finite random value, and η_t converges weakly on E to its law. The equation determines only the one-time laws η_t , so the weak limits of η_t are statements about the equation, whereas the almost-sure path statements concern the multi-time law of the constructed process, which is an artifact of the construction.*

Proof. The path product gives $X_t = X_0 e^{-F(t)} \mathcal{M}_t e^{A_t}$, with the last two factors bounded almost surely. This proves the first assertion. In the second case the fragmentation driver has finite total intensity $2F(\infty)$, so it has finitely many points almost surely. Together with $K_\infty < \infty$ this leaves only finitely many jumps of either type, each between positive finite sizes. The weak limits follow by bounded convergence applied to continuous tests. $\square$

5 Poisson comparison, transport, and logarithmic limits

5.1 An independent fragmentation reference and exact transport cost

For the independent reference, assume the daughter kernel is selfsimilar: there is a measurable family of measures B_t on $(0, 1)$ such that

$$b_{t,x} = (\theta \mapsto x\theta)_\# B_t, \quad B_t((0, 1)) = 2, \quad \int_{(0,1)} \theta B_t(d\theta) = 1. \tag{5.1}$$

Here $\#$ denotes pushforward. The fraction law $B_t/2$ is independent of the parent size. Under (5.1) the distribution function of $b_{t,x}/2$ at z is that of $B_t/2$ at z/x , so the quantile of $b_{t,x}/2$ equals x

times the quantile of $B_t/2$. In the construction of [theorem 4.1](#) the daughter chosen by the mark u is therefore $X_{s-}\Theta$ with Θ the quantile of $B_s/2$ at u : the retained fraction is a function of the fragmentation driver's point (s, u) alone, not of the state. Consequently, on the probability space of [theorem 4.1](#), the fragmentation times and fractions form a marked Poisson random measure $\mathbf{P}$ of intensity $2\sigma(t) dt (B_t/2)(d\theta) = \sigma(t) dt B_t(d\theta)$, independent of X_0 and of the coagulation driver. Define

$$Z_t = \log X_t, \quad \bar{Z}_t = Z_0 + \int_{(0,t] \times (0,1)} \log \theta \mathbf{P}(ds, d\theta), \quad \rho_t = \text{Law}(Z_t), \quad \bar{\rho}_t = \text{Law}(\bar{Z}_t). \quad (5.2)$$

The integral is a finite sum at each finite time, even with infinite daughter log moments. The increments of $\bar{Z}_t - Z_0$ are independent; they are stationary when σ is constant and $B_t = B$ is fixed, as in the setting below. If $\sigma = 0$ almost everywhere, the increments are identically zero regardless of B_t . Changes on null sets of times do not affect the reference law. This is exactly the number-normalized pure-fragmentation reference, and no separate solution theory is presupposed for it: with $\lambda = 0$, the linear equation (2.4) is solved directly by $\bar{n}_t = N_0 e^{F(t)} \text{Law}(e^{\bar{Z}_t})$, as the compensator formula for $\mathbf{P}$ shows, and $\bar{n}_t$ has count $N_0 e^{F(t)}$ and constant mass m because $\int \theta B_t(d\theta) = 1$.

For arbitrary probability measures on $\mathbb{R}$, define the extended transport cost

$$W_1(\alpha, \beta) = \inf_{\gamma \in \Gamma(\alpha, \beta)} \int_{\mathbb{R}^2} |z - w| \gamma(dz, dw) \in [0, \infty], \quad (5.3)$$

where Γ is the set of couplings. On $\mathcal{P}_1(\mathbb{R}) = \{\alpha : \int |z| \alpha(dz) < \infty\}$ this is the ordinary Wasserstein distance. A finite cost between two laws does not require either law to belong to $\mathcal{P}_1$. On the line the classical identity of Vallender [\[36\]](#),

$$W_1(\alpha, \beta) = \int_{\mathbb{R}} |F_\alpha(z) - F_\beta(z)| dz$$

with distribution functions F_α, F_β , holds in $[0, \infty]$ without moment assumptions, and it is attained by the ordered (quantile) coupling.

Theorem 5.1 (Moment-free exact transport identity). *Under (5.1), the coupling (5.2) satisfies $Z_t = \bar{Z}_t + A_t \geq \bar{Z}_t$. Thus ρ_t stochastically dominates $\bar{\rho}_t$, and*

$$\begin{aligned} W_1(\rho_t, \bar{\rho}_t) &= \mathbb{E} A_t = \int_0^t a(s) ds \\ &\leq \frac{H(0)^2}{mN_0} \int_0^t b(s) e^{-\kappa C(s)} ds \leq \frac{H(0)^2}{\kappa m N_0}. \end{aligned} \quad (5.4)$$

The cost increases to the finite constant $\mathbb{E} A_\infty$.

Proof. The path identity gives both the ordered coupling and the upper bound on transport cost. Optimality of ordered couplings is classical; we include a moment-free verification that avoids subtracting infinite means. Put $c_L(z) = \max(-L, \min(z, L))$. For every coupling (V, W) of $(\rho_t, \bar{\rho}_t)$, the one-Lipschitz property gives

$$\mathbb{E}|V - W| \geq \mathbb{E} c_L(V) - \mathbb{E} c_L(W) = \mathbb{E}[c_L(Z_t) - c_L(\bar{Z}_t)].$$

For real $z \geq w$, the difference $c_L(z) - c_L(w)$ is the length of $(w, z] \cap [-L, L]$, so it increases to $z - w$ as $L \rightarrow \infty$. Monotone convergence makes the last expectation tend to $\mathbb{E} A_t$. This proves the matching lower bound for every coupling, including ones of infinite cost. The estimates are [equations \(4.11\) and \(4.13\)](#). $\square$

Remark 5.2 (The boundary at parent dependence). For general $b_{t,x}$ the identity (4.10) and all of theorem 4.3 and corollary 4.4 remain valid. However, the realized fractions then depend on X_{s_j-} and hence on its coagulation history. Removing A_t while keeping those fractions need not produce a standalone fragmentation process. The independent Poisson reference and its limits in this section require (5.1).

The corresponding bounded-test log equation is useful for identifying the two mechanisms. For a bounded Borel ϕ on $\mathbb{R}$,

$$\begin{aligned} \langle \phi, \rho_t \rangle - \langle \phi, \rho_0 \rangle &= \int_0^t \sigma(s) \int_{\mathbb{R}} \int_{(0,1)} [\phi(z + \log \theta) - \phi(z)] B_s(d\theta) \rho_s(dz) ds \\ &\quad + \int_0^t \langle \phi, R_s \rangle ds, \end{aligned} \quad (5.5)$$

$$\langle \phi, R_s \rangle := \frac{\lambda(s)}{N(s)} \iint x [\phi(\log(x+y)) - \phi(\log x)] n_s(dx) n_s(dy). \quad (5.6)$$

This follows directly from (4.2). The signed measure R_s has mass zero and variation at most $2b(s)$; it is nonnegative on increasing tests, and $|\langle \phi, R_s \rangle| \leq \text{Lip}(\phi) a(s)$ for bounded Lipschitz tests. These are estimates of paired increments. They do not assert separate absolute log moments for the two positive measures in (5.6).

5.2 Constant-rate bounds and their critical specialization

For the remainder of the section, unless stated otherwise, let $\lambda > 0$ and $\sigma \geq 0$ be constant, put $b = \lambda m$, and take $B_t = B$ fixed. Write

$$r = 2\sigma, \quad \Theta \sim B/2, \quad Y = \log \Theta < 0, \quad \omega = \kappa(b + \sigma), \quad D_0 = \frac{\lambda H(0)^2}{\kappa(b + \sigma) N_0} \leq \frac{b}{\kappa(b + \sigma)}. \quad (5.7)$$

The symbol D_0 here is a transport bound, not relative entropy. The reference is the compound-Poisson process $\bar{Z}_t = Z_0 + \sum_{j=1}^{J_t} Y_j$, where J has rate r and the Y_j are independent copies of Y , independent of J and Z_0 . The limit A_∞ need not be independent of these variables.

Corollary 5.3 (All constant fragmentation rates). *With no logarithmic moment assumptions,*

$$W_1(\rho_t, \bar{\rho}_t) = \mathbb{E} A_t \leq D_0 (1 - e^{-\omega t}), \quad (5.8)$$

$$\mathbb{E}(A_\infty - A_t) \leq D_0 e^{-\omega t}. \quad (5.9)$$

At criticality $\sigma = b$, count is N_0 , $\omega = 2\kappa b$, and

$$D_0 = \frac{H(0)^2}{2\kappa m N_0} \leq \frac{1}{2\kappa}. \quad (5.10)$$

Proof. In (4.13), $a(t) \leq \lambda H(0)^2 e^{-\omega t} / N_0$. Integrate over $[0, t]$ or $[t, \infty)$. $\square$

The growing one-time cost (5.8) differs from the decreasing remaining correction (5.9). Neither estimate is a bound on raw-size moments or a claim that the constant D_0 is optimal.

5.3 Laws of large numbers and Gaussian limits

Pure-fragmentation logarithmic central limit theorems are classical. For example, Bertoin [4] studies homogeneous fragmentation using mass-weighted logarithmic sizes, and Doumic and Escobedo [13] gives refined critical-fragmentation asymptotics and a mass-weighted Gaussian limit. Their sampling weights differ from $B/2$ here. Our number-weighted reference is the classical many-to-one computation for finite-activity conservative binary fragmentation, and its limit is an ordinary compound-Poisson limit; the only new ingredient is the finite nonlinear correction of [theorem 4.3](#), which transfers that limit to the additive coagulation–fragmentation equation.

Theorem 5.4 (Logarithmic speed and functional central limit theorem). *Assume $\sigma > 0$ and fixed B . If $\mathbb{E}|Y| < \infty$, set $\mu = \mathbb{E}Y$. Then*

$$\frac{Z_t}{t} \longrightarrow r\mu \quad \text{almost surely.} \quad (5.11)$$

If $\nu_2 := \mathbb{E}Y^2 < \infty$, then, for each $S < \infty$,

$$\left(\frac{Z_{Ts} - r\mu Ts}{\sqrt{T}} \right)_{0 \leq s \leq S} \Longrightarrow (\sqrt{r\nu_2} W_s)_{0 \leq s \leq S} \quad (5.12)$$

in $D([0, S], \mathbb{R})$ with the Skorokhod J_1 topology, where W is standard Brownian motion. Here $D([0, S], \mathbb{R})$ is the space of right-continuous real paths with left limits; J_1 permits small changes in jump times through increasing continuous time changes. In particular,

$$\frac{Z_t - r\mu t}{\sqrt{t}} \Longrightarrow \mathcal{N}(0, r\nu_2). \quad (5.13)$$

No moment assumption on Z_0 is needed for these conclusions. The almost-sure statement (5.11) and the functional statement (5.12) concern the auxiliary process of [theorem 4.1](#), one specific coupling of the one-time laws; the equation itself determines only the one-time limits, such as (5.13) and the transport statements below.

If also $\mathbb{E}|Z_0| < \infty$, the one-time convergence (5.13) holds in ordinary W_1 , and

$$W_1(\text{Law}(Z_t/t), \delta_{r\mu}) \leq \frac{\mathbb{E}|Z_0| + D_0}{t} + \sqrt{\frac{r\nu_2}{t}}. \quad (5.14)$$

Under only $\mathbb{E}|Y| < \infty$ and $\mathbb{E}|Z_0| < \infty$, ordinary W_1 convergence to the speed still holds, without the displayed square-root rate.

Proof. Put $S_t^Y = \sum_{j \leq J_t} Y_j$. The strong laws for independent integrable variables and for the Poisson count give $J_t/t \rightarrow r$ and $J_t^{-1} \sum_{j \leq J_t} Y_j \rightarrow \mu$ almost surely. Since Z_0 is finite almost surely and $A_t \leq A_\infty$ almost surely, (4.10) proves (5.11).

We give the reduction of the functional assertion to the classical iid invariance principle, so its moment requirement is explicit. The increments $\xi_k = S_k^Y - S_{k-1}^Y - r\mu$ are iid with mean zero and variance $r\nu_2$. Donsker's theorem, Billingsley [8, Theorem 8.2], gives $X^N \Rightarrow \sqrt{r\nu_2} W$ on $C[0, 1]$ as the integer $N \rightarrow \infty$, where X^N is the linear interpolation of $N^{-1/2} \sum_{k \leq Nu} \xi_k$ between the grid points $u = k/N$; on $C[0, 1]$ the uniform and J_1 topologies agree. Rescaling to $[0, S]$ and passing from integer N to real T are one deterministic time change. Let Y_T be the linear interpolation of $T^{-1/2} \sum_{k \leq Ts} \xi_k$ between the grid points $s = k/T$ of $[0, S]$. With $N = [TS]$, $Y_T(s) = \sqrt{N/T} X^N(\theta_T s)$, where $\theta_T = T/N \rightarrow 1/S$ and $\theta_T s \leq 1$. The map $(x, \theta) \mapsto x(\theta \cdot)$ from $D([0, 1], \mathbb{R}) \times (0, 1/S]$ to $D([0, S], \mathbb{R})$ is continuous at every (x, θ) with x continuous, because J_1 convergence to a continuous limit is

uniform convergence and x is uniformly continuous; the continuous mapping theorem applied to (X^N, θ_T) therefore gives $Y_T \Rightarrow \sqrt{S}\sqrt{r\nu_2}W(\cdot/S)$ in $D([0, S], \mathbb{R})$, and $\sqrt{S}W(\cdot/S)$ is again a standard Brownian motion. To replace interpolation by the actual compound-Poisson path, let $V_k = \sum_{k-1 < u \leq k} |\Delta S_u^Y|$. Then V_k are iid and

$$\mathbb{E}V_k^2 = r\nu_2 + r^2(\mathbb{E}|Y|)^2 < \infty.$$

On each unit interval, the error between the centered process and its interpolation is bounded by $2(V_k + r|\mu|)$. Finite second moment implies $u^2\mathbb{P}(V_1 > u) \leq \mathbb{E}[V_1^2 \mathbf{1}_{\{V_1 > u\}}] \rightarrow 0$. For every $\varepsilon > 0$, a union bound therefore gives

$$\mathbb{P}\left(\max_{k \leq \lceil TS \rceil + 1} V_k > \varepsilon\sqrt{T}\right) \leq (\lceil TS \rceil + 1)\mathbb{P}(V_1 > \varepsilon\sqrt{T}) \rightarrow 0.$$

The functional limit follows for the reference process. Finally,

$$\sup_{0 \leq s \leq S} \frac{|Z_{Ts} - S_{Ts}^Y|}{\sqrt{T}} \leq \frac{|Z_0| + A_\infty}{\sqrt{T}} \rightarrow 0 \quad \text{almost surely.} \quad (5.15)$$

Uniform perturbations tending to zero also tend to zero in J_1 . This proves (5.12); evaluation at time one gives (5.13).

For the ordinary transport statements, the path identity first gives finite first log moments at every finite time:

$$\mathbb{E}|Z_t| \leq \mathbb{E}|Z_0| + rt\mathbb{E}|Y| + \mathbb{E}A_t < \infty. \quad (5.16)$$

The centered reference increment has $\mathbb{E}[(S_t^Y - r\mu t)^2]/t = r\nu_2$. These bounded second moments give uniform integrability of the first absolute moments of the scaled reference. Together with its weak Gaussian limit this gives ordinary W_1 convergence: by a quantile coupling the variables converge almost surely to the Gaussian, and uniform integrability upgrades that convergence to L^1 . Adding $Z_0/\sqrt{t}$ costs at most $\mathbb{E}|Z_0|/\sqrt{t}$; adding $A_t/\sqrt{t}$ costs at most $D_0/\sqrt{t}$. The triangle inequality proves the asserted ordinary transport CLT. Cauchy–Schwarz for $S_t^Y - r\mu t$ and the same coupling give (5.14).

For the speed with only a first jump moment, truncate the jumps at $Y^{(L)} = \max(Y, -L)$. The centered truncated sum divided by t tends to zero in L^1 by its finite variance. The L^1 difference between the original and truncated centered sums divided by t is at most $2r\mathbb{E}|Y - Y^{(L)}|$, uniformly in t . Let first $t \rightarrow \infty$ and then $L \rightarrow \infty$. Equation (4.10) finishes the proof. $\square$

The Gaussian variance parameter is $r\mathbb{E}Y^2$, the *raw* second moment of a jump, because the Poisson count fluctuates too. We do not assert convergence of $\text{Var}(Z_t)/t$: the hypotheses do not even require a second moment of Z_0 or A_∞ . Nor does a weak logarithmic CLT imply convergence to a lognormal density; for instance, with monodisperse data at size x_0 and equal splitting, the reference stays on the lattice $\log x_0 + (\log 2)\mathbb{Z}$ and the nonlinear size law stays atomic on $\{x_0 k 2^{-j} : k \geq 1, j \geq 0\}$.

Corollary 5.5 (Moment-free upper speed). *For $\sigma > 0$ and fixed B , without any log moments,*

$$\limsup_{t \rightarrow \infty} \frac{Z_t}{t} \leq -2\sigma \log 2 \quad \text{almost surely.}$$

If $\mathbb{E}(-Y) = \infty$, then $Z_t/t \rightarrow -\infty$ almost surely.

Proof. For each positive integer L , replace every Y_j by $Y_j^{(L)} = \max(Y_j, -L) \geq Y_j$. The strong law for these bounded jumps and $A_t \leq A_\infty$ give $\limsup Z_t/t \leq r\mathbb{E}Y^{(L)}$ on a common probability-one event. Monotone convergence of the negative parts lets $L \rightarrow \infty$. The resulting bound is $r\mathbb{E}Y$, allowing $-\infty$. For its universal upper bound, Jensen applied to $\max(\Theta, e^{-L})$ gives $\mathbb{E}Y^{(L)} \leq \log \mathbb{E} \max(\Theta, e^{-L})$; the right-hand side tends to $\log \mathbb{E}\Theta = -\log 2$. $\square$

For equal splitting, $\mu = -\log 2$ and $\nu_2 = (\log 2)^2$, so the speed and Gaussian variance parameter are respectively $-2\sigma \log 2$ and $2\sigma(\log 2)^2$. For deterministic complementary fractions v and $1-v$, they are $\sigma \log(v(1-v))$ and $\sigma[(\log v)^2 + (\log(1-v))^2]$.

Proposition 5.6 (Transfer on every diverging log scale). *Return to the controlled selfsimilar setting (5.1). Let $d_t > 0$ tend to infinity and let c_t be any deterministic centering. Then*

$$\mathbb{E} \left| \frac{Z_t - c_t}{d_t} - \frac{\bar{Z}_t - c_t}{d_t} \right| \leq \frac{\mathbb{E}A_\infty}{d_t} \rightarrow 0. \quad (5.17)$$

Every weak limit of the scaled reference laws therefore transfers to the corresponding scaled number laws. For deterministic paths $c_T(s)$ and scales $d_T \rightarrow \infty$, the same transfer holds in $D([0, S], \mathbb{R})$ with J_1 whenever the centered reference paths belong to that space and converge there. The one-time transfer concerns the laws ρ_t , determined by the equation; the path transfer concerns the auxiliary process of theorem 4.1, one specific coupling of the one-time laws, since the equation itself determines only the one-time limits.

Proof. The one-time bound is $0 \leq Z_t - \bar{Z}_t = A_t \leq A_\infty$. Markov's inequality and Slutsky's theorem transfer the weak limit. For paths the uniform difference is at most A_∞/d_T , which tends to zero almost surely and bounds the J_1 distance using the identity time change. $\square$

This proposition includes a stable limit when the specified reference satisfies the relevant classical domain-of-attraction assumptions. It supplies no such domain-of-attraction conclusion for arbitrary daughter laws, time-dependent marks, or controls.

5.4 Mean log size, geometric means, and relative entropy

Proposition 5.7 (Exact logarithmic identities). *In the constant-rate, fixed- B setting assume $\mathbb{E}|Z_0| < \infty$ and, when $\sigma > 0$, $\mathbb{E}|Y| < \infty$. Write $\ell = r\mathbb{E}Y$ when $\sigma > 0$, and $\ell = 0$ when $\sigma = 0$. Then*

$$\mathbb{E}Z_t = \mathbb{E}Z_0 + \ell t + \mathbb{E}A_t. \quad (5.18)$$

The geometric mean $G(t) := \exp(\int \log x \eta_t(dx))$ therefore satisfies

$$e^{-\ell t} G(t) \longrightarrow G_* := \exp(\mathbb{E}Z_0 + \mathbb{E}A_\infty) \in (0, \infty), \quad 0 \leq \log G_* - \log(e^{-\ell t} G(t)) \leq D_0 e^{-\omega t}. \quad (5.19)$$

The relative entropy of number against mass sampling is finite and obeys

$$\begin{aligned} D_{\text{KL}}(\eta_t \| \pi_t) &:= \int_E \log \left(\frac{d\eta_t}{d\pi_t} \right) d\eta_t = \log \frac{m}{N(t)} - \mathbb{E}Z_t \\ &= D_{\text{KL}}(\eta_0 \| \pi_0) + (b - \sigma - \ell)t - \mathbb{E}A_t. \end{aligned} \quad (5.20)$$

For $\sigma > 0$, $\mu = \mathbb{E}Y \leq -\log 2$, so its asymptotic linear growth rate is $b - \sigma - r\mu \geq b + \sigma(2\log 2 - 1) > 0$. At criticality this rate is $-2b\mu$.

Proof. Integrability follows from (5.16), with the jump sum absent if $\sigma = 0$. Taking expectations in the finite-cost path identity proves (5.18); the correction-tail bound proves (5.19). The sampling laws are mutually absolutely continuous, with $d\eta_t/d\pi_t = m/(N(t)x)$. Its logarithm is integrable by the established first log moment, so integration and equations (2.8) and (5.18) give (5.20). Finally Jensen's inequality gives $\mu \leq \log \mathbb{E}\Theta = -\log 2$. $\square$

No mass-weighted logarithmic moment such as $\int x|\log x|n_0(dx)$ is needed here. Without the stated first log moments we retain the paired transport identity, but make no finite mean-log, positive geometric-mean-prefactor, or finite-entropy claim. The entropy identity expresses the standard change of sampling weight; it is not an additional entropy-production estimate.

5.5 The pure additive endpoint and the classical Borel solution

Corollary 5.8 (Pure coagulation). *Let $\sigma = 0$ and $\lambda > 0$ be constant. Then Z_t increases to a finite real Z_∞ almost surely and is eventually constant. Writing $\rho_\infty = \text{Law}(Z_\infty)$, for $0 \leq s \leq t$,*

$$W_1(\rho_s, \rho_t) = \int_s^t a(u) du, \quad (5.21)$$

$$W_1(\rho_t, \rho_\infty) = \int_t^\infty a(u) du \leq D_0 e^{-\kappa b t}. \quad (5.22)$$

These statements require no initial log moment. In particular the number-normalized size laws have a proper weak limit on E . The transport identities and this weak limit are properties of the equation; the almost-sure monotone convergence concerns the constructed auxiliary process.

Proof. Now $Z_t = Z_0 + A_t$, and theorem 4.3 supplies the finite limit and finite total jumps. Apply the clipped-test optimality argument of theorem 5.1 to the ordered pairs (Z_t, Z_s) and (Z_∞, Z_t) . Their expected differences are the displayed integrals. Almost-sure convergence and continuity of the exponential give the size-law convergence. $\square$

The existence of a proper number-normalized limit at this endpoint has classical precedents. For $n_0 = \delta_1$, $m = 1$, and $\lambda = 1$, the monodisperse additive solution is

$$n_t(\{k\}) = e^{-t} \frac{[k(1 - e^{-t})]^{k-1}}{k!} e^{-k(1 - e^{-t})}, \quad k \geq 1, \quad t > 0. \quad (5.23)$$

This is the classical Golovin solution as recorded in Bertoin [5, equation (3)]; branching representations of the additive solution are developed in Deaconu and Tanré [10]. Its number law is $\text{Borel}(1 - e^{-t})$ and tends to $\text{Borel}(1)$, the proper law with probabilities $p_k = e^{-k} k^{k-1}/k!$. One can see that this critical law is proper as the total progeny of a Poisson-one Galton–Watson tree: its extinction probability is the smallest fixed point of $q = e^{q-1}$ in $[0, 1]$, namely one. Stirling's formula gives $p_k \sim (2\pi)^{-1/2} k^{-3/2}$, so its mean is infinite even though its first logarithmic moment is finite. At finite times the number mean size is $m/N(t) = e^t$. This classical example shows why the finite log transport limit does not control the arithmetic mean, and why the pure-coagulation endpoint itself is not a new limiting phenomenon.

6 The last auxiliary coagulation: tail bounds, daughter envelopes, and exact critical representations

Fix a solution in [definition 2.2](#) and its number process from [theorem 4.1](#). Retain the coagulation count K_t of [theorem 4.3](#), and define

$$Q_t = \frac{N(t)X_t}{m}, \quad L = \sup\{s > 0 : s \text{ is an auxiliary coagulation time}\}, \quad h(t) = \mathbb{P}(L > t), \quad (6.1)$$

where the supremum of the empty set is zero. Thus $\mathbb{E}Q_t = 1$ and $\mathbb{E}Q_t^p = \mathcal{A}_p(t)$. Although L is finite almost surely by [theorem 4.3](#), it is generally not a stopping time, and nothing below conditions at L . All conditional future laws mean the Markov construction restarted at a deterministic time t and state x in the prescribed environment n_s ; this specifies them even at states of zero probability under η_t . By the Markov property of the Poisson-random-measure construction, the regular conditional law of $(X_s)_{s \geq t}$ given X_t equals this restart law for η_t -almost every x , so averaging restart quantities against η_t gives unconditional probabilities.

6.1 Controlled first-event and last-event bounds

Theorem 6.1 (Controlled first-event and last-event bounds). *For $0 \leq t < T \leq \infty$, set*

$$u_{t,T} = 1 - \exp\left(-\int_t^T b(s) ds\right), \quad u_t = u_{t,\infty}. \quad (6.2)$$

Given $X_t = x$, write $q = N(t)x/m$. Then

$$\mathbb{P}(K_T > K_t \mid X_t = x) \leq 1 - e^{-qu_{t,T}} \leq \min(1, qu_{t,T}). \quad (6.3)$$

At $T = \infty$ the event means at least one coagulation after t . For every $0 < p < 1$,

$$\mathbb{P}(K_T > K_t) \leq u_{t,T}^p \mathcal{A}_p(t) \leq u_{t,T}^p \mathcal{A}_p(0) e^{-a_p B_c(t) - a_{1-p} F(t)}, \quad (6.4)$$

$$h(t) \leq u_t^p \mathcal{A}_p(0) e^{-a_p B_c(t) - a_{1-p} F(t)}. \quad (6.5)$$

Together with finite-horizon nonexplosion from [theorem 4.1](#), these bounds imply $L < \infty$ and $K_\infty < \infty$ almost surely, for all allowed controls. No logarithmic or second size moment is required.

Proof. Start a fragmentation-only process V_s at $V_t = x$, with the prescribed daughter kernel evaluated at its own current size, driven by the fragmentation clock and uniform innovations of the auxiliary construction. It has finitely many events on each finite interval, $0 < V_s \leq x$, and conditional daughter mean $V_{s-}/2$, so compensation and the integrating factor give

$$\mathbb{E}_{t,x} V_s = x e^{-[F(s) - F(t)]}, \quad \mathbb{E}_{t,x} \frac{N(s)V_s}{m} = q e^{-[B_c(s) - B_c(t)]}. \quad (6.6)$$

The cancellation in the second identity is why only remaining coagulation activity enters [\(6.3\)](#).

Until its first coagulation after t , the auxiliary process has state V_s and coagulation rate $\lambda(s)N(s)V_{s-}$. The coagulation and fragmentation Poisson measures are independent, so conditional on the whole fragmentation path the retained coagulation points before the first coagulation form a Poisson process with intensity $\lambda(s)N(s)V_s ds$. The first future coagulation is therefore the first crossing of an independent mean-one exponential threshold by the hazard

$$\mathcal{H}_{t,T} = \int_t^T b(s) \frac{N(s)V_s}{m} ds, \quad (6.7)$$

after which a partner is sampled and the full process resumes; the full future compensator is not identified with (6.7). Tonelli's theorem gives $\mathbb{E}_{t,x} \mathcal{H}_{t,T} = q \int_t^T b(s) e^{-[B_c(s) - B_c(t)]} ds = qu_{t,T}$, finite even at $T = \infty$. The event probability is exactly $\mathbb{E}_{t,x}(1 - e^{-\mathcal{H}_{t,T}})$, and Jensen's inequality for the concave map $z \mapsto 1 - e^{-z}$ gives (6.3). Strict concavity makes the first inequality an equality if and only if $\mathcal{H}_{t,T} = qu_{t,T}$ almost surely. Vanishing fragmentation on (t, T) is sufficient, but is not necessary: fragmentation after all coagulation activity in the window has ended does not affect the hazard. The second inequality is strict whenever $0 < qu_{t,T} < \infty$. Averaging over η_t , then using $\min(1, z) \leq z^p$ and theorem 3.3, gives the two tail bounds.

If $B_c(\infty) = \infty$, the coagulation exponent in (6.5) diverges; if $B_c(\infty) < \infty$, then $u_t \rightarrow 0$ and $\mathcal{A}_p(t) \leq 1$. In both cases $h(t) \rightarrow 0$, so almost surely there is a finite time after which no coagulation occurs, and nonexplosion on that finite horizon leaves finitely many coagulations before it. Hence $L < \infty$ and $K_\infty < \infty$. $\square$

The conditional bound (6.3) is thus instantaneously sharp as a uniform bound over the full controlled class; its averaged consequence need not have a sharp time exponent. Each right-hand side in equations (6.4) and (6.5) may be multiplied by the scalar factor

$$c_p = \max_{z>0} \frac{1 - e^{-z}}{z^p} < 1, \quad c_{1/2} \approx 0.638, \quad (6.8)$$

because the ratio tends to zero at both endpoints and its derivative changes sign exactly at the unique positive root of $z/(e^z - 1) = p$. This is sharp scalar domination, not a model asymptotic assertion.

Corollary 6.2 (Approximation of the entire future path). *Start a fragmentation-only process $V^{(t)}$ at the random state $V_t^{(t)} = X_t$, with daughter kernel $b_{s, V_{s-}^{(t)}}/2$. For $T \leq \infty$, the total variation distance between the laws of $(X_s)_{t \leq s \leq T}$ and $(V_s^{(t)})_{t \leq s \leq T}$ on the Borel space of càdlàg E -valued paths is at most*

$$u_{t,T}^p \mathcal{A}_p(0) e^{-a_p B_c(t) - a_{1-p} F(t)}. \quad (6.9)$$

Proof. Use common future fragmentation times and uniform innovations for both processes; after their states diverge each quantile map uses its own state, so each marginal keeps its daughter dynamics. The paths agree on the whole interval unless $K_T > K_t$, so any two path probabilities differ by at most $\mathbb{P}(K_T > K_t)$; apply (6.4). The coupling is defined for all future times, which covers $T = \infty$. $\square$

This reference starts from η_t , not η_0 . For parent-dependent daughters, common innovations do not mean common fractions after divergence, consistent with remark 5.2.

Corollary 6.3 (Time moments and rare event-bearing paths). *Let $b > 0$ and $\sigma \geq 0$ be constant, and put*

$$\chi_p = ba_p + \sigma a_{1-p} > 0. \quad (6.10)$$

Then $h(t) \leq \mathcal{A}_p(0) e^{-\chi_p t}$, and, for $0 < r < \chi_p$,

$$\mathbb{E} e^{rL} \leq 1 + \frac{r \mathcal{A}_p(0)}{\chi_p - r}. \quad (6.11)$$

For a fixed window length $\ell > 0$, let $D_t = K_{t+\ell} - K_t$ and $\omega_p = (1 - e^{-b\ell})^p \mathcal{A}_p(0)$. Then $\mathbb{E}D_t = b\ell$, $D_t = 0$ eventually almost surely, and

$$\mathbb{P}(D_t > 0) \leq \omega_p e^{-\chi_p t}, \quad (6.12)$$

$$\mathbb{E}[D_t \mid D_t > 0] \geq \frac{b\ell}{\omega_p} e^{\chi_p t}, \quad (6.13)$$

$$\mathbb{E}D_t^r \geq (b\ell)^r \omega_p^{1-r} e^{(r-1)\chi_p t} \quad (r > 1). \quad (6.14)$$

The last inequality allows an infinite left-hand side.

Proof. The time-tail bound is (6.5) with $u_t = 1$, and Tonelli applied to $e^{rL} - 1 = \int_0^L r e^{rs} ds$ proves (6.11). The count compensator has mean $\int_t^{t+\ell} b \mathbb{E}Q_s ds = b\ell$, whereas the finite last time makes D_t eventually zero on each path. The window bound is (6.4); dividing the positive mean by $\mathbb{P}(D_t > 0)$ gives (6.13), and Hölder gives $b\ell \leq (\mathbb{E}D_t^r)^{1/r} \mathbb{P}(D_t > 0)^{1-1/r}$. $\square$

The window counts are not uniformly integrable despite their almost-sure disappearance: their mean concentrates on rare auxiliary paths. No burst-size law, reactor count, or physical mass-tag event is asserted.

6.2 A classical pure-coagulation benchmark

For monodisperse data $n_0 = N_0 \delta_{x_0}$ and pure coagulation, write $b = \lambda m > 0$ and $\alpha = 1 - e^{-bt}$. The Golovin solution (5.23) [23], the marginal of the additive coalescent [1, 3], says after rescaling that X_t/x_0 has law Borel(α), whose generating function satisfies $G_\alpha(z) = z \exp\{\alpha(G_\alpha(z) - 1)\}$ by the Poisson branching decomposition of total progeny. Because $N(t) = N_0(1 - \alpha)$, the normalized state is $Q_t = (1 - \alpha)X_t/x_0$ and the no-future-event probability is exactly e^{-Q_t} . Hence

$$h(t) = 1 - G_\alpha(e^{-(1-\alpha)}), \quad -\log(1 - h(t)) = (1 - \alpha) + \alpha h(t). \quad (6.15)$$

This is a deduction from the Golovin–Borel solution as recorded by Bertoin [5, equation (3)], not a new solution. Since $h(t) \rightarrow 0$, subtracting $h(t)$ in (6.15) yields $h(t)^2(1/2 + O(h(t))) = (1 - \alpha)(1 - h(t))$, and therefore

$$h(t) \sim \sqrt{2} e^{-bt/2}. \quad (6.16)$$

Even this solvable endpoint has a faster tail than the certificate $e^{-a_{1/2}bt}$ from corollary 6.3 with $\sigma = 0$ and $p = 1/2$, where $a_{1/2} = \kappa/2$. Sharpness of a moment differential inequality does not imply sharpness of its derived last-event exponent.

6.3 Daughter-law envelopes at constant rates

From here on $b = \lambda m > 0$ and $\sigma \geq 0$ are constant; the competing daughter kernels retain all the time and parent dependence allowed in assumption 2.1. For a restart at time t and normalized state q , let $g_t(q)$ be the probability of a future auxiliary coagulation and $w_t(q) = 1 - g_t(q)$. By the first-event construction these do not depend on future partner laws, though they may depend on t through the daughter kernel.

Set $d = \sigma - b$ and $J_s = \int_0^s e^{du} du$. For $\sigma > 0$, let τ be exponential of rate σ and P a Poisson process of rate 2σ , and define

$$Z_{\text{ext}} = bJ_\tau, \quad Z_{\text{eq}} = b \int_0^\infty e^{ds} 2^{-P_s} ds, \quad (6.17)$$

$$w_{\text{ext}}(q) = \mathbb{E}e^{-qZ_{\text{ext}}}, \quad w_{\text{eq}}(q) = \mathbb{E}e^{-qZ_{\text{eq}}}. \quad (6.18)$$

For $\sigma = 0$ define both hazards to equal one, so both transforms equal e^{-q} . The first benchmark is the hazard of a comparison process growing at rate d until a reset to zero at rate σ ; the zero-reset rule is not an admissible daughter law and is never asserted to be one. The second is the hazard of equal splitting. Tonelli gives $\mathbb{E}Z_{\text{ext}} = \mathbb{E}Z_{\text{eq}} = 1$, since $e^{ds}\mathbb{P}(\tau > s) = e^{-bs}$ and $e^{ds}\mathbb{E}2^{-P_s} = e^{-bs}$. The second moment of Z_{ext} is finite exactly when $\sigma < 2b$; only first moments are used below.

At criticality, $d = 0$, $Z_{\text{ext}} = b\tau$ is a mean-one exponential variable, and $w_{\text{ext}}(q) = (1 + q)^{-1}$. For equal splitting the holding times are independent exponentials of rate $2b$, so with independent mean-one exponentials E_k ,

$$Z_{\text{eq}} \stackrel{d}{=} S := \sum_{k=1}^{\infty} 2^{-k} E_k, \quad \Psi(q) := \mathbb{E}e^{-qS} = \prod_{k=1}^{\infty} (1 + q/2^k)^{-1}. \quad (6.19)$$

This is the Poissonian exponential functional of Bertoin et al. [7, equations (1.3), (1.6), Theorem 1.1(i)], whose Theorem 1.1(i) is stated for parameter values in $(0, 1)$; in their notation S has law $I^{(1/2)}/2$. The product converges locally uniformly since $\sum_k \log(1 + q/2^k) \leq q$, and independence of the holding times gives

$$\mathbb{E}S = 1, \quad \text{Var } S = \frac{1}{3}, \quad (2 + q)\Psi(q) = 2\Psi(q/2), \quad \Psi(q) \leq \frac{1}{1 + q}, \quad (6.20)$$

the last because $\prod_k (1 + q/2^k) \geq 1 + q \sum_k 2^{-k} = 1 + q$. The Laplace representation makes Ψ strictly decreasing, strictly convex, and completely monotone, with $\Psi(q) \rightarrow 0$ as $q \rightarrow \infty$. A large- q form and a rational certificate for $\Psi(1)$ are in [appendix B](#).

Theorem 6.4 (Conditional envelopes at all constant rates). *For every allowed daughter kernel, every deterministic $t \geq 0$, and every $q > 0$,*

$$w_{\text{eq}}(q) \leq w_t(q) \leq w_{\text{ext}}(q), \quad 1 - w_{\text{ext}}(q) \leq g_t(q) \leq 1 - w_{\text{eq}}(q). \quad (6.21)$$

At criticality $\sigma = b$ these read

$$\Psi(q) \leq w_t(q) \leq \frac{1}{1 + q}, \quad \frac{q}{1 + q} \leq g_t(q) \leq 1 - \Psi(q), \quad (6.22)$$

and for equal splitting $g_t(q) = g(q) := 1 - \Psi(q)$. Both envelopes are instantaneously sharp as uniform bounds over the stated class: equal splitting attains the lower bound on w_t , and the positive binary daughter measures

$$b_{s,x}^{(\varepsilon)} = \delta_{\varepsilon x} + \delta_{(1-\varepsilon)x}, \quad 0 < \varepsilon < 1/2, \quad (6.23)$$

approach the upper bound as $\varepsilon \downarrow 0$. At $\sigma = 0$ the envelopes coincide for every daughter kernel.

Proof. For the fragmentation-only future write $Y_s = N(t + s)V_{t+s}/m$, $Y_0 = q$. Between events it has drift dY_s ; at rate 2σ it is multiplied by a fraction Θ with conditional mean $1/2$. By (6.6),

$$\mathbb{E}Y_s = qe^{-bs}, \quad \mathcal{H}_s = \int_0^s bY_v dv, \quad w_t(q) = \mathbb{E}e^{-\mathcal{H}_\infty}, \quad \mathbb{E}\mathcal{H}_\infty = q. \quad (6.24)$$

Both benchmark hazards are strictly positive with mean one, so their transforms w are decreasing, convex, and continuously differentiable on $[0, \infty)$ (from the right at zero, by dominated convergence using the finite mean), with

$$w(0) = 1, \quad |w'(q)| \leq 1, \quad 0 \leq 1 - w(q) \leq q. \quad (6.25)$$

Benchmark equations. For $\sigma > 0$ and $q \geq 0$,

$$dqw'_{\text{ext}}(q) + \sigma[1 - w_{\text{ext}}(q)] - bqw_{\text{ext}}(q) = 0, \quad (6.26)$$

$$dqw'_{\text{eq}}(q) + 2\sigma[w_{\text{eq}}(q/2) - w_{\text{eq}}(q)] - bqw_{\text{eq}}(q) = 0; \quad (6.27)$$

for $\sigma = 0$ both hold with $d = -b$ and $w = e^{-q}$. For the first, $w_{\text{ext}}(q) = \int_0^\infty \sigma e^{-\sigma s} e^{-qbJ_s} ds$, and $\mathbb{E}Z_{\text{ext}} = 1$ justifies differentiation under the integral. Since $\partial_s e^{-qbJ_s} = -qb(1 + dJ_s)e^{-qbJ_s}$ and $e^{-\sigma s} e^{-qbJ_s} \rightarrow 0$ as $s \rightarrow \infty$, integration by parts gives

$$1 - w_{\text{ext}}(q) = qb \int_0^\infty e^{-\sigma s} (1 + dJ_s) e^{-qbJ_s} ds = \frac{qb}{\sigma} w_{\text{ext}}(q) - \frac{qd}{\sigma} w'_{\text{ext}}(q),$$

which is (6.26). For the second, let τ_1 be the first jump time of P , exponential of rate 2σ . The strong Markov property of P at τ_1 gives $Z_{\text{eq}} = bJ_{\tau_1} + e^{d\tau_1} Z'_{\text{eq}}/2$ in law, with Z'_{eq} a copy of Z_{eq} independent of τ_1 . Hence

$$w_{\text{eq}}(q) = \int_0^\infty 2\sigma e^{-2\sigma s} e^{-qbJ_s} w_{\text{eq}}(qe^{ds}/2) ds.$$

If $d = 0$ this reads $(2\sigma + bq)w_{\text{eq}}(q) = 2\sigma w_{\text{eq}}(q/2)$, which is (6.27) and the recurrence in (6.20). If $d \neq 0$, substitute $r = qe^{ds}$:

$$w_{\text{eq}}(q) = \frac{2\sigma}{|d|} q^{2\sigma/d} e^{bq/d} \int_{I_q} r^{-2\sigma/d-1} e^{-br/d} w_{\text{eq}}(r/2) dr,$$

with $I_q = (q, \infty)$ for $d > 0$ and $I_q = (0, q)$ for $d < 0$. Differentiating this product by the fundamental theorem of calculus, w_{eq} being continuous, gives $dq w'_{\text{eq}}(q) = (2\sigma + bq)w_{\text{eq}}(q) - 2\sigma w_{\text{eq}}(q/2)$, which is (6.27).

Comparison. For any convex w with $w(0) = 1$ and every admissible conditional fraction law, Jensen's inequality and the chord on $[0, q]$ give

$$w(q/2) \leq \mathbb{E}[w(q\Theta) \mid s, q] \leq \frac{1}{2}[1 + w(q)]. \quad (6.28)$$

Let $\mathcal{G}_s^{\text{frag}} f(q) = dqf'(q) + 2\sigma\{\mathbb{E}[f(q\Theta) \mid t + s, q] - f(q)\}$ be the fragmentation-only generator in the Y coordinate, with the actual conditional fraction law at absolute time $t + s$. The benchmark equations and (6.28) give

$$(\mathcal{G}_s^{\text{frag}} - bq)w_{\text{ext}}(q) \leq 0, \quad (\mathcal{G}_s^{\text{frag}} - bq)w_{\text{eq}}(q) \geq 0.$$

Dynkin's formula applies to $e^{-\mathcal{H}_s} w(Y_s)$ for either benchmark transform: the process lies in $[0, 1]$; between jumps $Y_s > 0$ follows the deterministic flow and $w \in C^1$ on $[0, \infty)$ with $|w'| \leq 1$ and $Y_s \leq qe^{d_+ s}$, $d_+ = \max(d, 0)$, so the drift term is integrable on finite horizons; and $\mathbb{E} \int_0^s 2\sigma |w(Y_v \Theta) - w(Y_v)| dv \leq 2\sigma s < \infty$ because $0 \leq w \leq 1$. Hence $e^{-\mathcal{H}_s} w_{\text{ext}}(Y_s)$ is a supermartingale and $e^{-\mathcal{H}_s} w_{\text{eq}}(Y_s)$ a submartingale. By (6.25), $\mathbb{E}|e^{-\mathcal{H}_s} w(Y_s) - e^{-\mathcal{H}_s}| \leq \mathbb{E}Y_s = qe^{-bs} \rightarrow 0$, while $e^{-\mathcal{H}_s} \rightarrow e^{-\mathcal{H}_\infty}$ boundedly. Taking expectations and letting $s \rightarrow \infty$ proves (6.21); no monotonicity of normalized size, stationarity of the competing kernel, or daughter log moment is used. The critical forms (6.22) are the case $d = 0$ with the benchmarks computed above.

Sharpness. For equal splitting, $Y_s = qe^{ds} 2^{-P_s}$, so the lower bound on w_t is attained. For (6.23), use independent clocks of rate σ for the two marks, common for all ε . Let τ_ε be the first ε -mark and P'_s the count of the other marks. The hazard before τ_ε is

$$\mathcal{H}_\varepsilon^{\text{pre}} = bq \int_0^{\tau_\varepsilon} e^{ds} (1 - \varepsilon)^{P'_s} ds \longrightarrow qZ_{\text{ext}} \quad \text{in } L^1,$$

by pathwise convergence and domination by qZ_{ext} . Immediately after τ_ε , $Y_{\tau_\varepsilon} = q\varepsilon e^{d\tau_\varepsilon}(1 - \varepsilon)^{P'_{\tau_\varepsilon}}$, and the conditional expected remaining hazard is Y_{τ_ε} by (6.24) and the fresh clocks at this stopping time. Independence of the clocks gives

$$\mathbb{E}Y_{\tau_\varepsilon} = q\varepsilon\sigma \int_0^\infty e^{-(b+\sigma\varepsilon)s} ds = \frac{q\varepsilon\sigma}{b + \sigma\varepsilon} \rightarrow 0. \quad (6.29)$$

Thus the full hazard converges in L^1 to qZ_{ext} , and since e^{-z} is one-Lipschitz on $[0, \infty)$ the survival probability tends to $w_{\text{ext}}(q)$. When $\sigma = 0$ the hazard is deterministically q . $\square$

Convex inequalities with generator verification are established comparison methods; ordered-generator comparisons for time-inhomogeneous Markov processes appear in Rüschendorf et al. [34]. The conclusion orders future-event probabilities; it asserts neither convex order of hazards nor an order of total coagulation counts.

Remark 6.5 (Exponential functionals of subordinators). At criticality with parent-independent fractions Θ , $Y_s = qe^{-\xi s}$, where ξ is the compound-Poisson subordinator with jump rate $2b$ and jump law that of $-\log \Theta$. The future hazard $qb \int_0^\infty e^{-\xi s} ds$ is an exponential functional of a Lévy process in the sense of Bertoin and Yor [6], and w_t is its Laplace transform. This is why the Poissonian exponential functional of Bertoin et al. [7] appears for equal splitting, where ξ is $\log 2$ times a Poisson process of rate $2b$ and $b \int e^{-\xi s} ds$ has the law of S . In this language theorem 6.4 compares $\mathbb{E}e^{-qbI_\xi}$ over jump laws of ξ with $\mathbb{E}e^{-\Delta\xi} = 1/2$: it is the convex-order statement (6.28) between the deterministic fraction $1/2$, the law of Θ , and the degenerate limit concentrated on $\{0, 1\}$.

Corollary 6.6 (Overlap coefficients at all constant rates). *Write*

$$O(t) = 1 - \|\eta_t - \pi_t\|_{\text{TV}} = \int_E \min\left(1, \frac{N(t)x}{m}\right) \eta_t(dx) = \mathbb{E} \min(1, Q_t), \quad c_{b,\sigma} = 1 - w_{\text{ext}}(1) > 0. \quad (6.30)$$

Then

$$c_{b,\sigma}O(t) \leq h(t) \leq O(t), \quad (6.31)$$

with $c_{b,0} = 1 - e^{-1}$ and $c_{b,b} = 1/2$. For equal splitting at criticality, moreover,

$$c_*O(t) \leq h(t) \leq O(t), \quad c_* = 1 - \Psi(1) = 0.580577558204892\dots \quad (6.32)$$

Each lower coefficient and the common upper coefficient one are instantaneously sharp as uniform constants over the stated class of daughter kernels and initial populations.

Proof. The overlap identity follows from $d\pi_t/d\eta_t = Q_t$ and the convention (2.5). The increasing concave function $f(q) = 1 - w_{\text{ext}}(q)$ vanishes at zero, so its chord on $[0, 1]$ and monotonicity on $[1, \infty)$ give $f(q) \geq f(1) \min(1, q)$. Average the lower envelope in (6.21) over η_t for the lower bound and use (6.3) with $u_t = 1$ for the upper bound. At criticality $f(1) = 1/2$; for equal splitting the same chord argument applies to g , with $g(1) = c_*$.

For the lower constants, monodisperse data give $Q_0 = 1$ and $O(0) = 1$; equal splitting attains c_* , and the ε -daughters approach $c_{b,\sigma}$. For the upper constant use the mean-one two-point laws

$$\nu_{p,s} = (1-p)\delta_s + p\delta_{r(p,s)}, \quad r(p,s) = \frac{1 - (1-p)s}{p}, \quad 0 < p < 1, \quad 0 < s < 1, \quad (6.33)$$

where δ_x is a Dirac measure. Under equal splitting at criticality with $Q_0 \sim \nu_{\varepsilon,\varepsilon^2}$, $O(0) = (1-\varepsilon)\varepsilon^2 + \varepsilon \sim \varepsilon$ and $h(0) = (1-\varepsilon)g(\varepsilon^2) + \varepsilon g(r) \sim \varepsilon$, because $g(q) \leq q$ and $g(r) \rightarrow 1$ as $r \sim \varepsilon^{-1} \rightarrow \infty$. $\square$

All sharpness preparations are realizable populations: for any finitely supported positive mean-one law μ , the measure $n_0 = N_0(q \mapsto mq/N_0)_\# \mu$ has count N_0 , mass m , and finite second moment, so [theorem 2.3](#) supplies a solution with $Q_0 \sim \mu$. The two-sided bounds make vanishing of the overlap equivalent to disappearance of future coagulation in this auxiliary representation. These constants compare probabilities at arbitrary times and populations; they do not identify a decay exponent.

6.4 Exact critical representations of the last-event law

Assume from here on that $b = \sigma > 0$, so $N(t) = N_0$. Let μ_t be the law of $Q_t = N_0 X_t / m$, and let Q'_t denote an independent variable with law μ_t whenever it appears with Q_t . The generator of Q is

$$\mathcal{G}_t f(q) = 2b\{\mathbb{E}[f(q\Theta) \mid t, q] - f(q)\} + bq \int_E [f(q+v) - f(q)] \mu_t(dv), \quad (6.34)$$

with the original daughter kernel expressed in normalized size in the first term, so $\mathbb{E}\Theta = 1/2$ conditionally. The formulas below are exact identities in terms of the unknown law μ_t : representations, not closed-form evaluations.

Theorem 6.7 (Equal-split tail, density, and Laplace representation). *For equal splitting and Ψ from (6.19),*

$$h(t) = 1 - \int_E \Psi(q) \mu_t(dq), \quad (6.35)$$

$$h'(t) = -b\mathbb{E}[Q_t \Psi(Q_t + Q'_t)]. \quad (6.36)$$

Here h' is continuous on $[0, \infty)$, the value at zero being a right derivative. The law of L has an atom $\mathbb{E}\Psi(Q_0)$ at zero and density $b\mathbb{E}[Q_t \Psi(Q_t + Q'_t)]$ on $(0, \infty)$. For the size Laplace transform $\phi_t(z) = \int_E e^{-zq} \mu_t(dq)$, $z \geq 0$,

$$\partial_t \phi_t(z) = 2b[\phi_t(z/2) - \phi_t(z)] + b[1 - \phi_t(z)] \partial_z \phi_t(z), \quad (6.37)$$

with the right derivative in z at zero, and with an independent S from (6.19),

$$h(t) = 1 - \mathbb{E}_S \phi_t(S), \quad -h'(t)/b = \mathbb{E}_S [-\partial_z \phi_t(S) \phi_t(S)]. \quad (6.38)$$

All these formulas hold in the finite-count, finite-mass solution class without higher size moments.

Equation (6.37) is the classical Laplace-transform reduction of the additive-kernel Smoluchowski equation [3, 23] with the dilation term $2b[\phi_t(z/2) - \phi_t(z)]$ added by equal splitting; it is recorded for the mixture identities, not as a new equation.

Proof. The conditional no-future-event probability is $\Psi(q)$ by [theorem 6.4](#), so averaging proves (6.35). Apply (6.34) to the bounded test Ψ ; the recurrence in (6.20) cancels the fragmentation term against the loss part of the coagulation term:

$$\mathcal{G}_t \Psi(q) = 2b[\Psi(q/2) - \Psi(q)] + bq \int_E [\Psi(q+v) - \Psi(q)] \mu_t(dv) = bq \int_E \Psi(q+v) \mu_t(dv).$$

By $\Psi(q) \leq (1+q)^{-1}$, $q\Psi(q+v) \leq q\Psi(q) \leq 1$, and the coagulation term before cancellation is bounded by b . The bounded-test forward equation (4.2) therefore yields, for $s \leq t$,

$$h(t) - h(s) = -b \int_s^t \iint_{E^2} q\Psi(q+v) \mu_u(dq) \mu_u(dv) du. \quad (6.39)$$

The solution is continuous in total variation by [definition 2.2](#), and the constant normalization at criticality makes μ_t , hence its product law, continuous in total variation; the bounded integrand in (6.39) thus has an expectation continuous on $[0, \infty)$, proving the derivative assertion. Since $L < \infty$ almost surely, this derivative and $1 - h(0) = \mathbb{E}\Psi(Q_0)$ exhaust the law of L .

The first size moment gives $\partial_z \phi_t(z) = -\mathbb{E}[Q_t e^{-zQ_t}]$, including the right derivative -1 at zero. At fixed $z > 0$ both e^{-zq} and qe^{-zq} are bounded; the coagulation contribution to the bounded-test equation for ϕ is $b\mathbb{E}[Q_t e^{-zQ_t}](\phi_t(z) - 1) = b[1 - \phi_t(z)]\partial_z \phi_t(z)$, and fragmentation contributes $2b[\phi_t(z/2) - \phi_t(z)]$. Continuity of these bounded expectations gives the pointwise time derivative in (6.37); at $z = 0$ both sides vanish. Finally the Laplace representation of Ψ , independence of Q_t, Q'_t, S , and Tonelli give $\mathbb{E}\Psi(Q_t) = \mathbb{E}_S \phi_t(S)$ and $\mathbb{E}[Q_t \Psi(Q_t + Q'_t)] = \mathbb{E}_S \{\mathbb{E}[Q_t e^{-SQ_t}] \mathbb{E}[e^{-SQ'_t}]\}$, which is (6.38). $\square$

The density counts an event followed by no later coagulation; the expected rate of all coagulations remains b . The scalar mixture (6.38) does not by itself give an evolution equation in h alone.

Theorem 6.8 (Last-event density for general critical daughters). *Allow time- and parent-dependent daughters. With the restarted survival function w_t of [section 6.3](#), the law of L on $(0, \infty)$ is absolutely continuous, with density*

$$j(t) = b\mathbb{E}[Q_t w_t(Q_t + Q'_t)] \quad \text{for almost every } t > 0, \quad (6.40)$$

and atom $\mathbb{E}w_0(Q_0)$ at zero. For every $t \geq s \geq 0$,

$$h(t) \geq h(s)e^{-b(t-s)}. \quad (6.41)$$

The coefficient b is instantaneously sharp as a uniform constant over initial populations, already within equal splitting.

Proof. The map $(t, q) \mapsto w_t(q)$ is jointly measurable: with Y started at q at time t and driven by the fragmentation Poisson measure through the measurable quantile map, $w_t(q) = \mathbb{E} \exp(-\int_0^\infty bY_s ds)$ is the expectation of a composition of measurable maps of (t, q) and the Poisson points, obtained as the monotone limit of finite-horizon survival probabilities. Enumerate the auxiliary coagulation times as τ_i .

Two steps give the density. (i) By the strong Markov property at the stopping time τ_i , the Poisson increments after τ_i are independent of $\mathcal{F}_{\tau_i}$, so the conditional probability given $\mathcal{F}_{\tau_i}$ that no coagulation follows τ_i is $w_{\tau_i}(Q_{\tau_i})$, evaluated at the post-event state; this conditions at a genuine stopping time, not at L . (ii) For a Borel set $A \subset (0, T]$, pathwise exactly one coagulation has no successor when $L \in A$ and none otherwise, so $\mathbb{P}(L \in A) = \mathbb{E} \sum_i \mathbf{1}_{\{\tau_i \in A\}} w_{\tau_i}(Q_{\tau_i})$, an integrable sum since the expected number of coagulations on $(0, T]$ is bT . The marked coagulation point process has compensator $bQ_{t-}\mu_t(dv)dt$ on the partner mark v , and the integrand $w_t(Q_{t-} + v)$ is predictable in t , so

$$\mathbb{P}(L \in A) = \int_A b \iint q w_t(q + v) \mu_t(dq) \mu_t(dv) dt,$$

where the pre-event marginal is μ_t for almost every t because the prescribed jump intensities have no fixed-time atoms. This proves (6.40), local absolute continuity of h , and the atom at zero by the definition of w_0 .

The critical envelopes (6.22) imply

$$j(t) \leq b\mathbb{E} \frac{Q_t}{1 + Q_t + Q'_t} \leq b\mathbb{E} \frac{Q_t}{1 + Q_t} \leq b\mathbb{E} g_t(Q_t) = bh(t).$$

Hence $h' = -j \geq -bh$ almost everywhere; multiplication by e^{bt} and absolute continuity give (6.41).

For sharpness take equal splitting and $Q_0 \sim \nu_{\varepsilon^2, \varepsilon}$ from (6.33), with large atom $r = r(\varepsilon^2, \varepsilon)$. Since $g'(0) = 1$, $h(0) = (1 - \varepsilon^2)g(\varepsilon) + \varepsilon^2g(r) \sim \varepsilon$, while the pair of small states gives $\mathbb{E}[Q_0\Psi(Q_0 + Q'_0)] \geq (1 - \varepsilon^2)^2\varepsilon\Psi(2\varepsilon) \sim \varepsilon$. The proved upper bound is $h(0)$, so $-h'(0)/(bh(0)) \rightarrow 1$: any coefficient smaller than b fails for some such initial law, using the right derivative at zero. $\square$

Corollary 6.9 (Exponential-moment bracket at criticality). *For every allowed critical daughter kernel,*

$$h(0)e^{-bt} \leq h(t) \leq \mathcal{A}_{1/2}(0)e^{-\kappa bt}, \quad \kappa = 3 - 2\sqrt{2}, \quad (6.42)$$

and

$$\mathbb{E}e^{rL} < \infty \quad (0 < r < \kappa b), \quad \mathbb{E}e^{rL} = \infty \quad (r \geq b). \quad (6.43)$$

Proof. Use theorem 6.8 and corollary 6.3, with $\chi_{1/2} = 2ba_{1/2} = \kappa b$. The lower envelope $g_t(q) \geq q/(1+q)$ and $Q_0 > 0$ give $h(0) > 0$. The identity $\mathbb{E}e^{rL} = 1 + r \int_0^\infty e^{rt}h(t) dt$, allowing infinity, gives divergence even at $r = b$. $\square$

These are proved bounds; they do not identify the large-time exponent or the exponential-moment abscissa within $[\kappa b, b]$. The next statement reduces that question to the deterministic equation.

Proposition 6.10 (Reduction of the last-event exponent to the overlap). *For every critical daughter kernel, with $O(t)$ the deterministic number–mass overlap of (6.30), $\frac{1}{2}O(t) \leq h(t) \leq O(t)$ for all $t \geq 0$. Consequently $\liminf_{t \rightarrow \infty} (-\log h(t))/t = \liminf_{t \rightarrow \infty} (-\log O(t))/t$, the same holds for the limits superior, and $\sup\{r : \mathbb{E}e^{rL} < \infty\} = \sup\{r : \int_0^\infty e^{rt}O(t) dt < \infty\}$. All these exponents lie in $[\kappa b, b]$ by (6.42) and (3.6).*

Proof. The two-sided bound is corollary 6.6 at $\sigma = b$. A constant factor changes neither exponential rates nor the finiteness of $\int_0^\infty e^{rt}h(t) dt$, which determines $\mathbb{E}e^{rL}$ by the identity in the preceding proof. $\square$

The calendar-time exponent of the last auxiliary coagulation is therefore a property of n_t alone: the exponential decay rate of $\int \min(1, N_0x/m) \eta_t(dx)$. The open question is purely about the deterministic equation.

Remark 6.11 (Conjecture for monodisperse equal-split data). For monodisperse critical data with equal splitting we conjecture that $O(t)$ and $h(t)$ decay with exponent $b/2$, the pure-coagulation Golovin–Borel exponent of (6.16) [5, 23]; the certified bracket (6.42) confines the exponent to $[\kappa b, b]$ with $\kappa b \approx 0.17b$. The nine retained particle runs of section 7 point toward the conjectured value: the fitted decay rates of the finite-system half moment on $[4, 10]$ listed in table 3 are $0.23b$ – $0.28b$ for $n = 10^3$, $0.30b$ – $0.37b$ for $n = 10^4$, and $0.43b$ – $0.50b$ for $n = 10^5$, increasing with n and well above κb ; an additional exploratory Marcus–Lushnikov run with $n = 2 \times 10^4$ and 5×10^4 , not archived, gave log-slopes of $O(t)$ on $[4, 9]$ of $0.54b$ and $0.57b$. This is numerical evidence subject to finite-size effects (the mass ceiling of section 7 keeps $M_{1/2} \geq n^{-1/2}$ in the unit-mass runs) and to a short fitting window; it is not a proof, and $\mathcal{A}_{1/2}$ bounds O above rather than equalling it.

Remark 6.12 (A two-point obstruction to uniform scalar dissipation). For equal splitting and a positive mean-one law μ put

$$\mathfrak{h}(\mu) = \int_E [1 - \Psi(q)]\mu(dq), \quad \mathfrak{d}(\mu) = \iint_{E^2} q\Psi(q+v)\mu(dq)\mu(dv),$$

so that $h(0) = \mathfrak{h}(\mu_0)$ and $h'(0) = -b\mathfrak{d}(\mu_0)$ by theorem 6.7. One might hope to close the bracket (6.42) by an instantaneous scalar inequality $h' \leq -cbh^\beta$ holding uniformly over initial data. For

$\beta < 1$ this is impossible for a trivial reason: it forces h to reach zero in finite time, contradicting $h(t) \geq h(0)e^{-bt} > 0$. For $\beta \geq 1$, with $\beta = 1$ the relevant case, a two-point family excludes it: for $\varepsilon \downarrow 0$ let

$$q_\varepsilon^- = \exp[-(\log(1/\varepsilon))^3], \quad q_\varepsilon^+ = \frac{1 - (1 - \varepsilon)q_\varepsilon^-}{\varepsilon}, \quad \mu_\varepsilon = \nu_{\varepsilon, q_\varepsilon^-} = (1 - \varepsilon)\delta_{q_\varepsilon^-} + \varepsilon\delta_{q_\varepsilon^+}. \quad (6.44)$$

Since $q_\varepsilon^+ \sim \varepsilon^{-1}$ and $g(q_\varepsilon^+) \rightarrow 1$, $\varepsilon g(q_\varepsilon^+) \leq \mathfrak{h}(\mu_\varepsilon) \leq \varepsilon + q_\varepsilon^-$, so $\mathfrak{h}(\mu_\varepsilon) \sim \varepsilon$. Separating the first coordinate into its two values, monotonicity of Ψ and $\varepsilon q_\varepsilon^+ \leq 1$ give $\mathfrak{d}(\mu_\varepsilon) \leq q_\varepsilon^- + \Psi(q_\varepsilon^+)$. The first k product factors give, for every integer $k \geq 1$,

$$\Psi(R) \leq 2^{k(k+1)/2} R^{-k}, \quad (6.45)$$

so for fixed β and $k > \beta$ both $q_\varepsilon^-/\varepsilon^\beta$ and $\Psi(q_\varepsilon^+)/\varepsilon^\beta$ tend to zero:

$$\inf_{\substack{\mu \text{ supported on two positive points} \\ \int q\mu(dq)=1}} \frac{\mathfrak{d}(\mu)}{\mathfrak{h}(\mu)^\beta} = 0 \quad (\beta > 0). \quad (6.46)$$

Realizing μ_ε as initial data by the scaling after [corollary 6.6](#), and using continuity of h' to extend a strict violation at zero to a positive time interval, refutes every uniform inequality $h' \leq -cbh^\beta$. By the large- q form [\(B.1\)](#) the computation excludes more than powers: along this family $\mathfrak{d} \leq \exp[-(1 + o(1))(\log(1/\mathfrak{h}))^2/(2 \log 2)]$.

The mechanism is a deferred huge atom: almost all number sits at a size whose own coagulation hazard is negligible, while the rare huge particle will certainly coagulate, but not yet. The family refutes only uniform-in-initial-data instantaneous bounds of the form $h' \leq -cbh^\beta$; it says nothing about data-dependent or asymptotic scalar bounds, and it contradicts neither the upper tail certificate, whose prefactor carries the separate information $\mathcal{A}_{1/2}(0)$, nor the conjecture above. Every μ_ε has finite moments of every order. The small atom q_ε^- decays faster than every power of ε yet stays representable in double precision (about 10^{-232} at $\varepsilon = 3 \times 10^{-4}$), which keeps the illustrative floating-point evaluations in `supplement/last_coagulation_exact.py` inside the positive-size model; $e^{-1/\varepsilon}$ would underflow to zero there.

7 Finite populations: count accuracy and mass-distribution discrepancy

The critical continuum equation conserves particle count while its mass law escapes every fixed size window. A finite system has an additional constraint: no particle can exceed the system's entire mass. This ceiling separates the finite mass distribution from a growing-time continuum approximation, even while the count prediction remains accurate. The comparison below starts from exactly the same empirical measure; it contains no initial sampling error.

The result should be read against the convergence literature for Marcus–Lushnikov particle systems. Hydrodynamic limits on a fixed time horizon are classical: Norris [\[31\]](#) proves such a limit for the stochastic coalescent, and Aldous [\[2\]](#) surveys the mean-field theory. For the additive kernel the stochastic time scale is logarithmic in the population size: Aldous and Pitman [\[1\]](#) obtain the standard additive coalescent by starting the n -cluster additive coalescent at time $-\frac{1}{2} \log n$, so that clusters carrying a positive fraction of the total mass appear at times of order $\frac{1}{2} \log n$ after the start from n unit clusters (with $b = 1$ their pair rates coincide with those of the process below without fragmentation). Discrete deterministic mean-field models are surveyed by Wattis [\[37\]](#). [Theorem 7.2](#)

is a growing-time statement of a different kind. It compares one finite realization with the continuum solution started from the same empirical measure; it is not a propagation-of-chaos result, and it does not attempt the first divergence time or the behaviour of the largest cluster.

7.1 The physical particle process and its autonomous count

Fix $\lambda, m > 0$ and $b = \lambda m$. For each integer $n \geq 1$, start with $L_0^n \geq 1$ deterministic positive masses whose sum is nm . Each unordered pair $i < j$ merges at rate $\lambda(x_i + x_j)/n$. Each particle fragments at rate b into two positive complementary masses. Its split distribution may be any measurable function of its parent size. Write b_x for the resulting expected daughter measure. Eventwise mass conservation is required here; the larger deterministic class (2.2) alone would not specify such a physical process. Set

$$\mu_t^n = \frac{1}{n} \sum_{i=1}^{L^n(t)} \delta_{x_i(t)}, \quad c_n = \frac{L_0^n}{n}, \quad \pi_t^n(dx) = \frac{x\mu_t^n(dx)}{m}, \quad (7.1)$$

and assume $\sup_n c_n < \infty$. Let v_t^n be the solution of [theorem 2.3](#) with initial measure $v_0^n = \mu_0^n$, the same b_x , and critical rates. This solution exists for every n since its initial measure has finite support; no uniform initial second-moment bound is needed. Its count is c_n and its mass is m . Denote its mass law by $\bar{\pi}_t^n(dx) = xv_t^n(dx)/m$.

Proposition 7.1 (Exact count law and concentration). *The particle process is nonexplosive, conserves mass nm pathwise, and has the autonomous count rates*

$$\ell \longrightarrow \ell + 1 : b\ell, \quad \ell \longrightarrow \ell - 1 : b(\ell - 1), \quad \ell \geq 1. \quad (7.2)$$

For deterministic L_0^n ,

$$\begin{aligned} \mathbb{E}L^n(t) &= L_0^n + bt, \\ \text{Var } L^n(t) &= b(2L_0^n - 1)t + b^2t^2, \end{aligned} \quad (7.3)$$

$$\mathbb{E} \sup_{t \leq T} |L^n(t)/n - c_n|^2 \leq \frac{8b(2L_0^n - 1)T + 10b^2T^2}{n^2} \quad (T \geq 0). \quad (7.4)$$

Consequently count per n converges uniformly to its continuum prediction in L^2 on every deterministic horizon $T_n = o(n)$.

Proof. In a configuration with ℓ particles,

$$\sum_{i < j} (x_i + x_j) = (\ell - 1) \sum_i x_i = (\ell - 1)nm.$$

Thus coagulation has total rate $b(\ell - 1)$ and fragmentation has rate $b\ell$. After k events, $\ell \leq L_0^n + k$, so the event count is dominated by a pure-birth chain with rate $2b(L_0^n + k)$. This chain is nonexplosive (the reciprocal rates have divergent sum) and has finite moments of every fixed order on finite time intervals. The usual sequential exponential-clock construction therefore gives the full process, with only finitely many events on each finite interval. Every event preserves strictly positive masses and their sum.

The generator on ℓ and ℓ^2 gives b and $4b\ell - b$. The compensated process $M_t = L^n(t) - L_0^n - bt$ is a square-integrable martingale with

$$\langle M \rangle_t = \int_0^t b(2L^n(s) - 1) ds, \quad \mathbb{E}M_T^2 = b(2L_0^n - 1)T + b^2T^2. \quad (7.5)$$

Here $\langle M \rangle$ denotes predictable quadratic variation. The generator identities prove (7.3); all calculations may first be stopped at bounded count and then unstopped using the preceding domination. Doob's inequality gives $\mathbb{E} \sup_{t \leq T} |M_t + bt|^2 \leq 8\mathbb{E}M_T^2 + 2b^2T^2$, which is (7.4). Since $L_0^n = O(n)$, its right-hand side tends to zero for $T_n = o(n)$. $\square$

The finite count has drift b , and the normalized count has drift b/n . This is the correction from excluding self-pairs. Concentration is absolute; relative accuracy and the persistence of order- n particles follow, for example, if $c_n \rightarrow c > 0$, but need not follow when $c_n \rightarrow 0$. The exact probability generating function and the classical branching-with-immigration diffusion of Feller [18] and Kawazu and Watanabe [26] on the later time scale $t = n\tau$ are derived in [appendix C](#).

7.2 A logarithmic-time mass-CDF discrepancy

For probabilities on E , use the cumulative-distribution distance

$$d_{\text{CDF}}(\alpha, \beta) = \sup_{R>0} |\alpha((0, R]) - \beta((0, R])| \leq 1.$$

This avoids a purely atomic-versus-continuous obstruction in total variation.

Theorem 7.2 (Mass discrepancy with simultaneous count accuracy). *For every $C > 1/(b \log 2)$ there is a fixed $p \in (0, 1)$ such that $\eta := bC\kappa_p - (1 - p) > 0$. At $T_n = C \log n$, every realization satisfies*

$$d_{\text{CDF}}(\pi_{T_n}^n, \bar{\pi}_{T_n}^n) \geq 1 - c_n^{1-p} n^{-\eta} \rightarrow 1, \quad (7.6)$$

whereas $\mathbb{E} \sup_{t \leq T_n} |L^n(t)/n - c_n|^2 \rightarrow 0$. The same discrepancy lower bound holds, trivially and for the same reason, with $\mathbb{E}\pi_{T_n}^n$ in place of $\pi_{T_n}^n$. The estimates are uniform over all such initial configurations with a common upper bound on c_n , and over all admissible split laws.

The particle dynamics enter (7.6) only through the total-material ceiling $\pi_t^n((0, nm]) = 1$, which holds for any mass-conserving finite system. The discrepancy bound is therefore a corollary of [corollary 3.5](#) for the continuum solution started from the empirical measure. The dynamics are used only in the accompanying count statement, through [proposition 7.1](#).

Proof. The physical mass ceiling gives $\pi_t^n((0, nm]) = 1$ on every path. By [corollary 3.5](#) and Hölder's inequality,

$$M_p(v_t^n) \leq c_n^{1-p} m^p e^{-b\kappa_p t}.$$

Since $x \leq (nm)^{1-p} x^p$ on $(0, nm]$,

$$\bar{\pi}_t^n((0, nm]) \leq (nc_n)^{1-p} e^{-b\kappa_p t}. \quad (7.7)$$

The limit $\kappa_p/(1-p) \rightarrow \log 2$ as $p \uparrow 1$ permits a fixed $p < 1$ with $\eta > 0$. Substitute $t = T_n$ in (7.7) and compare at $R = nm$. Expectation preserves the physical ceiling, so the same comparison applies to $\mathbb{E}\pi_t^n$. Count accuracy follows from [proposition 7.1](#) because $\log n = o(n)$. $\square$

The deterministic equation puts almost all material in sizes larger than the entire finite system can contain. The mass-CDF bound is the continuum escape of [corollary 3.5](#) read against the ceiling; it uses nothing about the particle dynamics beyond mass conservation. This proves failure of a uniform approximation through these logarithmic horizons in mass CDF. The constant is sufficient: the theorem neither locates the first breakdown time nor proves accuracy at earlier growing times. It is not a propagation-of-chaos theorem. A minimum indivisible particle size would change the splitting rule and invalidate the autonomous count law once sufficiently small particles appear. With the total-variation convention (2.5), the same lower bound also holds for $\|\pi_{T_n}^n - \bar{\pi}_{T_n}^n\|_{\text{TV}}$.

Remark 7.3 (The threshold constant and the certificate). The threshold $1/(b \log 2)$ comes from the $p \uparrow 1$ end of the fractional-moment bound, where $\kappa_p/(1-p) \rightarrow \log 2$. It is a sufficient threshold for mass-CDF discrepancy. A separate second-moment comparison is available for equal splitting: testing (2.4) with $f(x) = x^2$ gives $M'_2 = (2b-b/2)M_2$, so $M_2(v_t^n) = M_2(v_0^n)e^{3bt/2}$. The mass-weighted mean size $M_2(v_t^n)/m$ reaches the ceiling nm at

$$t_*^n = \frac{2}{3b} \log \frac{nm^2}{M_2(v_0^n)}.$$

If $M_2(v_0^n)$ is bounded above and below by positive constants independent of n , then $t_*^n = (2/3b) \log n + O(1)$. The theorem does not impose this initial-moment scaling. Since the finite system has $M_2(\mu_t^n) \leq nm^2$, its second moment differs from the continuum second moment for $t > t_*^n$. A small amount of mass at very large sizes can produce this difference, so it does not establish a macroscopic mass-CDF discrepancy or determine its first occurrence. Moreover $\eta = bC\kappa_p - (1-p)$ depends on the excess $C - 1/(b \log 2)$: expanding κ_p at $p = 1$ shows that the best η vanishes like the square of this excess. The certificate $1 - c_n^{1-p}n^{-\eta}$ can therefore be weak near the threshold. At time 10 with $b = 1$, the best value over p is about 2×10^{-5} for $n = 10^3$ and zero for $n = 10^4$ and 10^5 (section 7.3).

There is a related relative-moment obstruction. The ceiling gives $M_p(\mu_t^n) \geq m(nm)^{p-1}$ and hence

$$\frac{M_p(\mu_{T_n}^n)}{M_p(v_{T_n}^n)} \geq c_n^{p-1}n^\eta \rightarrow \infty. \quad (7.8)$$

The denominator is positive by positive conserved mass. This is a divergent relative error, with no asserted nonvanishing absolute moment error.

The finite correction can also be identified before any comparison estimate. Let $\mathcal{D}_p(\mu)$ be the continuum weak drift for x^p , evaluated at an empirical measure $\mu = n^{-1} \sum_i \delta_{x_i}$. Its coagulation part includes the diagonal $i = j$, whereas the physical generator $\mathcal{G}_n$ only permits distinct particles. Thus

$$\mathcal{G}_n M_p(\mu) = \mathcal{D}_p(\mu) + \frac{\lambda(2-2^p)}{n} M_{p+1}(\mu), \quad 0 < p < 1. \quad (7.9)$$

Indeed the included diagonal is $\lambda(2^p - 2)n^{-2} \sum_i x_i^{p+1}$; its subtraction gives (7.9). Fragmentation is linear and already has exactly the continuum drift. The factor $1/n$ does not ensure uniform smallness when higher moments grow. At $p = 0$ the same calculation gives the count correction $\lambda m/n = b/n$.

7.3 Nine finite-particle realizations

The numerical experiment uses $\lambda = m = b = 1$, equal splitting, and n initial unit masses, for $n \in \{10^3, 10^4, 10^5\}$ and seeds 11, 29, 47 at each size. All nine runs and all ten snapshot times 0, 0.5, 1, 2, 3, 4, 5, 6, 8, 10 are retained in the supplied 90-row CSV; the runs processed 6,588,932 events. These simulations describe the finite process only. No continuum solution is computed, and no theorem is tested.

The sampler is an exact event construction in real arithmetic. With L particles it draws an exponential waiting time of rate $2L - 1$, fragments a uniform parent with probability $L/(2L - 1)$, and otherwise merges a particle chosen proportionally to mass with a second chosen uniformly among the other $L - 1$; the resulting unordered pair rate is $(x_i + x_j)/n$. At $L = 1$ fragmentation is certain. A Fenwick tree supplies mass sampling and updates in $O(\log L)$ time. Masses are `C++ long double`.

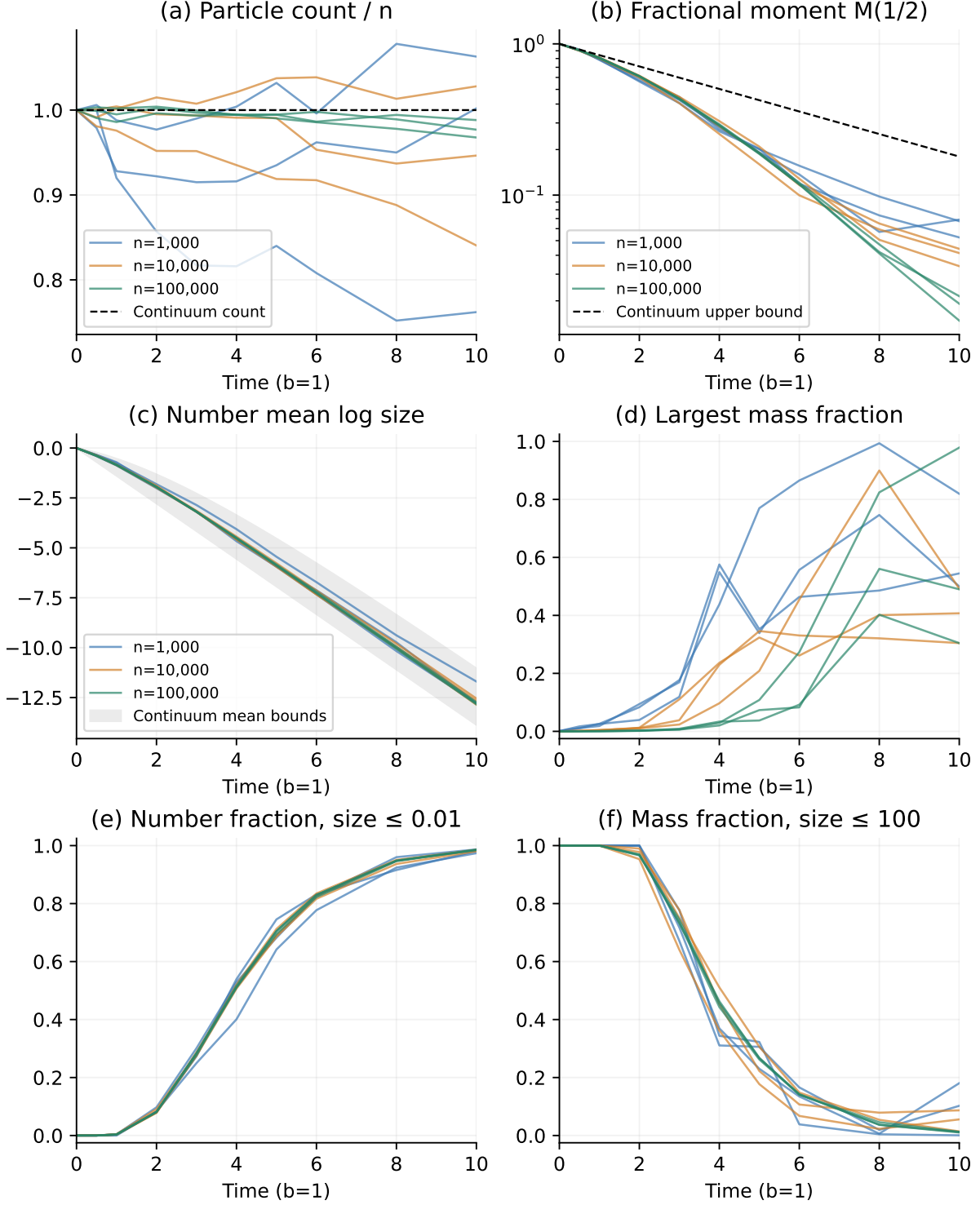


Figure 1: All nine finite-particle runs, with three seeds in each color. The dashed count and half-moment curves and the shaded mean-log interval are continuum references, not pathwise finite-system bounds. Number fractions and log means use the actual count L ; mass fractions use total physical mass n ; $M_{1/2} = n^{-1} \sum_i \sqrt{x_i}$. The figure is generated directly from the supplied 90-row CSV.

Initial n	$L(10)/n$	$M_{1/2}(10)$	Largest mass fraction	Number fraction $x \leq 0.01$
10^3	0.762–1.063	0.0525–0.0688	0.500–0.820	0.974–0.987
10^4	0.8405–1.028	0.0339–0.0441	0.304–0.492	0.981–0.986
10^5	0.96772–0.98823	0.0148–0.0215	0.304–0.979	0.985–0.986

Table 2: Minima and maxima over the three declared seeds at time 10. These ranges are descriptive, not confidence intervals.

Initial n	Seed 11	Seed 29	Seed 47
10^3	0.228	0.275	0.236
10^4	0.311	0.368	0.302
10^5	0.496	0.432	0.455

Table 3: Fitted decay rate $-\log(A_{1/2}(10)/A_{1/2}(4))/6$ of the finite half moment on $t \in [4, 10]$ with $b = 1$, from the saved CSV. The certified continuum coefficient is $\kappa \approx 0.172$.

Figure 1 and table 2 show number concentration near small sizes alongside highly variable material concentration in a few large particles. At time 10 the count is within 3.3% of its continuum value for $n = 10^5$. The largest absolute relative deviations for $n = 10^3$ and $n = 10^4$ are 23.8% and 15.95%, respectively. The run with $n = 10^4$, seed 47, ends 3.59 exact standard deviations below its count mean and is included without selection. Three seeds do not reliably estimate the count distribution or mass-tail variability. The dashed half-moment reference is the continuum bound $e^{-\kappa t}$, equal to 0.17983 at time 10; the shaded mean-log interval is $[-2t \log 2, -2t \log 2 + (1 - e^{-2\kappa t})/(2\kappa)]$ from theorem 5.1 and the half-moment estimate. All nine finite log means fall inside this interval, but that fact does not turn the interval into a stochastic bound. The mass-discrepancy certificate of theorem 7.2 at this horizon is about 1.95×10^{-5} for $n = 10^3$ and zero for the two larger sizes, whose threshold times $\log n / \log 2$ are 9.97, 13.29, and 16.61 (remark 7.3). These runs therefore neither establish a nearly maximal continuum discrepancy nor measure a first breakdown time.

The half-moment panel does give a rough empirical decay rate. For the finite system write $A_{1/2}(t) = M_{1/2}(\mu_t^n)$. The fitted rate $-\log(A_{1/2}(10)/A_{1/2}(4))/6$ on $t \in [4, 10]$, computed from the saved CSV, is listed in table 3. It is 0.23–0.28 for $n = 10^3$, 0.30–0.37 for $n = 10^4$, and 0.43–0.50 for $n = 10^5$, increasing with n , against the certified continuum coefficient $\kappa = 3 - 2\sqrt{2} \approx 0.172$. This supports the conjecture in remark 6.11 that the true critical exponent for these data is near $b/2$. The caveats are finite-size effects, including the ceiling and the $1/n$ correction (7.9), and only three seeds per size.

The self-contained supplement includes the C++ source, standard-library Python runner, recorded data and compiler metadata, figure script, and build commands. Built-in checks exercise selection intervals, resizing, removal, pair rates, count-generator identities, and conservation over 10,000 events. Each snapshot checks $L = n + \text{births} - \text{deaths}$, the mass ceiling, and agreement of direct and tree mass sums, with relative mass tolerance 10^{-10} . The archived normalized mass is printed as one at every snapshot. Equal splitting preserves dyadic masses in real arithmetic, but a bound on their size range does not bound the number of significant bits needed for exact arithmetic. Floating-point rounding can occur in merges and cumulative sums even in these runs; the printed mass values do not certify exact conservation. Longer runs with extreme size ratios can lose small weights in cumulative sums, and the uniform draws have 53 bits of resolution. Different floating-point platforms or C++ library implementations can change a path with the same seed. The supplement also records the prior independent structural and count-law checks and their reproducible commands;

they are implementation diagnostics, not extra research trajectories.

8 What bulk observables hide, and what fails beyond the additive kernel

Count neutrality has an exact algebraic classification at fixed mass. It explains why constant count, mass, and mean size can coexist with rapid broadening. A second observable removes this universal ambiguity. Beyond the additive kernel, the same classification also leads to a finite-count nonstationarity theorem, while fractional-moment monotonicity can fail.

8.1 Universal count neutrality and observable rigidity

In this subsection $K : E^2 \rightarrow [0, \infty)$ is measurable, symmetric and finite pointwise, and $S : E \rightarrow [0, \infty)$ is measurable and finite pointwise. The expected daughter measure b_x satisfies (2.2). For a finitely supported preparation v , define its count vector field by

$$\mathcal{R}(v) = \int S(x)v(\mathrm{d}x) - \frac{1}{2} \iint K(x, y)v(\mathrm{d}x)v(\mathrm{d}y).$$

This definition concerns initial rates and needs no existence theorem for the general kernel. Fix a prescribed material mass $m > 0$. The preparation class below fixes this mass while allowing count to vary.

Theorem 8.1 (The complete count-neutral class). *The identity $\mathcal{R}(v) = 0$ holds for every nonnegative finitely supported v with $M_1(v) = m$ if and only if there is a nonnegative measurable function d , finite pointwise, such that*

$$K(x, y) = xd(y) + yd(x), \quad S(x) = md(x). \quad (8.1)$$

For this class, any mass-conserving trajectory at mass m whose weak count balance has locally integrable event rates has constant count.

Proof. The preparation $(m/x)\delta_x$ gives $S(x) = mK(x, x)/(2x)$. Define $d(x) = S(x)/m$. For $x \neq y$, substitute $v = (m/(2x))\delta_x + (m/(2y))\delta_y$ and use the diagonal identities. After cancellation the result is $K(x, y) = xd(y) + yd(x)$; the same relation already holds on the diagonal. Conversely, for (8.1),

$$\mathcal{R}(v) = (m - M_1(v)) \int d(x)v(\mathrm{d}x). \quad (8.2)$$

This proves the equivalence. Along a trajectory with the stated event integrability, the weak count balance integrates the right-hand side of (8.2), which vanishes by mass conservation. $\square$

The full-time assertion uses the weak equation and conserved mass, rather than inferring a trajectory from an initial derivative. The fixed-mass preparation class is essential: a class fixing both mass and count admits only one monodisperse size and cannot support the proof below.

Proposition 8.2 (An additional moment forces zero kinetics). *Fix any $0 < p < 1$. If the count vector field vanishes for every finitely supported preparation of mass m , and the p th-moment vector field vanishes for every monodisperse preparation of mass m , then $K = S = 0$ pointwise. The same conclusion holds if the second moment replaces the p th moment. In particular one may use the half moment.*

Proof. By [theorem 8.1](#), the rates have form (8.1) with $d \geq 0$. At $v = (m/x)\delta_x$ the coagulation contribution to the p th-moment vector field is $m^2 d(x) x^{p-1} (2^p - 2)$. Jensen's inequality bounds fragmentation by $m^2 d(x) x^{p-1} (2^{1-p} - 1)$. Therefore their sum is at most

$$-\kappa_p m^2 d(x) x^{p-1}. \quad (8.3)$$

If this vector field is zero, then $d(x) = 0$ for every x . For the second moment, Jensen gives $\int z^2 b_x(dz) \geq x^2/2$, so the corresponding vector field is at least $3m^2 x d(x)/2$. Its universal vanishing again forces $d = 0$. Every displayed integral is finite for these monodisperse preparations because daughters lie below their parent. $\square$

This is a result about a class of preparations at fixed mass, not identification of arbitrary rates from a single observed trajectory. Only monodisperse preparations enter the moment hypothesis, and the proof of [theorem 8.1](#) uses only preparations with at most two atoms, so the count hypothesis can be weakened to that class as well. Nonnegative physical rates are used in the conclusion from (8.3).

8.2 Broadening and entropy production with fixed bulk outputs

Return to the additive member $d \equiv \lambda > 0$, with critical selection $b = \lambda m$. The daughter law may now depend measurably on time and size, as in [assumption 2.1](#). In the finite-second-moment solution class already constructed, second-moment and entropy balances require no extra assumptions. Use $\mathcal{H}(t) = \int x \log x n_t(dx)$ for the entropy moment, to avoid confusing it with the half moment H used earlier.

Proposition 8.3 (Two unbounded observables). *For the critical solution of [theorem 2.3](#), M_2 and $\mathcal{H}$ are finite and locally absolutely continuous, and almost everywhere*

$$\frac{3}{2} \lambda m M_2(t) \leq M_2'(t) \leq 2 \lambda m M_2(t), \quad (8.4)$$

$$\mathcal{H}'(t) \geq \lambda m^2 \log 2. \quad (8.5)$$

In particular,

$$M_2(0) e^{3\lambda m t/2} \leq M_2(t) \leq M_2(0) e^{2\lambda m t}, \quad \mathcal{H}(t) \geq \mathcal{H}(0) + \lambda m^2 t \log 2. \quad (8.6)$$

Equal splitting gives exactly $M_2(t) = M_2(0) e^{3\lambda m t/2}$. The entropy production constant is sharp as a uniform instantaneous bound: monodisperse data with equal splitting attain it at time zero.

Proof. The unbounded-test justification is given in [appendix D](#); it establishes the complete integrated weak balances using only locally bounded M_2 . The coagulation contribution to M_2' is $2\lambda m M_2$. By Jensen and $z < x$,

$$x^2/2 \leq \int z^2 b_{t,x}(dz) \leq x^2,$$

so the fragmentation contribution lies between $-bM_2/2$ and zero. This proves (8.4) and its equal-split identity.

For $f(x) = x \log x$, put $u = x/(x+y)$ and $h(u) = -u \log u - (1-u) \log(1-u)$. Then $\Delta f(x, y) = (x+y)h(u)$. Divide (3.1) by $1-p$ and let $p \uparrow 1$ at fixed positive x, y . Differentiability of the scalar powers gives

$$(x+y)\Delta f(x, y) \geq 4(\log 2)xy, \quad \text{equivalently} \quad h(u) \geq 4(\log 2)u(1-u). \quad (8.7)$$

Estimates of this subadditivity type for concave tests are standard in the coagulation–fragmentation literature; see Escobedo et al. [17]. Thus coagulation produces at least $2\lambda m^2 \log 2$. Jensen’s inequality for $b_{t,x}/2$ and the increase of $f(z)/z = \log z$ give

$$-x \log 2 \leq \int f(z) b_{t,x}(dz) - f(x) \leq 0.$$

Fragmentation loses at most $bm \log 2$, proving (8.5). Integrating proves (8.6). Both scalar inequalities are equalities at a monodisperse equal-split initial state. The right derivative there is justified at the end of appendix D, so sharpness is a statement about actual solutions as well as their vector fields. $\square$

Expected daughter count two and expected daughter mass x suffice for these bounds; an eventwise binary law is not needed. For any fixed initial measure with $0 < M_2(0) < \infty$, compare the equal-split solution with the static model $K = S = 0$ from that same measure. At every time both models have identical count N_0 , mass m , and mean size m/N_0 , yet their second moments differ by $M_2(0)(e^{3\lambda m t/2} - 1)$. At a fixed positive time this difference can be made arbitrarily large by increasing λ . Consequently these three bulk outputs alone cannot bound the second-moment error uniformly over the class. For each fixed λ the second moment remains finite at every finite time; this example has no finite-time gelation claim. The entropy inequality provides another strictly increasing observable in the same accepted solution class.

8.3 A power-kernel nonstationarity theorem

Consider a separate time-independent model,

$$K(x, y) = xy^\alpha + yx^\alpha, \quad S(x) = sx^\alpha, \quad 0 < \alpha \leq 1, \quad s \geq 0. \quad (8.8)$$

Let B_x be the pushforward of b_x by $z \mapsto z/x$, so $B_x((0, 1)) = 2$ and $\int \theta B_x(d\theta) = 1$. These fraction measures may depend measurably on the parent. A stationary measure means a nonnegative measure v whose coagulation–fragmentation vector field is zero for every bounded Borel test, with finite total event integrals.

Theorem 8.4 (No finite-count equilibrium for power kernels). *For (8.8) and every such daughter law, there is no stationary measure v with finite positive count N and finite positive mass m . No negative moment, logarithmic moment, or moment above order one is required.*

Proof. Suppose v is such a measure and put $D = M_\alpha(v)$. The inequality $x^\alpha \leq 1+x$ gives $0 < D < \infty$; the total coagulation and fragmentation event rates are mD and sD . Testing stationarity with one gives $(s - m)D = 0$, hence $s = m$. The case $s = 0$ is already excluded because $m > 0$.

At $s = m$ the integrated population vector field equals $\int \mathcal{L}f dv$, where

$$\mathcal{L}f(x) = x \int y^\alpha [f(x+y) - f(x)] v(dy) + mx^\alpha \int [f(\theta x) - f(x)] B_x(d\theta). \quad (8.9)$$

To verify this, symmetry combines the coagulation gains and the two losses in the full weak field into $\iint xy^\alpha f(x+y) dv dv - D \int xf dv - m \int x^\alpha f dv$. Fragmentation adds $m \int x^\alpha \int f(\theta x) B_x(d\theta) v(dx) - m \int x^\alpha f dv$, exactly as in (8.9). The total jump rate of this representation is

$$q(x) = xD + 2mx^\alpha, \quad \int q(x) v(dx) = 3mD. \quad (8.10)$$

Divide its nonnegative gain kernel by $q(x) > 0$ to obtain a Markov transition kernel $P(x, dz)$, and set $\zeta(dx) = q(x)v(dx)/(3mD)$. Bounded-test stationarity is exactly the finite-measure identity $\zeta P = \zeta$.

The singular test $g(x) = x^{-\alpha}$ need not be integrable under v , but it is integrable under the rate-weighted probability:

$$3mD \int g d\zeta = DM_{1-\alpha}(v) + 2mN < \infty. \quad (8.11)$$

Test $\zeta P = \zeta$ first with $g \wedge k$ and let $k \uparrow \infty$. Monotone convergence gives $\int Pg d\zeta = \int g d\zeta < \infty$. Both gain and loss are therefore integrable, and subtraction is legitimate:

$$\int \mathcal{L}g dv = 0. \quad (8.12)$$

No jump-process construction or time-dependent population solution has been used. Even if some daughter inverse moments are initially infinite, the hypothetical stationary identity forces their integrated gain to be finite by (8.11).

Jensen for $B_x/2$ gives $\int \theta^{-\alpha} B_x(d\theta) \geq 2^{1+\alpha}$, so fragmentation in (8.12) contributes at least $2mN(2^\alpha - 1)$. Write its coagulation contribution as $-J$, where

$$J = \iint xy^\alpha [x^{-\alpha} - (x+y)^{-\alpha}] v(dx) v(dy) \geq 0.$$

We claim $J \leq \alpha mN$. Symmetrize the integrand, set $r = x + y$ and $u = x/r$, and denote the symmetrized integrand by $rC_\alpha(u)/2$, where

$$C_\alpha(u) = u(1-u)^\alpha(u^{-\alpha} - 1) + (1-u)u^\alpha((1-u)^{-\alpha} - 1).$$

Concavity of z^α gives $u^{-\alpha} - 1 \leq \alpha(u^{-1} - 1)$ and the counterpart with $1 - u$. Thus

$$C_\alpha(u) \leq \alpha[(1-u)^{1+\alpha} + u^{1+\alpha}] \leq \alpha.$$

Integrating $r/2$ gives mN , proving the claim. It follows that

$$\int \mathcal{L}g dv \geq [2(2^\alpha - 1) - \alpha]mN \geq (2 \log 2 - 1)\alpha mN > 0, \quad (8.13)$$

contradicting (8.12). □

At $\alpha = 1$, $D = m$ and $\zeta = xv/m$ is the mass probability itself. The embedded transition coagulates with probability $1/3$ by adding an independent mass-sampled size and fragments with probability $2/3$ using $B_x/2$. Since $\int x^{-1}\zeta(dx) = N/m < \infty$, stationarity would imply

$$3\mathbb{E}(1/X) = \mathbb{E}[1/(X+Y)] + 2\mathbb{E}[1/(X\Theta)] \geq 4\mathbb{E}(1/X),$$

a direct contradiction; here X, Y are independent with law ζ and $\Theta \mid X$ has law $B_X/2$. This illustrates why rate weighting makes the singular test admissible.

8.4 Failure of fractional-moment monotonicity beyond the additive kernel

The preceding proof does not transfer the additive model's monotone positive moments. With equal splitting, take $v_R = \delta_1 + R^{-1}\delta_R$, $R > 1$, of mass two, and choose $s = m = 2$. Direct substitution gives, for $0 < p < 1$,

$$\begin{aligned}\mathcal{D}_p(v_R) &= A_p(1 + R^{\alpha+p-1}) + (1 + R^{\alpha-1})[(1 + R)^p - 1 - R^p], \\ A_p &= 2^p + 2^{2-p} - 4 = \frac{(2^p - 2)^2}{2^p} > 0.\end{aligned}\tag{8.14}$$

For verification, the two diagonal coagulation terms are $(2^p - 2)(1 + R^{\alpha+p-1})$, fragmentation is $2(2^{1-p} - 1)(1 + R^{\alpha+p-1})$, and the cross coagulation term is the second term in (8.14). If $p > 1 - \alpha$, the first term tends to infinity, while the negative second term remains bounded: $-1 < (1 + R)^p - R^p - 1 < 0$ and $1 + R^{\alpha-1} \leq 2$. Nothing is proved for $p \leq 1 - \alpha$: there the first term stays bounded and this family does not settle the sign of $\mathcal{D}_p(v_R)$.

In particular the half moment can have positive drift for every $\alpha > 1/2$. An explicit half-moment counterexample is $\alpha = 1$, $p = 1/2$, $R = 64$:

$$\mathcal{D}_{1/2}(v_{64}) = 27\sqrt{2} + 2\sqrt{65} - 54 > \frac{151}{500} > 0.\tag{8.15}$$

Indeed $(1414/1000)^2 < 2$ and $(8062/1000)^2 < 65$, and substitution gives the rational lower bound. The supplement independently evaluates the atomic double sum and certifies this inequality by rational arithmetic. These are vector-field counterexamples. If a local mass-conserving solution with the required initial moment derivative is separately established, they give that derivative; no time-dependent existence assertion for (8.8) is made here. Since count and mass are neutral at these data, the same sign obstruction applies to a universal initial decay inequality for the normalized affinity.

The finite-count qualification is necessary. Tran and Van [35, Proposition 4.3 and Remark 4.4 of arXiv version 3, whose numbering we follow] construct critical stationary Bernstein transforms for $K(x, y) = xy$ and a constant binary breakup kernel, which in our notation means selection $x/2$ and uniform daughter fractions. Their mass-one convention maps to $K = 2xy$, $S = mx$ at mass m by $n_t = mc_{2mt}$. Writing their stationary transform as $F(q) = \int (1 - e^{-qx})c(dx)$ and $G = 1 - F'$, their identity is

$$\frac{G(q)}{(1 - G(q))^3} = Cq, \quad C > 0.$$

It implies $F'(0) = 1$ and $F(q) \sim \frac{3}{2}C^{-1/3}q^{2/3} \rightarrow \infty$: indeed $G(q) \rightarrow 1$ and $F'(q) = 1 - G(q) \sim (Cq)^{-1/3}$, which can be integrated using two-sided asymptotic bounds. Monotone convergence in the Bernstein representation then gives infinite count. These known equilibria do not contradict theorem 8.4.

For $0 < \alpha \leq 1$, that theorem establishes only stationary exclusion. It does not supply mass conservation, boundary convergence, logarithmic scaling, or a final auxiliary coagulation event for time-dependent power kernels. For $\alpha > 1$, both the finite flux and the concavity estimate in its proof can fail under count and mass assumptions alone; that regime is outside the result.

9 Late-time Fourier identification and a pointwise sampling bound

Throughout this section $\lambda > 0$ and $\sigma \geq 0$ are constants, $b = \lambda m$, and the selfsimilar daughter measure in (5.1) is a fixed measure B . Thus $B((0, 1)) = 2$ and $\int \theta B(d\theta) = 1$. We assume a solution

in [definition 2.2](#); no initial or daughter logarithmic moment is required. The well-posed class in [theorem 2.3](#) supplies such solutions when $M_2(0) < \infty$. Define the log-size characteristic function and fragmentation exponent by

$$\chi_t(k) = \int_E e^{ik \log x} \eta_t(dx), \quad \psi(k) = \sigma \int_{(0,1)} (\theta^{ik} - 1) B(d\theta), \quad k \in \mathbb{R}. \quad (9.1)$$

The symbol χ_t is a log-size characteristic function, not a raw-size Laplace transform. The exponent ψ is continuous, vanishes at zero, and has nonpositive real part. Its finite jump measure is $\nu = \sigma(\log)_\# B$, supported on $(-\infty, 0)$.

Fourier and Mellin inversion are established tools for fragmentation inference. The exact pure-fragmentation transform evolution appears in Doumic and Escobedo [[13](#), equations (12)–(14)]; inverse recovery from asymptotic profiles and from short-time observations is developed by Doumic et al. [[14](#), [15](#)]. Most directly, Garnier [[21](#), Section 2.2] cancels an unknown factor with two-time characteristic-function quotients and uses a distinguished logarithm for decompounding. Here the task is to justify this cancellation asymptotically in a nonlinear equation and bound the coagulation error. The limiting factor need not be a characteristic function. Hoang et al. [[24](#), Sections 3.1.1–3.1.4] already develops log-size Fourier estimation from independent samples of an asymptotic growth–fragmentation profile, including denominator regularization. That model has size growth and division without coagulation.

9.1 Factorization with an exponentially small nonlinear error

Set, consistently with ([5.7](#)),

$$\omega = \kappa(b + \sigma) > 0, \quad a_0 = \frac{\lambda H(0)^2}{N_0}, \quad \delta(k) = \omega + \operatorname{Re} \psi(k), \quad \Xi(k) = \frac{|k|a_0}{\delta(k)} \quad (\delta(k) > 0). \quad (9.2)$$

The condition $\delta(k) > 0$ defines an admissible frequency set containing a neighborhood of zero; it can have several connected components. Since $1 - \cos u \leq u^2/2$,

$$\operatorname{Re} \psi(k) \geq -\frac{\sigma k^2}{2} \int_{(0,1)} (\log \theta)^2 B(d\theta), \quad (9.3)$$

so if $\sigma > 0$ and the fraction law has a finite second logarithmic moment, the set $\{\delta > 0\}$ contains

$$|k| < \left(\frac{2\omega}{\sigma \int (\log \theta)^2 B(d\theta)} \right)^{1/2}. \quad (9.4)$$

When $\sigma = 0$, the set is all of $\mathbb{R}$. At criticality with equal splitting, $\sigma = b$, $\omega = 2\kappa b$, and $\int (\log \theta)^2 B(d\theta) = 2(\log 2)^2$, so ([9.4](#)) reads $|k| < \sqrt{2\kappa}/\log 2 \approx 0.85$. In this case $\operatorname{Re} \psi(k) = 2b[\cos(k \log 2) - 1]$. With $\theta_0 = \arccos(1 - \kappa)$, the full admissible set is therefore

$$\{k : \delta(k) > 0\} = \bigcup_{j \in \mathbb{Z}} \left(\frac{2\pi j - \theta_0}{\log 2}, \frac{2\pi j + \theta_0}{\log 2} \right).$$

Its component containing zero is $|k| < \theta_0/\log 2 \approx 0.86$. These bands certify factorization; the nonvanishing amplitude needed for quotient identification is guaranteed below only on a neighborhood of zero. The admissible set is inherited from the rate ω in ([3.5](#)), whose asymptotic optimality is not asserted. The factorization may hold at more frequencies, but it is proved only where $\delta > 0$.

Theorem 9.1 (Nonlinear Fourier factorization). *For every real k with $\delta(k) > 0$, there is an amplitude $C_\infty(k)$ such that, for all $t \geq 0$,*

$$|\chi_t(k) - e^{t\psi(k)}C_\infty(k)| \leq \Xi(k)e^{-\omega t}, \quad (9.5)$$

$$|e^{-t\psi(k)}\chi_t(k) - C_\infty(k)| \leq \Xi(k)e^{-\delta(k)t}. \quad (9.6)$$

The set $\{\delta > 0\}$ is open and contains zero, and C_∞ is continuous on it. There is a compact interval I with zero in its interior on which δ and $|C_\infty|$ are bounded below by positive constants. Also $C_\infty(0) = 1$.

Proof. Use the real and imaginary parts of e^{ikz} in (5.5). In the locally absolutely continuous sense,

$$\begin{aligned} \partial_t \chi_t(k) &= \psi(k)\chi_t(k) + r_t(k), \\ r_t(k) &= \frac{\lambda}{N(t)} \iint x[e^{ik \log(x+y)} - e^{ik \log x}]n_t(dx)n_t(dy). \end{aligned} \quad (9.7)$$

The paired increment bound $|e^{iku} - e^{ikv}| \leq |k||u - v|$ and $\log(1 + y/x) \leq \sqrt{y/x}$ give, by (3.5),

$$|r_t(k)| \leq |k| \frac{\lambda H(t)^2}{N(t)} \leq |k| a_0 e^{-\omega t}. \quad (9.8)$$

Variation of constants therefore defines the absolutely convergent integral

$$C_\infty(k) = \chi_0(k) + \int_0^\infty e^{-s\psi(k)} r_s(k) ds. \quad (9.9)$$

Subtracting its tail proves (9.6); multiplying by $e^{t \operatorname{Re} \psi(k)}$ proves (9.5). Since ψ is continuous with $\psi(0) = 0$, the set $\{\delta > 0\}$ is open and contains zero. On each compact subset of it, δ is bounded below by a positive constant and Ξ is bounded, so $e^{-t\psi}\chi_t$, which is continuous in k , converges uniformly to C_∞ by (9.6). Thus the amplitude is continuous on $\{\delta > 0\}$. At zero it equals one, so a compact neighborhood of zero gives the nonvanishing. $\square$

The amplitude includes both the unknown initial law and the accumulated nonlinear interactions. Equation (9.9) gives no independence interpretation for it and does not assert positive definiteness. The notation distinguishes it from the controlled clock $C(t)$ in (2.7).

9.2 Exact-data identification from local frequencies

Corollary 9.2 (Late-time quotients). *Let I be the interval of theorem 9.1. Then $\chi_t(k) \neq 0$ for all sufficiently large t , uniformly in $k \in I$, and for every $h > 0$*

$$R_t(k) := \frac{\chi_{t+h}(k)}{\chi_t(k)} \longrightarrow e^{h\psi(k)} \quad (9.10)$$

uniformly on I . Whenever $\Xi(k)e^{-\delta(k)t} \leq |C_\infty(k)|/2$,

$$|R_t(k) - e^{h\psi(k)}| \leq \frac{2\Xi(k)}{|C_\infty(k)|} e^{h \operatorname{Re} \psi(k)} (1 + e^{-\delta(k)h}) e^{-\delta(k)t}. \quad (9.11)$$

Fix $h > 0$. There is a compact interval $I' \subset I$ with zero in its interior on which the continuous logarithm of the limiting quotient, anchored at zero by $\log 1 = 0$, recovers ψ :

$$\psi(k) = \frac{1}{h} \log \left(\lim_{t \rightarrow \infty} R_t(k) \right), \quad k \in I'. \quad (9.12)$$

Proof. Write $e^{-t\psi(k)}\chi_t(k) = C_\infty(k) + \epsilon_t(k)$. The denominator has modulus at least $|C_\infty(k)|/2$ under the stated condition. Subtract $e^{h\psi(k)}$ from the quotient and bound $|\epsilon_{t+h}(k) - \epsilon_t(k)|$ using (9.6). Uniform bounds on Ξ , δ^{-1} , and $|C_\infty|^{-1}$ on I give the uniform conclusion. Choose $I' \subset I$ so small that $e^{h\psi(k)}$ stays near one and $|h \operatorname{Im} \psi(k)| < \pi$ on I' . The principal logarithm there is continuous and equals $h\psi(k)$; equivalently it is the unique continuous branch anchored at zero. $\square$

The interval I depends on the kinetics and the initial data through ω , a_0 , and the lower bound for $|C_\infty|$; the interval I' depends in addition on h . These quantities are unknown in an identification problem, and no computable procedure for choosing I or I' from observations is offered here. The corollary is a proof of principle.

For completeness, the analytic uniqueness ingredient needs no log moments. It is the classical uniqueness theorem for boundary values of bounded holomorphic functions in a half-plane, in a Paley–Wiener-type form: the transform of a finite measure on a half-line extends to a bounded holomorphic function in a half-plane, and a bounded holomorphic function whose boundary values vanish on a set of positive length is identically zero (a theorem of F. and M. Riesz, extended by Lusin and Privalov). Continuity up to the boundary allows the short reflection proof below.

Lemma 9.3 (One-sided transform uniqueness). *Let ν_1, ν_2 be finite measures on $(-\infty, 0)$. If*

$$\int (e^{iky} - 1) \nu_1(dy) = \int (e^{iky} - 1) \nu_2(dy)$$

on a nonempty open real interval, then $\nu_1 = \nu_2$.

Proof. Put $\Delta = \nu_1 - \nu_2$ and $\mu = \Delta - \Delta(\mathbb{R})\delta_0$. This is a finite signed measure supported on $(-\infty, 0]$, and its Fourier transform vanishes on the specified interval. The function

$$f(z) = \int e^{izy} \mu(dy), \quad \operatorname{Im} z < 0,$$

is holomorphic in the lower half-plane and continuous up to the real boundary. Indeed, the exponential is bounded by one there; on compact subsets of the open half-plane its damping controls every power of $|y|$, justifying differentiation without moments. Its zero boundary values are real. Schwarz reflection across a smaller segment extends f holomorphically through that segment. The extension vanishes on the segment, so the identity theorem makes it zero on an open subset and then throughout the lower half-plane. Boundary continuity gives a zero Fourier transform on all of $\mathbb{R}$. Uniqueness of Fourier transforms of finite signed measures yields $\mu = 0$. Restriction to $(-\infty, 0)$ gives $\Delta = 0$. $\square$

Theorem 9.4 (Structural identification). *Ideal exact knowledge of the normalized size laws at the pairs of times $(t_j, t_j + h)$, for one fixed lag $h > 0$ and some sequence $t_j \rightarrow \infty$, determines $\nu = \sigma(\log)_\# B$ uniquely. If $\sigma > 0$, it determines*

$$\sigma = \frac{1}{2} \nu((-\infty, 0)), \quad B = \sigma^{-1}(\exp)_\# \nu. \quad (9.13)$$

If $\sigma = 0$, it identifies the zero selection rate but cannot identify an unused B . With additionally known conserved mass m and observed unnormalized count growth r_N , it determines

$$r_N = \frac{\log N(t+h) - \log N(t)}{h} = \sigma - \lambda m, \quad \lambda = \frac{\sigma - r_N}{m}. \quad (9.14)$$

Neither the initial distribution nor its characteristic function needs to be known.

Proof. Since the limit in (9.10) exists, the observations along t_j give (9.12) on the open neighborhood $\text{int } I'$ of zero. For two candidate models, intersect their neighborhoods; lemma 9.3 identifies their jump measures. Their common total mass is 2σ , proving (9.13) and the zero-rate case. The count identity (2.8) proves (9.14). $\square$

A fixed change of size unit multiplies both χ_t and χ_{t+h} by the same phase, which cancels in their quotient. Unknown initial data cancel through C_∞ . Absolute mass and unnormalized count growth are separate observations in (9.14). The theorem uses a continuum of exact frequencies in the window I' ; finitely many Fourier values do not supply the one-sided uniqueness conclusion, and the passage from ψ on I' to ν is an analytic continuation from a short interval, whose ill-posedness is quantified below. The measure B records expected daughters and does not specify additional correlations or genealogical information absent from the population balance. The strict support assumption excludes zero log jumps. If one instead allows B on $(0, 1]$ with the same count and mass constraints, an atom at one contributes nothing directly to the exponent. It is nevertheless identified by those constraints. Indeed, the visible measure $\nu_- := \sigma(\log)_\#(B|_{(0,1)})$ is determined by lemma 9.3, and

$$\sigma = \int_{(-\infty, 0)} (1 - e^y) \nu_-(dy), \quad z := \sigma B(\{1\}) = 2\sigma - \nu_-(\mathbb{R}).$$

The integral has a bounded integrand. For $\sigma > 0$ these formulas recover $B = \sigma^{-1}[(\exp)_\# \nu_- + z\delta_1]$. Thus strict reduction is not necessary for this algebraic identification when both daughter constraints are known. The forward solution framework in this paper retains its stated support assumption.

The holomorphic exponent satisfies the mass identity $\psi(-i) = \sigma \int (\theta - 1) B(d\theta) = -\sigma$. This is an identity for the uniquely determined exponent, not an instruction to continue noisy observations to a complex argument. Identification from the window rests on lemma 9.3, that is, on analytic continuation from a short interval, and such continuation is exponentially ill-posed. Quantitatively, let $g = \psi_1 - \psi_2$ for two admissible models with the same σ . Then g is holomorphic in the open lower half-plane, continuous up to the real axis, and bounded there by 8σ . If $|g| \leq \varepsilon$ on I' , the two-constants theorem gives

$$|g(z)| \leq \varepsilon^{\varpi(z)} (8\sigma)^{1-\varpi(z)},$$

where $\varpi(z)$ is the harmonic measure of I' seen from z in the lower half-plane. This exponent tends to zero as z approaches any real frequency outside I' , so the data on I' control no real frequency outside I' without further assumptions. If in addition $\int \theta^{-a} B(d\theta) < \infty$ for some $a > 0$, so that both exponents extend holomorphically to the strip $|\text{Im } z| < a$, the same theorem in the strip slit along I' gives an exponent that decays exponentially with the distance from I' . The resulting bound is then informative only up to distances of order $\log \log(1/\varepsilon)$ from I' . The norm discontinuity that follows is one symptom of this ill-posedness. In fact, take $p \in (0, 1) \setminus \{1/2\}$ and $B_p = \delta_p + \delta_{1-p}$. As $q \rightarrow p$ with the four daughter atoms distinct, the exponents for B_q converge uniformly on each bounded real-frequency interval, while $\|B_q - B_p\|_{\text{var}} = 4$. Thus inversion to the daughter measure in variation norm is not continuous under this data norm, even with fixed positive σ and both daughter constraints. This does not exclude stability in weaker metrics or under additional regularity assumptions, but any such stability must be proved separately; nothing here supplies it.

9.3 Independent sampling: a pointwise sampling bound and observation time

Fix one frequency $k \in I$ and a lag $h > 0$. At the predetermined times t and $t + h$, observe n independent draws from each of the deterministic continuum number laws η_t and η_{t+h} , with the two samples also independent. These are hypothetical observations of continuum laws, not

the interacting particles of [section 7](#), and not auxiliary paths from [section 4](#). Write $U = \chi_t(k)$, $V = \chi_{t+h}(k)$, and let $\widehat{U}, \widehat{V}$ be the corresponding empirical averages of $e^{ik \log X}$. Put $\widehat{R} = \widehat{V}/\widehat{U}$ when defined. For $0 < \varsigma < 1$, set

$$\varepsilon_n = 2\sqrt{\frac{\log(8/\varsigma)}{n}}. \quad (9.15)$$

Proposition 9.5 (Pointwise sampling bound for the ratio). *With probability at least $1 - \varsigma$, both empirical characteristic function errors are at most ε_n . On this event, whenever $|\widehat{U}| > \varepsilon_n$, the exact finite-time ratio is defined and*

$$|\widehat{R} - R_t(k)| \leq \frac{\varepsilon_n(1 + |\widehat{R}|)}{|\widehat{U}| - \varepsilon_n}. \quad (9.16)$$

Let $\delta = \delta(k) > 0$ be as in [\(9.2\)](#), so that $\omega - \delta = -\operatorname{Re} \psi(k) \geq 0$. Choose $\underline{C} > 0$, $\Upsilon < \infty$, and $t_0 \geq 0$ so that for $t \geq t_0$,

$$|U| \geq \frac{\underline{C}}{2} e^{-(\omega - \delta)t}, \quad |R_t(k) - e^{h\psi(k)}| \leq \Upsilon e^{-\delta t}. \quad (9.17)$$

Such constants exist by [theorem 9.1](#) and [corollary 9.2](#). If additionally $\varepsilon_n \leq (\underline{C}/4)e^{-(\omega - \delta)t}$, then on the same event $\widehat{U} \neq 0$ and

$$|\widehat{R} - e^{h\psi(k)}| \leq \Upsilon e^{-\delta t} + \frac{4(2 + \Upsilon)}{\underline{C}} \varepsilon_n e^{(\omega - \delta)t}. \quad (9.18)$$

For all sufficiently large n , the deterministic schedule

$$t_n = \frac{\log(1/\varepsilon_n)}{\omega} \quad (9.19)$$

satisfies these hypotheses and gives

$$|\widehat{R} - e^{h\psi(k)}| \leq \left[\Upsilon + \frac{4(2 + \Upsilon)}{\underline{C}} \right] \varepsilon_n^{\delta/\omega} \quad (9.20)$$

with probability at least $1 - \varsigma$.

Proof. Hoeffding's bounded-variable inequality [\[25\]](#), applied to each real and imaginary component in $[-1, 1]$ at threshold $\varepsilon/\sqrt{2}$, and a union bound over all four components give

$$\mathbb{P}(|\widehat{U} - U| > \varepsilon \text{ or } |\widehat{V} - V| > \varepsilon) \leq 8e^{-n\varepsilon^2/4}.$$

Substitute [\(9.15\)](#). On the resulting event, $|U| \geq |\widehat{U}| - \varepsilon_n$, and

$$\widehat{R} - R_t = \frac{\widehat{R}(U - \widehat{U}) - (V - \widehat{V})}{U}$$

proves [\(9.16\)](#). Under the stronger signal condition, $|\widehat{U}| \geq (\underline{C}/4)e^{-(\omega - \delta)t}$ and $|R_t| \leq 1 + \Upsilon$, because $|e^{h\psi(k)}| \leq 1$. Using instead

$$\widehat{R} - R_t = \frac{\widehat{V} - V - R_t(\widehat{U} - U)}{\widehat{U}}$$

gives the second term in [\(9.18\)](#); add the deterministic bias. Finally, at [\(9.19\)](#) both $e^{-\delta t_n}$ and $\varepsilon_n e^{(\omega - \delta)t_n}$ equal $\varepsilon_n^{\delta/\omega}$, so both terms are constant multiples of this quantity. It tends to zero, which verifies the signal condition eventually; also $t_n \rightarrow \infty$. $\square$

The pointwise sampling bound (9.16) controls sampling error to the exact finite-time ratio; the bias in (9.11) must still be added to estimate $e^{h\psi(k)}$. Its strict denominator condition is required. At fixed confidence, (9.20) is an upper bound of order $n^{-\delta/(2\omega)}$, attained at a time of order $(\log n)/(2\omega)$. This is a sufficient balance of the proved bounds, not a minimax result or an optimal experimental-design theorem. Both the schedule and the exponent use the certified decay coefficient $\omega = \kappa(b + \sigma)$ for H^2/N in (3.5). Its asymptotic optimality is not asserted. For the Golovin benchmark of pure additive coagulation from monodisperse data [23], H^2/N decays roughly like e^{-bt} up to polynomial factors, against the certified $\omega = \kappa b \approx 0.17b$. This benchmark shows that a stronger decay estimate can permit an earlier observation time, but does not prove faster decay throughout the model class. If a stronger exponential forcing bound is established for a given model, the balance can be recomputed with that bound. At frequencies with $\operatorname{Re} \psi(k) < 0$, increasing the decay coefficient improves the resulting sampling exponent $\delta/(2\omega)$. When $\delta = \omega$, that is, when $\operatorname{Re} \psi(k) = 0$, the sampling term does not grow with time and the same schedule remains sufficient. The sampling exponent is then already $1/2$, including in the pure-coagulation benchmark, and a stronger decay bound does not increase it. The constants depend on unknown kinetics and initial data; using the schedule requires their relevant values or prior class bounds. No adaptive choice of time or frequency is proved, and no computable procedure selects the frequency interval I ; the proposition is a proof of principle under a hypothetical observation model. A local logarithm separated from its branch cut transfers pointwise ratio convergence to the exponent, but supplies no full noisy daughter inversion. The logarithmic observation time in sample size also does not validate an estimator based on a single finite reactor: the continuum obstruction in theorem 7.2 concerns a different population parameter and a different observation law.

10 Discussion

The additive equation can hide persistent kinetics from bulk count, mass, and mean size while separating number and mass sampling almost completely. Fractional moments quantify this distinction and also control a different object: the accumulated nonlinear displacement of an auxiliary number path. Its finite log correction explains why fragmentation governs the large-scale logarithmic limits and the late-time Fourier quotient. Its almost-sure finite coagulation count is compatible with infinite expected activity because increasingly rare paths carry large event counts. It is not a statement that the physical population eventually stops coagulating.

Several qualifications are part of the mathematical conclusions. The moment coefficients are sharp only as instantaneous uniform constants; the Golovin–Borel solution and the archived critical simulations decay several times faster, so the estimates are Lyapunov bounds rather than rates. The equal-split daughter envelopes and overlap constants are extremal over their stated classes, but the critical calendar-time exponent of the last-coagulation tail is not determined. The envelopes reduce that exponent to the decay rate of the deterministic number–mass overlap, a question about the equation alone, and the simulations suggest the value $b/2$. Proving that rate, or any asymptotic rate for the fractional moments at criticality, is the main open problem this paper leaves. Parent-dependent daughters retain the controlled path bounds, but do not yield the independent translation-invariant fragmentation reference without additional structure.

The finite-population statement shows that checking count alone can miss a nearly maximal error in the material distribution. Its particle input is only the material ceiling of a finite vessel, so it is a corollary of the moment bound. It gives a sufficient logarithmic horizon for this failure, and does not identify the first failure time or prove accuracy before it. The exact count law, its later diffusion limit, and the retained simulations answer different questions from propagation of a

full size distribution. A growing-time approximation for specific bounded statistics would require a separate argument adapted to those statistics.

The inverse application likewise separates exact identifiability from stable estimation. A nonvanishing local amplitude cancels the unknown initial law, and one-sided support determines the jump measure from exact frequencies on a short low-frequency interval; that step is analytic continuation and is severely ill-posed. The empirical sampling bound makes attenuation of the denominator explicit and provides a sufficient balance between coagulation bias and sampling noise, at a certified rate whose asymptotic optimality is not asserted. It neither adapts the observation time to unknown kinetics nor supplies stable analytic continuation. Establishing recovery of an entire daughter law from noisy data requires additional assumptions and regularization. Connecting such an estimator to measurements in a finite vessel also requires an observation model that accounts for particle dependence and continuum error. These are distinct tasks beyond the factorization proved here.

The preparation algebra in the supplementary note offers a complementary lesson: what can be identified depends on which preparations and observables are available. Preserving material loading does not remove the complete count-neutral family. Same-shape dilution can separate linear and quadratic initial-rate terms when coefficients remain unchanged, but an unknown source leaves an additional ambiguity with only two concentrations. Distributional information, physical preparation constraints, and the sampling law must therefore be specified together.

A Existence and uniqueness for measurable daughter kernels

This appendix proves [theorem 2.3](#). In addition to providing a solution class for the estimates in the paper, the argument explains why mere measurability of the parent-dependent daughter law causes no gap. All the approximations converge in weighted variation, which controls bounded measurable tests directly.

A.1 Weak kernel integrals and a comparison lemma

Throughout this appendix, $w(x) = 1 + x$, so that $\|\mu\|_w = \int w d|\mu|$ is the weighted variation [\(2.5\)](#). If G_t is a signed measure kernel with $\int_0^T \|G_t\|_w dt < \infty$, its integral is the measure defined on Borel sets by integration in time. Countable additivity follows from dominated convergence for the variation kernels, and

$$\left\| \int_s^t G_u du \right\|_w \leq \int_s^t \|G_u\|_w du. \quad (\text{A.1})$$

The same convention applies to other positive weights. This construction does not require strong measurability of G_t as a Banach-space-valued function. For instance, $t \mapsto \delta_t$ need not be strongly measurable in total variation, even though all its time integrals on Borel sets are well defined. We therefore use weak kernel integrals, rather than a Bochner differential equation on measures.

We first establish the norm estimate used below. Suppose a signed measure curve d_t satisfies, in weak kernel form,

$$d_t = d_0 + \int_0^t \{H_s - a_s d_s\} ds, \quad (\text{A.2})$$

where $a_s(x) \geq 0$ is jointly measurable and, for every finite $R, T > 0$, there is a nonnegative $h_{R,T} \in L^1(0, T)$ such that $\sup_{0 < x \leq R} a_s(x) \leq h_{R,T}(s)$ for almost every $s \in (0, T)$. Assume also that $t \mapsto d_t(A)$ is measurable for each Borel A and that both H_s and $a_s d_s$ have integrable weighted variation on $(0, T)$. The loss integrating factor gives

$$d_t(dx) = e^{-\int_0^t a_u(x) du} d_0(dx) + \int_0^t e^{-\int_s^t a_u(x) du} H_s(dx) ds. \quad (\text{A.3})$$

Here is the proof. By Tonelli's theorem, $(s, x) \mapsto \int_s^t a_u(x) du$ is jointly measurable, and it is finite for every x because $a_\cdot(x) \leq h_{x,T}$ on $(0, T)$. The exponential factors are therefore Borel functions with values in $[0, 1]$, and the right-hand side of (A.3) is a finite signed measure on E .

Variations of setwise measurable curves are measurable in time, by a countable supremum: if $t \mapsto \rho_t(A)$ is measurable for each Borel A , then for every Borel A and bounded Borel $\phi \geq 0$,

$$\int_A \phi d|\rho_t| = \sup_{B \in \mathcal{B}_0} \left[\int_{A \cap B} \phi d\rho_t - \int_{A \setminus B} \phi d\rho_t \right], \quad (\text{A.4})$$

where $\mathcal{B}_0$ is the countable algebra of finite unions of intervals with rational endpoints. Indeed each bracket is at most the left side, and the Hahn sets of $\phi\rho_t$ are approximated in $\phi|\rho_t|$ -measure by sets from the generating algebra $\mathcal{B}_0$. Monotone convergence in ϕ extends this to unbounded Borel $\phi \geq 0$, in particular to $\|\rho_t\|_w$.

Fix R, T and restrict the output coordinate to $(0, R]$: for Borel $A \subset (0, R]$, (A.2) reads $d_t(A) = d_0(A) + \int_0^t \{H_s(A) - (a_s d_s)(A)\} ds$. The restriction applies only to the output coordinate; it does not discard any contribution to H_s from outside that set. Two facts follow from the assumptions. First, the restricted curve is bounded in variation: by (A.1) with weight one, $\sup_{t \leq T} |d_t|((0, R]) \leq \|d_0\|_w + \int_0^T (\|H_s\|_w + \|a_s d_s\|_w) ds < \infty$. Second, on $(0, R]$ the loss is a multiplication operator bounded by an integrable function: $|a_s \rho|((0, R]) \leq h_{R,T}(s) |\rho|((0, R])$ for every finite signed measure ρ on $(0, R]$ and almost every s . Consider the linear Volterra equation

$$\rho_t(A) = d_0(A) + \int_0^t H_s(A) ds - \int_0^t (a_s \rho_s)(A) ds, \quad A \subset (0, R] \text{ Borel}, \quad t \leq T, \quad (\text{A.5})$$

in the class of curves ρ of finite signed measures on $(0, R]$ such that $t \mapsto \rho_t(A)$ is measurable for each Borel A and $\sup_{t \leq T} |\rho_t|((0, R]) < \infty$. The restriction of d belongs to this class and solves (A.5). Uniqueness holds in this class: the difference δ of two solutions satisfies $\delta_t = -\int_0^t a_s \delta_s ds$, hence $|\delta_t|((0, R]) \leq \int_0^t h_{R,T}(s) |\delta_s|((0, R]) ds$, and the bounded function $s \mapsto |\delta_s|((0, R])$, measurable by (A.4), vanishes identically by Gronwall's inequality. For existence, let ρ_t be the restriction to $(0, R]$ of the right-hand side of (A.3). It is a finite signed measure with $|\rho_t|((0, R]) \leq |d_0|(E) + \int_0^T |H_s|(E) ds$, and $t \mapsto \rho_t(A)$ is measurable as an integral of a jointly measurable integrand. For Borel $A \subset (0, R]$, the scalar identity

$$\int_r^t a_s(x) e^{-\int_r^s a_u(x) du} ds = 1 - e^{-\int_r^t a_u(x) du},$$

valid because $a_\cdot(x)$ is integrable on $(0, T)$, and Fubini's theorem, applicable because the integrands are dominated by $h_{R,T}(s)|d_0|(A)$ and $h_{R,T}(s)|H_r|(A)$, give

$$\begin{aligned} \int_0^t (a_s \rho_s)(A) ds &= \int_A [1 - e^{-\int_0^t a_u(x) du}] d_0(dx) + \int_0^t \int_A [1 - e^{-\int_r^t a_u(x) du}] H_r(dx) dr \\ &= d_0(A) + \int_0^t H_r(A) dr - \rho_t(A). \end{aligned}$$

So ρ solves (A.5), and by uniqueness ρ_t is the restriction of d_t to $(0, R]$. Since both sides of (A.3) are finite signed measures on E , letting $R \rightarrow \infty$ through the integers and using countable additivity gives the identity on every Borel set. This proves (A.3) for every curve d satisfying the assumptions of this paragraph; in particular, uniqueness is asserted among setwise measurable curves of finite signed measures whose restrictions to bounded sets are bounded in variation on bounded time intervals.

Define the nonnegative majorant

$$z_t(dx) = e^{-\int_0^t a_u(x) du} |d_0|(dx) + \int_0^t e^{-\int_s^t a_u(x) du} |H_s|(dx) ds. \quad (\text{A.6})$$

Then $|d_t| \leq z_t$. Tonelli's theorem and the scalar loss identity give

$$\langle w, z_t \rangle + \int_0^t \langle w a_s, z_s \rangle ds = \|d_0\|_w + \int_0^t \|H_s\|_w ds. \quad (\text{A.7})$$

In particular the integrated loss of z is finite, without any higher moment assumption on z .

Lemma A.1 (Weighted comparison). *In the preceding setting, suppose P_s is a positive linear kernel operator and r_s a signed kernel such that*

$$|H_s| \leq P_s |d_s| + |r_s|, \quad P_s^* w(x) - a_s(x) w(x) \leq c(s) w(x), \quad (\text{A.8})$$

where $c \geq 0$ is locally integrable, $s \mapsto \langle P_s^* w - a_s w, |d_s| \rangle$ is measurable, and all terms paired with $|d_s|$ have finite time integrals. Then

$$\|d_t\|_w \leq e^{\int_0^t c(u) du} \left[\|d_0\|_w + \int_0^t \|r_s\|_w ds \right]. \quad (\text{A.9})$$

Proof. Since $z_s \geq |d_s|$, (A.7) implies

$$\begin{aligned} \|d_t\|_w &\leq \|d_0\|_w + \int_0^t \langle P_s^* w - a_s w, |d_s| \rangle ds + \int_0^t \|r_s\|_w ds \\ &\leq \|d_0\|_w + \int_0^t c(s) \|d_s\|_w ds + \int_0^t \|r_s\|_w ds. \end{aligned}$$

The function $s \mapsto \|d_s\|_w$ is measurable by (A.4) and locally bounded by (A.7), so Gronwall's inequality yields (A.9). Thus no differentiation of a norm or of a Hahn sign is needed. $\square$

A.2 Stability for the additive equation

Let n, v be two nonnegative curves with locally bounded count and second moment, satisfying the population equation with the same coefficients, possibly with an additional signed forcing term r_t in their difference. Put $d = n - v$ and $\mu = (n + v)/2$. Denote the moments of μ by $\bar{N}, \bar{m}, \bar{M}_2$; the two curves need not have equal mass. Polarization of the quadratic coagulation operator gives, against a test f ,

$$\langle f, Q_t(n) - Q_t(v) \rangle = \lambda(t) \iint (x + y) \Delta f(x, y) \mu_t(dx) d_t(dy). \quad (\text{A.10})$$

Write this difference and the fragmentation difference as $H_t - a_t d_t$, where

$$a_t(y) = \lambda(t)(\bar{m}(t) + \bar{N}(t)y) + \sigma(t). \quad (\text{A.11})$$

The terms in H_t are the coagulation gain, the signed cross loss $-\lambda(t)\mu_t(dx) \int (x + y) d_t(dy)$, the fragmentation gain, and r_t . Their variations are bounded by $P_t |d_t| + |r_t|$, where for a nonnegative measure ξ and Borel A ,

$$\begin{aligned} (P_t \xi)(A) &= \lambda(t) \iint (x + y) [\mathbf{1}_A(x + y) + \mathbf{1}_A(x)] \mu_t(dx) \xi(dy) \\ &\quad + \sigma(t) \int b_{t,y}(A) \xi(dy). \end{aligned} \quad (\text{A.12})$$

Although the cross loss is replaced by a positive term, keeping the direct loss $a_t d_t$ preserves the cancellation needed for stability:

$$\begin{aligned} P_t^* w(y) - a_t(y) w(y) &= \lambda(t) \int (x + y) [w(x + y) + w(x) - w(y)] \mu_t(dx) + \sigma(t) \\ &= \lambda(t) [\bar{m} + 2\bar{M}_2 + y(\bar{N} + 2\bar{m})] + \sigma(t) \leq c(t) w(y), \end{aligned} \quad (\text{A.13})$$

where we may choose

$$c(t) = \lambda(t)[\bar{N}(t) + 3\bar{m}(t) + 2\bar{M}_2(t)] + \sigma(t). \quad (\text{A.14})$$

All event integrals in weighted variation are finite under the stated second-moment bounds. For example, the absolute weighted coagulation increment is bounded by $(x+y)[w(x+y)+w(x)+w(y)] = (x+y)[3+2(x+y)]$. Also $a_t(y) \leq \lambda(t)(\bar{m}(t) + R\bar{N}(t)) + \sigma(t)$ for $0 < y \leq R$, providing the required integrable local bound. The measurability hypothesis of [lemma A.1](#) holds because $P_t^*w - a_t w$ in [\(A.13\)](#) is a polynomial in y whose coefficients are $\lambda(t)$, $\sigma(t)$, and moments of the setwise measurable curve μ_t ; more generally, for a product-type kernel $g(x, y) = \sum_i \phi_i(x)\psi_i(y)$ with finitely many Borel terms, $s \mapsto \iint g \mu_s(dx) d_s(dy)$ and the same integral against $|d_s|$ are finite sums of products of measurable functions of s , by setwise measurability and [\(A.4\)](#). All kernels in [\(A.10\)](#) and [\(A.12\)](#) that are paired with w are of this type. [Lemma A.1](#) therefore proves the stability estimate

$$\|n_t - v_t\|_w \leq e^{\int_0^t c(s) ds} \left[\|n_0 - v_0\|_w + \int_0^t \|r_s\|_w ds \right]. \quad (\text{A.15})$$

Solutions in the sense of [definition 2.2](#) are defined through bounded tests only. [Lemma A.1](#) needs [\(A.2\)](#) only on Borel sets, which is [\(2.4\)](#) with indicator tests applied to n and v and subtracted; the integrability of the weighted variations of H_s and $a_s d_s$ follows from the locally bounded second moments and the absolute event bound just displayed. The identities for tests with $|f| \leq w$, obtained from the Borel-set form by truncating f and dominated convergence with that bound, enter only through [\(A.1\)](#), which gives the weighted-variation continuity of $t \mapsto n_t$. In particular [\(A.15\)](#) proves uniqueness in the solution class of [theorem 2.3](#) once existence is established.

A.3 Bounded kernels with a finite third initial moment

First assume $\int (1+x^3)n_0(dx) < \infty$. For integers $r \geq 1$, replace $x+y$ in the coagulation rate by

$$k_r(x, y) = \min(x+y, r), \quad (\text{A.16})$$

while retaining offspring size $x+y$ and the original fragmentation kernel. In the space with norm $\|\mu\|_3 = \int (1+x^3) d|\mu|$, the resulting coagulation vector field is locally Lipschitz, with a time coefficient bounded on each ball by a constant times $r\lambda(t)$. Indeed $1+(x+y)^3 \leq 4[(1+x^3) + (1+y^3)]$, and the bilinear difference identity gives the estimate. Fragmentation is a bounded operator with norm at most $3\sigma(t)$, since

$$\int_E (1+z^3) b_{t,x}(dz) \leq 2+x^3 \leq 2(1+x^3). \quad (\text{A.17})$$

Picard iteration using weak kernel integrals constructs a local solution continuous in this norm, on a time interval whose length depends only on r , the rates, and the radius H of the ball in $\|\cdot\|_3$ containing the initial measure. Here are the relevant measurability and positivity details. A continuous weighted-variation path is a measurable measure kernel; composing it with the given daughter kernel and the coagulation map preserves setwise measurability. Inequality [\(A.1\)](#) makes each integral iterate continuous in norm and gives the usual contraction estimate with an integrable time coefficient. On a nonnegative ball of radius H , the loss rate is at most $r\lambda(t)H + \sigma(t)$. Add this scalar multiple of the measure to the vector field and solve its subtraction by an integrating factor. The remaining map has nonnegative gain, maps a sufficiently short-time nonnegative ball into itself, and is a contraction there. Since the existence time depends only on H , r , and the rates, the construction can be restarted from any time at which $\|n_t^r\|_3$ is finite. Iterating it gives a nonnegative maximal solution n^r , whose existence time is either infinite or a time at which $\|n_t^r\|_3$ is unbounded on the approach.

The tests $1, x, x^2, x^3$ are valid for this bounded kernel in the weighted space just used. Conservation of daughter mass and exact coagulation mass cancellation yield

$$M_1^r(t) = m, \quad N^r(t) \leq N_0 e^{F(t)}. \quad (\text{A.18})$$

Since $z < x$ and the daughter first moment is x , $\int z^q b_{t,x}(dz) \leq x^q$ for $q \geq 1$. Using $k_r \leq x + y$, the polynomial coagulation increments give

$$(M_2^r)' \leq 2\lambda(t)mM_2^r, \quad (\text{A.19})$$

$$(M_3^r)' \leq 3\lambda(t)[mM_3^r + (M_2^r)^2] \leq 6\lambda(t)mM_3^r. \quad (\text{A.20})$$

The final inequality follows from $(M_2^r)^2 \leq mM_3^r$ by Cauchy–Schwarz. Thus on each finite interval $[0, T]$ the count, second moment, and third moment are bounded uniformly in r . Since $\|n_t^r\|_3 = N^r(t) + M_3^r(t)$ then stays bounded on every finite interval, the local existence time is bounded below along the maximal solution, which therefore extends to the whole half-line.

Let Q denote the full additive coagulation operator and Q^r its cutoff. Treat n^r as a solution of the full equation with forcing $e^r = Q^r(n^r) - Q(n^r)$. With $s = x + y$ its weighted norm satisfies

$$\begin{aligned} \|e_t^r\|_w &\leq \frac{\lambda(t)}{2} \iint s \mathbf{1}_{\{s>r\}} (3 + 2s) n_t^r(dx) n_t^r(dy) \\ &\leq \frac{\lambda(t)}{2r} \iint (3s^2 + 2s^3) n_t^r(dx) n_t^r(dy). \end{aligned} \quad (\text{A.21})$$

The two double moments are $2N^r M_2^r + 2m^2$ and $2N^r M_3^r + 6mM_2^r$ respectively. They are uniformly bounded on $[0, T]$. Applying (A.15) to n^r, n^q , with forcing $e^r - e^q$, therefore gives

$$\sup_{t \leq T} \|n_t^r - n_t^q\|_w \leq C_T(r^{-1} + q^{-1}), \quad (\text{A.22})$$

where C_T is independent of r, q . The coefficient in the stability estimate is uniform because it requires only the common second-moment bound, not a bound on the cutoff size.

Completeness gives a nonnegative limit n_t in weighted variation, uniformly on each finite interval. Its count and mass pass to the limit in this norm; its second and third moments obey the preceding upper bounds by monotone truncation and lower semicontinuity. For every bounded Borel test, the coagulation map is continuous in this convergence: polarization and $|\Delta f| \leq 3\|f\|_\infty$ bound its difference by a constant times $\lambda(t)\|n_t^r - n_t\|_w$ when the background counts and masses are bounded. Fragmentation is likewise continuous because $|\mathcal{F}_t f| \leq 3\|f\|_\infty$. Equation (A.21) removes the forcing. Dominated convergence in time passes the complete weak equation, including for merely Borel parent-dependent daughters. The resulting n has all the properties claimed in [theorem 2.3](#) for this initial class.

A.4 Finite second initial moment

For the general initial condition of [theorem 2.3](#), let $n_0^j = \mathbf{1}_{(0,j]} n_0$. Each has finite third moment, and $\|n_0^j - n_0\|_w \rightarrow 0$. The solutions just constructed have masses $m_j \leq m$, initial counts at most N_0 , and the common bounds

$$N^j(t) \leq N_0 e^{F(t)}, \quad M_2^j(t) \leq M_2(0) e^{2m \int_0^t \lambda(s) ds}. \quad (\text{A.23})$$

The unforced estimate (A.15) now makes n^j Cauchy uniformly on $[0, T]$ in weighted variation, with no uniform third moment requirement. The same passage to bounded Borel tests gives a solution

with initial measure n_0 , mass $m_j \rightarrow m$, and the second-moment bound (2.6). Uniqueness has already been proved.

No tightness argument is needed: the limits were taken in the Banach space of finite signed measures on E with the norm $\|\cdot\|_w$, so n_t is itself a finite measure on E with the limiting mass, and no boundary atom can arise. Finally testing with 1 gives (2.8), completing the proof of theorem 2.3.

B The classical product: explicit remainder and a rational certificate

The product in (6.19) is the established Poisson exponential-functional transform of Bertoin et al. [7]. The calculations here give an explicit truncation error and a useful large-argument form for this particular normalization. They concern the conditional size argument q , not the calendar-time tail $h(t)$.

Proposition B.1 (Periodic large-size correction). *For $q \geq 1$, write $\log_2 q = n + \theta$ with integer $n \geq 0$ and $0 \leq \theta < 1$; the hypothesis $q \geq 1$ is what makes $n \geq 0$. Then*

$$\log \Psi(q) = -\frac{(\log q)^2}{2 \log 2} + \frac{1}{2} \log q - \mathcal{P}(\theta) + \mathcal{R}(q), \quad (\text{B.1})$$

where

$$\mathcal{P}(\theta) = \frac{\log 2}{2}(\theta - \theta^2) + \sum_{j=0}^{\infty} \log(1 + 2^{-j-\theta}) + \sum_{\ell=1}^{\infty} \log(1 + 2^{\theta-\ell}), \quad (\text{B.2})$$

$$\mathcal{R}(q) = \sum_{r=0}^{\infty} \log(1 + q^{-1}2^{-r}), \quad 0 \leq \mathcal{R}(q) \leq 2/q. \quad (\text{B.3})$$

The function $\mathcal{P}$ extends to a bounded Lipschitz one-periodic function on $\mathbb{R}$, with Lipschitz constant at most $\frac{9}{2} \log 2$. More precisely,

$$\frac{2}{q} - \frac{2}{3q^2} \leq \mathcal{R}(q) \leq \frac{2}{q} - \frac{2}{3q^2} + \frac{8}{21q^3}. \quad (\text{B.4})$$

The bounds (B.3) and (B.4) hold for every $q > 0$.

Proof. Put $a = n + \theta = \log_2 q$. Split $-\log \Psi(q)$ at $k = n$ and factor 2^{a-k} from each of the first n factors. This gives

$$\begin{aligned} -\log \Psi(q) &= \log 2 \left(na - \frac{n(n+1)}{2} \right) + \sum_{j=0}^{n-1} \log(1 + 2^{-j-\theta}) + \sum_{\ell=1}^{\infty} \log(1 + 2^{\theta-\ell}) \\ &= \frac{\log 2}{2}(a^2 - a + \theta - \theta^2) + \sum_{j=0}^{\infty} \log(1 + 2^{-j-\theta}) + \sum_{\ell=1}^{\infty} \log(1 + 2^{\theta-\ell}) - \mathcal{R}(q). \end{aligned}$$

The removed completion is $\sum_{j=n}^{\infty} \log(1 + 2^{-j-\theta}) = \mathcal{R}(q)$; this proves (B.1), also when $n = 0$. For $\theta \in [0, 1]$ the two series are uniformly dominated by geometric series, so their sums are continuous. At zero and one, their combined values both equal $\log 2 + 2 \sum_{j \geq 1} \log(1 + 2^{-j})$, and the polynomial term vanishes. Hence the endpoint values agree, giving the periodic extension. Its boundedness follows from continuity on $[0, 1]$. The termwise derivatives of the two series are bounded by $(\log 2)2^{-j}$ and $(\log 2)2^{1-\ell}$, whose sums are $2 \log 2$ each, and the polynomial term has derivative bounded by

$\frac{1}{2} \log 2$; so $\mathcal{P}$ is continuously differentiable on $[0, 1]$ with $|\mathcal{P}'| \leq \frac{9}{2} \log 2$. A continuous one-periodic function that is Lipschitz on $[0, 1]$ is Lipschitz on $\mathbb{R}$ with the same constant.

The inequalities $0 \leq \log(1+x) \leq x$ give (B.3). For $x \geq 0$, $x - x^2/2 \leq \log(1+x) \leq x - x^2/2 + x^3/3$: both differences vanish at zero and have derivatives $x^2/(1+x)$ and $x^3/(1+x)$. Sum these with $x = q^{-1}2^{-r}$; the geometric sums are $2/q$, $4/(3q^2)$, and $8/(7q^3)$, proving (B.4). Neither remainder bound uses $q \geq 1$; that hypothesis enters only through $n \geq 0$ in the decomposition (B.1). $\square$

In particular the leading term is a negative square of $\log q$, with a bounded periodic correction. This agrees with the faster-than-every-power estimate (6.45), which was sufficient for the scalar dissipation counterexample.

Proposition B.2 (Finite-product certificate). *For $q \geq 0$ and integer $k \geq 1$, put*

$$P_k(q) = \prod_{j=1}^k (1 + q/2^j)^{-1}.$$

Then

$$P_k(q)e^{-q/2^k} \leq \Psi(q) \leq P_k(q). \quad (\text{B.5})$$

If $q/2^k < 1$, the lower bound $P_k(q)(1 - q/2^k)$ is positive and uses only rational operations when q is rational. In particular

$$0.580577558204892402290043892297 < c_* < 0.580577558204892402290043892298. \quad (\text{B.6})$$

Proof. The omitted log sum lies between zero and $\sum_{j>k} q2^{-j} = q2^{-k}$. Exponentiating proves (B.5); $e^{-x} \geq 1 - x$ gives its rational version. At $q = 1$, $k = 128$, form the exact rational number $P = \prod_{j=1}^{128} 2^j/(2^j + 1)$. If ℓ and u are the two terminating decimals in (B.6), respectively, integer cross-multiplication verifies

$$\ell < 1 - P \leq c_* \leq 1 - P(1 - 2^{-128}) < u.$$

The following complete Python calculation reproduces the finite arithmetic part with the standard library, without floating point:

```
from fractions import Fraction as F
P = F(1)
for k in range(1, 129):
    P *= F(2**k, 2**k + 1)
lo = F("0.580577558204892402290043892297")
hi = F("0.580577558204892402290043892298")
assert lo < 1 - P
assert 1 - P * (1 - F(1, 2**128)) < hi
```

$\square$

The accompanying standard-library script `supplement/last_coagulation_exact.py` additionally writes outward-rounded decimal bounds and floating-point evaluations of (6.44). Its saved JSON distinguishes the rational certificate from those illustrative evaluations. The proofs of asymptotics and scalar obstruction do not depend on the floating-point examples.

C Exact count generating function and the classical diffusion limit

The count component in [proposition 7.1](#) is a critical linear branching process with immigration after shifting by one. Its generating function and diffusion are classical objects: the limiting diffusion is the branching diffusion of Feller [18], and the diffusion limit of branching processes with immigration is due to Kawazu and Watanabe [26]; birth–death immigration approximations to Feller diffusions are discussed, for example, by Giorno and Nobile [22]. We provide the specialization and path argument to fix the factors in the present physical normalization.

Proposition C.1 (Generating function and count-scale limit). *Let $Y_t = L^n(t) - 1$. For $y \in \{0, 1, \dots\}$ and $0 \leq z \leq 1$,*

$$\mathbb{E}[z^{Y_t} \mid Y_0 = y] = \frac{u_t(z)^y}{1 + bt(1 - z)}, \quad u_t(z) = 1 - \frac{1 - z}{1 + bt(1 - z)}. \quad (\text{C.1})$$

If $c_n = L_0^n/n \rightarrow c \geq 0$, then, on every finite interval $[0, T]$,

$$Z_n(\tau) := L^n(n\tau)/n \implies Z(\tau) \quad \text{in } D([0, T], [0, \infty)) \text{ with its } J_1 \text{ topology}, \quad (\text{C.2})$$

where Z has continuous paths and solves, in the weak sense,

$$dZ = b d\tau + \sqrt{2bZ} dW, \quad Z(0) = c. \quad (\text{C.3})$$

Its transition law is determined by

$$\mathbb{E}_c e^{-\theta Z(\tau)} = \frac{1}{1 + b\tau\theta} \exp\left(-\frac{c\theta}{1 + b\tau\theta}\right), \quad \theta \geq 0, \quad (\text{C.4})$$

and $\mathbb{E}_c Z(\tau) = c + b\tau$, $\text{Var}_c Z(\tau) = 2bc\tau + b^2\tau^2$.

Proof. The shifted chain has birth rate $b(Y + 1)$ and death rate bY . Realize it as independent critical birth–death families, each with per-individual birth and death rates b , together with immigration at Poisson rate b . The generating function for one family started from one individual solves $\partial_t u = b(u - 1)^2$, $u_0(z) = z$, by conditioning on its first event. Solving gives the displayed u_t . The y initial families contribute $u_t(z)^y$. Poisson immigration contributes

$$\exp\left(b \int_0^t [u_s(z) - 1] ds\right) = [1 + bt(1 - z)]^{-1},$$

proving (C.1). Nonexplosion from [proposition 7.1](#) justifies the family construction on finite intervals.

Substitute $t = n\tau$, $z = e^{-\theta/n}$, and $y = L_0^n - 1$ into (C.1); multiply by $e^{-\theta/n}$ to account for the shift. Since $n(1 - e^{-\theta/n}) \rightarrow \theta$, the resulting transform converges to (C.4). For fixed τ, θ this convergence is uniform when the scaled starting count stays in a bounded set. The limiting transform is continuous in that starting count. The Markov property, compact truncation of starting states, and this uniform transition convergence identify all finite-dimensional laws. Equivalently one may iterate the conditional Laplace transforms; products of exponentials form a convergence-determining class for nonnegative finite-dimensional vectors.

To verify path tightness, the scaled martingale decomposition is

$$Z_n(\tau) = c_n + b\tau + \mathcal{M}_n(\tau), \quad \langle \mathcal{M}_n \rangle_\tau = \int_0^\tau (2bZ_n(s) - b/n) ds. \quad (\text{C.5})$$

Since Z_n is a nonnegative submartingale,

$$\mathbb{P}\left(\sup_{\tau \leq T} Z_n(\tau) > R\right) \leq \frac{c_n + bT}{R}.$$

This proves compact containment. Stop at $\tau_R^n = \inf\{\tau : Z_n(\tau) > R\}$. The stopped bracket has rate at most $2bR$ before this time. For stopping times $S_n \leq U_n \leq T$ with $U_n - S_n \leq \delta$, the martingale isometry gives

$$\mathbb{E}|\mathcal{M}_n(U_n \wedge \tau_R^n) - \mathcal{M}_n(S_n \wedge \tau_R^n)|^2 \leq 2bR\delta.$$

The stopped drift changes by at most $b\delta$. Chebyshev's inequality, followed by $\delta \downarrow 0$ and then $R \uparrow \infty$, verifies the stopping-time tightness criterion for D ; see Billingsley [8, Section 16]. The jumps of Z_n have size $1/n$, so every subsequential limit is continuous. Finite-dimensional convergence therefore identifies the whole path limit in (C.2).

For an explicit construction, let B_1, B_2 be independent standard Brownian motions and set

$$Z(\tau) = \frac{b}{2} \left[(\sqrt{2c/b} + B_1(\tau))^2 + B_2(\tau)^2 \right]. \quad (\text{C.6})$$

Itô's formula gives drift b and martingale quadratic variation $2bZ(\tau) d\tau$. Normalize the radial Brownian increment when $Z > 0$ and choose a fixed unit direction when $Z = 0$; its stochastic integral has quadratic variation τ and hence is Brownian, giving (C.3). Rotational invariance of the two-dimensional Brownian increments shows that (C.6) is Markov and that its transitions depend only on its current squared radius. A Gaussian integral then gives (C.4) with the current state in place of c . These are precisely the limiting transitions already obtained. The same Gaussian calculation gives the displayed mean and variance. Thus the limit is a rescaled squared Bessel process of dimension two; it is independent of daughter law and initial mass allocation once b and c are fixed. $\square$

Order-one count fluctuations emerge on time n , whereas theorem 7.2 already forces a mass-CDF discrepancy on a logarithmic time scale. The classical diffusion is therefore a description of a later count scale, not an explanation that requires count collapse at the earlier mass-discrepancy horizon.

D Second-moment and entropy balances under finite second moments

We justify the tests used in proposition 8.3 directly in definition 2.2, assuming the locally bounded second moments of theorem 2.3. There is no additional logarithmic integrability assumption: $|x \log x| \leq e^{-1} + x^2$ implies finite entropy at every finite time.

For $R \geq 1$, let f be either x^2 or $x \log x$ and replace its tail by its tangent at R :

$$f_R(x) = \begin{cases} f(x), & 0 < x \leq R, \\ f(R) + f'(R)(x - R), & x > R. \end{cases} \quad (\text{D.1})$$

The function $f_R(x) - f'(R)x$ is bounded on E . Apply the bounded weak equation to this function and add the conserved first-moment balance. That balance is (2.4) with $f = x$, which is the trivial identity $m - m = 0$: its left side vanishes by the constant first moment in definition 2.2, and its right side vanishes identically because $\Delta x = 0$ and $\mathcal{F}_t x = 0$ by daughter mass conservation. So nothing beyond definition 2.2 is used. Linearity then gives the integrated weak identity for f_R without appealing to an unproved unbounded-test rule.

Both f_R and $f - f_R$ extend continuously to zero with value zero and are convex. For any such convex function a , $a(x)/x$ is nondecreasing; hence $a(x+y) \geq a(x) + a(y)$ and $\int a(z)b_{t,x}(dz) \leq a(x)$. Applying these facts to both convex functions proves

$$0 \leq \Delta f_R(x, y) \leq \Delta f(x, y), \quad (\text{D.2})$$

$$0 \leq f_R(x) - \int f_R(z)b_{t,x}(dz) \leq f(x) - \int f(z)b_{t,x}(dz). \quad (\text{D.3})$$

For fixed x , f_R agrees with f on the whole daughter interval once $R \geq x$, so the daughter increments converge pointwise as well.

For $f = x^2$, the respective majorants are

$$(x+y)\Delta f(x, y) = 2xy(x+y), \quad 0 \leq f(x) - \int f(z)b_{t,x}(dz) \leq x^2.$$

Their population integrals are $4mM_2(t)$ and at most $M_2(t)$. For $f = x \log x$, writing the merger increment through binary entropy gives

$$0 \leq (x+y)\Delta f(x, y) \leq (\log 2)(x+y)^2, \quad 0 \leq f(x) - \int f(z)b_{t,x}(dz) \leq x \log 2.$$

The latter bound follows from Jensen against $b_{t,x}/2$. The double integral of $(x+y)^2$ equals $2N(t)M_2(t) + 2m^2$. All these majorants, multiplied by their locally integrable rates, are integrable on every finite time interval. At the endpoints, $0 \leq f_R \leq x^2$ in the first case and $|f_R| \leq e^{-1} + x^2$ in the second. Dominated convergence in the complete integrated balance therefore proves both unbounded-test identities and local absolute continuity. Explicitly, for the rates of [assumption 2.1](#),

$$M'_2(t) = 2\lambda(t)mM_2(t) + \sigma(t) \int \left[\int z^2 b_{t,x}(dz) - x^2 \right] n_t(dx), \quad (\text{D.4})$$

$$\begin{aligned} \mathcal{H}'(t) &= \frac{\lambda(t)}{2} \iint (x+y)\Delta(x \log x)(x, y) n_t(dx) n_t(dy) \\ &\quad + \sigma(t) \int \left[\int z \log z b_{t,x}(dz) - x \log x \right] n_t(dx), \end{aligned} \quad (\text{D.5})$$

for almost every t . At criticality λ is constant and $\sigma = b$, which gives the forms used in [proposition 8.3](#).

For the instantaneous sharpness assertion, take time-independent equal splitting and monodisperse initial data. The just-proved second-moment balance gives continuity of $M_2(t)$ at zero, while the solution is continuous in total variation. These imply $\int (1+x^2)|n_t - n_0|(dx) \rightarrow 0$ as $t \downarrow 0$: first truncate x^2 on $(0, R]$ and use total-variation convergence; the remaining tail integrals are uniformly small as $t \downarrow 0$ because their total second moments converge to $M_2(0)$. More explicitly, subtract the integrals of $\min(x^2, R^2)$ from the convergent second moments, and then control the x^2 tail beyond $\sqrt{2}R$ by twice that remainder. The entropy coagulation integrand is bounded by $2(\log 2)(x^2 + y^2)$, so its double integral is continuous at zero in this weighted variation. The equal-split fragmentation increment is exactly $-x \log 2$ and is also continuous. Thus the integrated entropy balance has a right derivative at zero, equal to the monodisperse vector-field value $\lambda m^2 \log 2$.

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